User Guide and How To Run

Installing, running and using IntelliMetrics AI, with screenshots of every page

Project title	IntelliMetrics AI: A Governed Semantic Layer with TARA, an AI Assistant for Trusted Business Metrics
Document	User Guide and How To Run
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Institution	KL University
Academic year	2026-27
Date	26-Oct-2026 (IST)
Status	University project: academic capstone with synthetic demonstration data

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1 About this guide

IntelliMetrics AI is a **university project** (academic capstone) with synthetic demonstration data. This guide explains how to run it on a Windows computer, how to sign in, what each page does, and a ten-minute walkthrough that reproduces the main results. All screenshots were captured from the running application by the automated script `frontend/e2e-docs/screens.spec.ts`.

2 What you need

Requirement	Version used in development	Needed for
Python	3.11	Back end
Node.js and npm	Node 22	Front end
A modern browser	Chrome or Edge	Using the application
Docker Desktop	not used on the development machine	Optional: the Compose route (Section 5)

Everything else (database, analytics engine, background jobs) is installed by the commands below. **No internet connection is needed to run the application**, except to download the packages the first time and to show the stock photographs on the public pages (a gradient is shown instead when they cannot load). **No paid service, account or API key is required.**

3 Run it without Docker (recommended, tested route)

This is the route that was run and tested for this report. It uses SQLite for application data and a background thread for dataset processing, so Redis and PostgreSQL are not needed.

3.1 Step 1: back end

Open **PowerShell** in the project folder:

```
cd backend
python -m venv .venv
.venv\Scripts\activate
pip install -r requirements.txt
python -m app.seed
uvicorn app.main:app --reload --port 8000
```

`python -m app.seed` creates the database and the demonstration data (about 2.7 seconds on the development machine). Leave the window open. The API documentation is at http://localhost:8000/docs and the health check at http://localhost:8000/health.

3.2 Step 2: front end

Open a **second** PowerShell window:

```
cd frontend
npm install
npm run dev
```

Open http://localhost:5173. The development server forwards `/api` requests to port 8000.

3.3 Step 3: reset the demonstration at any time

```
scripts\reset-demo.ps1      # Windows
./scripts/reset-demo.sh     # Git Bash / Linux / macOS
```

This deletes the local database and uploaded files and seeds again. Stop the back end first, because Windows locks the SQLite file while it is running.

4 Run it with Docker Compose (not executed in development)

```
cp .env.example .env     # set SECRET_KEY and POSTGRES_PASSWORD
docker compose up --build
```

Open http://localhost:3000. This starts PostgreSQL, Redis, the API, a Celery worker and an Nginx front end. **This path was written but has not been run**, because Docker is not installed on the development machine; the services it starts were each verified separately (see the Testing Report). If a step fails, the Deployment Guide has a troubleshooting table.

5 Configuration

Settings are environment variables, listed in full in the Deployment Guide. The ones a student is likely to touch:

Variable	Default	Meaning
`AI_PROVIDER`	`none`	`none` = TARA's offline deterministic mode (no key needed). `openai` or `gemini` plus the matching `*_API_KEY` turns on live AI planning
`AUTO_SEED`	`true` in `.env.example`	Seeds demo data when the database is empty
`USE_CELERY`	`false`	`true` uses Celery and Redis for dataset processing
`DATABASE_URL`	local SQLite file	Use a `postgresql://` URL for PostgreSQL
`MAX_UPLOAD_MB`	500	Upload size limit

If no AI key is set, every TARA screen says that it is in **offline deterministic mode**. It still answers all the suggested questions below, because the questions are interpreted by a rule-based planner and the numbers always come from the data.

6 Demonstration accounts

All accounts are marked **DEMO**, and the login page has a *Demo accounts* panel that fills the form when clicked.

Role	Person (fictional)	Email	Password	Company
Admin	Aarav Menon	`admin@intellimetrics.demo`	`Admin@12345`	Bharat Retail Pvt Ltd
Data Analyst	Diya Reddy	`analyst@intellimetrics.demo`	`Analyst@12345`	Bharat Retail Pvt Ltd
Business User	Rohan Sharma	`business@intellimetrics.demo`	`Business@12345`	Bharat Retail Pvt Ltd
Reviewer	Meera Iyer	`reviewer@intellimetrics.demo`	`Reviewer@12345`	Bharat Retail Pvt Ltd
Admin (isolation demo)	Suresh Naidu	`admin@sagarfoods.demo`	`Admin@12345`	Sagar Foods & Beverages Ltd
Analyst (isolation demo)	Kavya Raju	`analyst@sagarfoods.demo`	`Analyst@12345`	Sagar Foods & Beverages Ltd

The two companies are separate *workspaces*. Sign in as Suresh Naidu to confirm that none of Bharat Retail's metrics, datasets or conversations are visible.

6.1 What each role can do

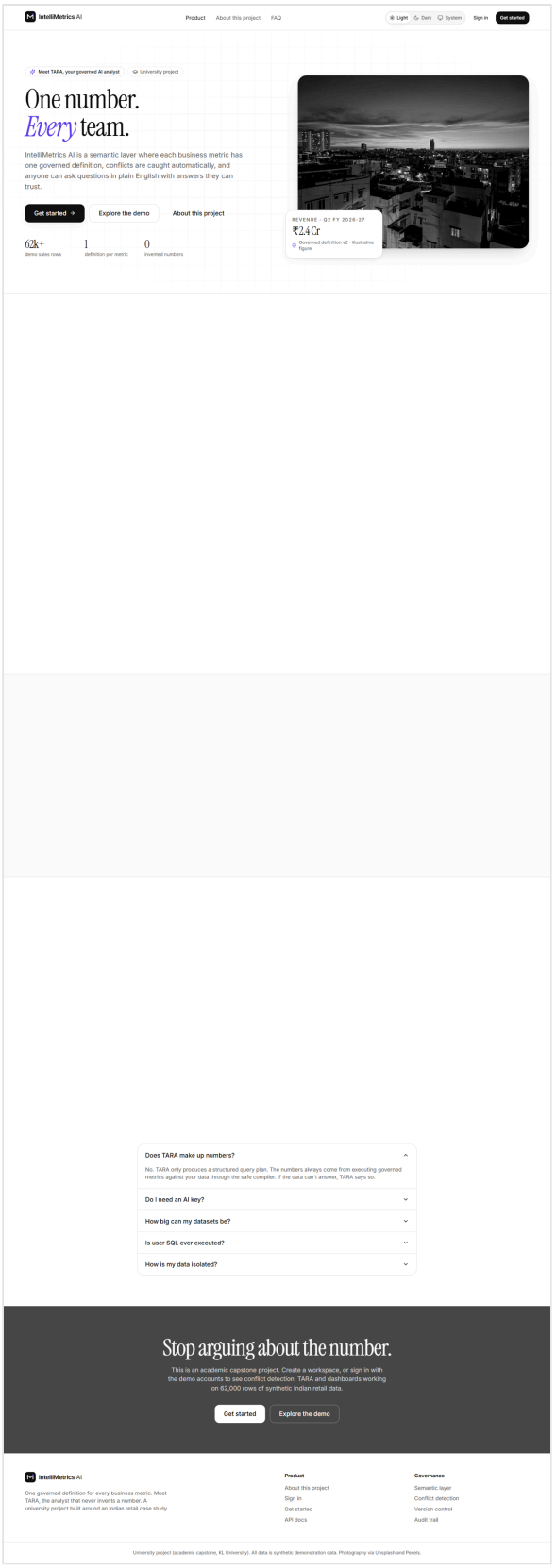
Capability	Admin	Data Analyst	Business User	Reviewer
View metrics catalog, datasets, dashboards (`catalog.view`)	Yes	Yes	Yes	Yes
Upload datasets / manage connections (`dataset.write`)	Yes	Yes	-	-
Delete datasets and connections (`dataset.delete`)	Yes	-	-	-
Create and edit metrics (`metric.write`)	Yes	Yes	-	-
Approve / reject / publish / deprecate metrics (`metric.review`)	Yes	-	-	Yes
Roll back a metric (creates a new version) (`metric.rollback`)	Yes	-	-	Yes
Use TARA metric generator (`ai.generate`)	Yes	Yes	-	-

Capability	Admin	Data Analyst	Business User	Reviewer
Ask TARA (natural-language BI) (`chat.ask`)	Yes	Yes	Yes	Yes
View conflicts (`conflict.view`)	Yes	Yes	Yes	Yes
Run conflict detection (`conflict.detect`)	Yes	Yes	-	Yes
Assign / review / resolve conflicts (`conflict.manage`)	Yes	-	-	Yes
Formula Lab: validate and execute (`formula.run`)	Yes	Yes	-	Yes
Create / edit dashboards (`dashboard.write`)	Yes	Yes	Yes	-
View performance evaluation (`evaluation.view`)	Yes	Yes	-	Yes
Run performance evaluation (`evaluation.run`)	Yes	Yes	-	-
Export reports (`report.export`)	Yes	Yes	Yes	Yes
Manage users and roles (`user.manage`)	Yes	-	-	-
Manage workspace and settings (`workspace.manage`)	Yes	-	-	-
View audit log (`audit.view`)	Yes	-	-	-

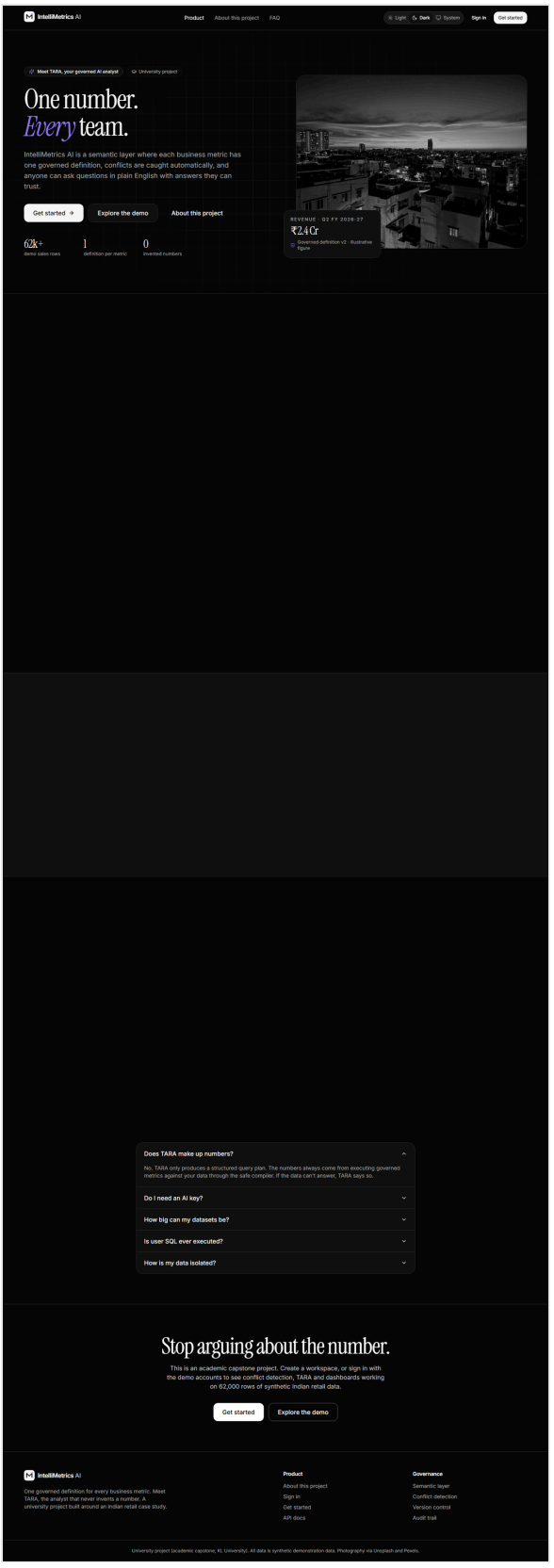
7 Tour of the application

7.1 Public pages

Landing page

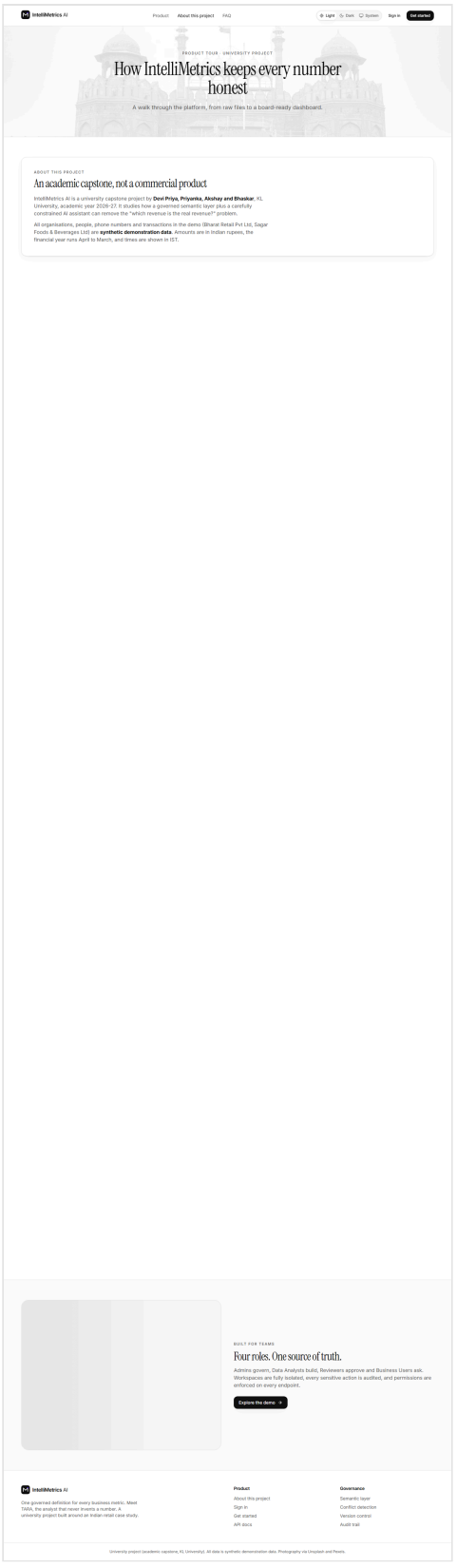


Landing page, light mode

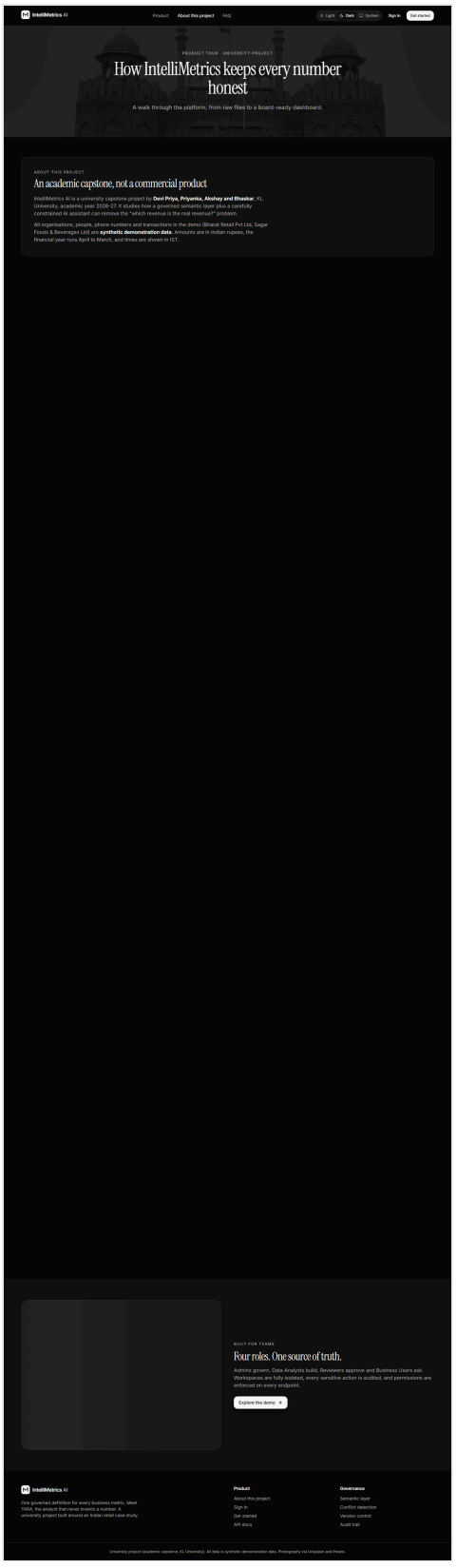


Landing page, dark mode

About / product tour

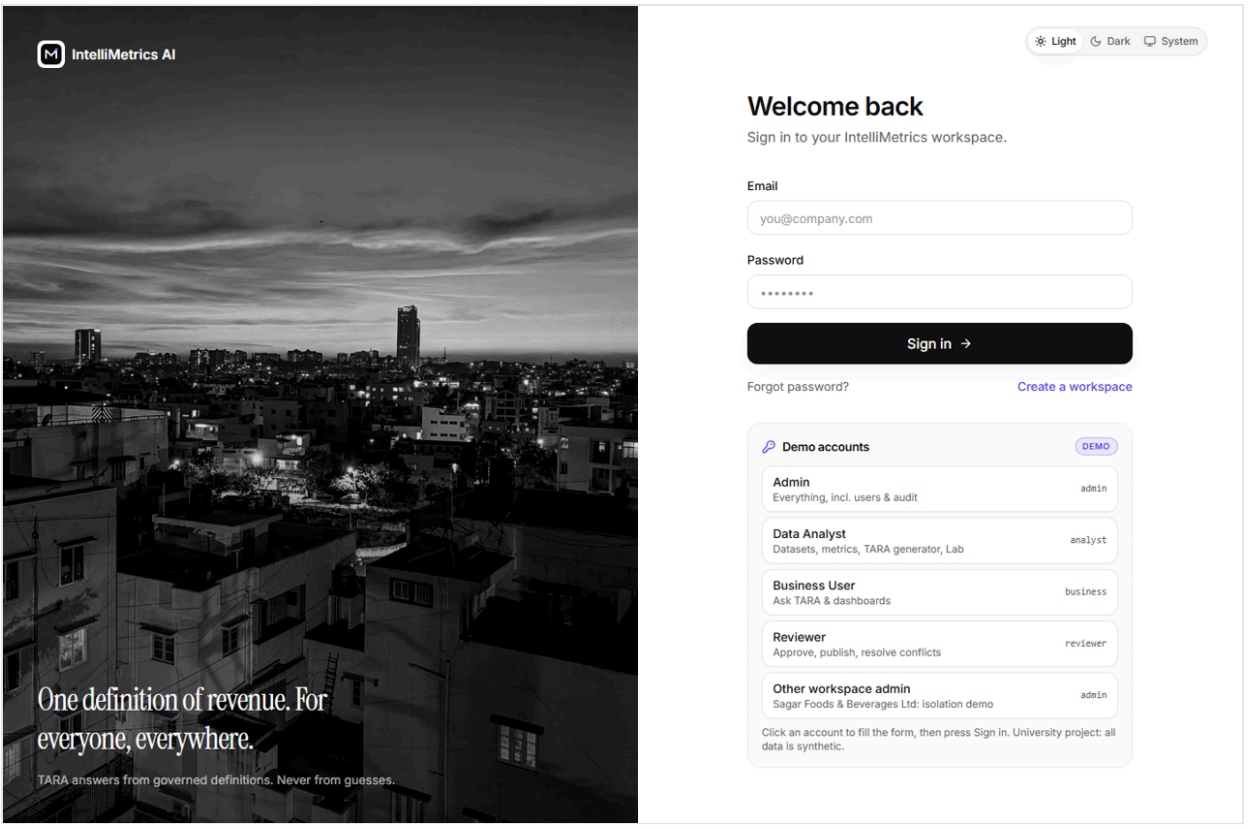


About / product tour, light mode

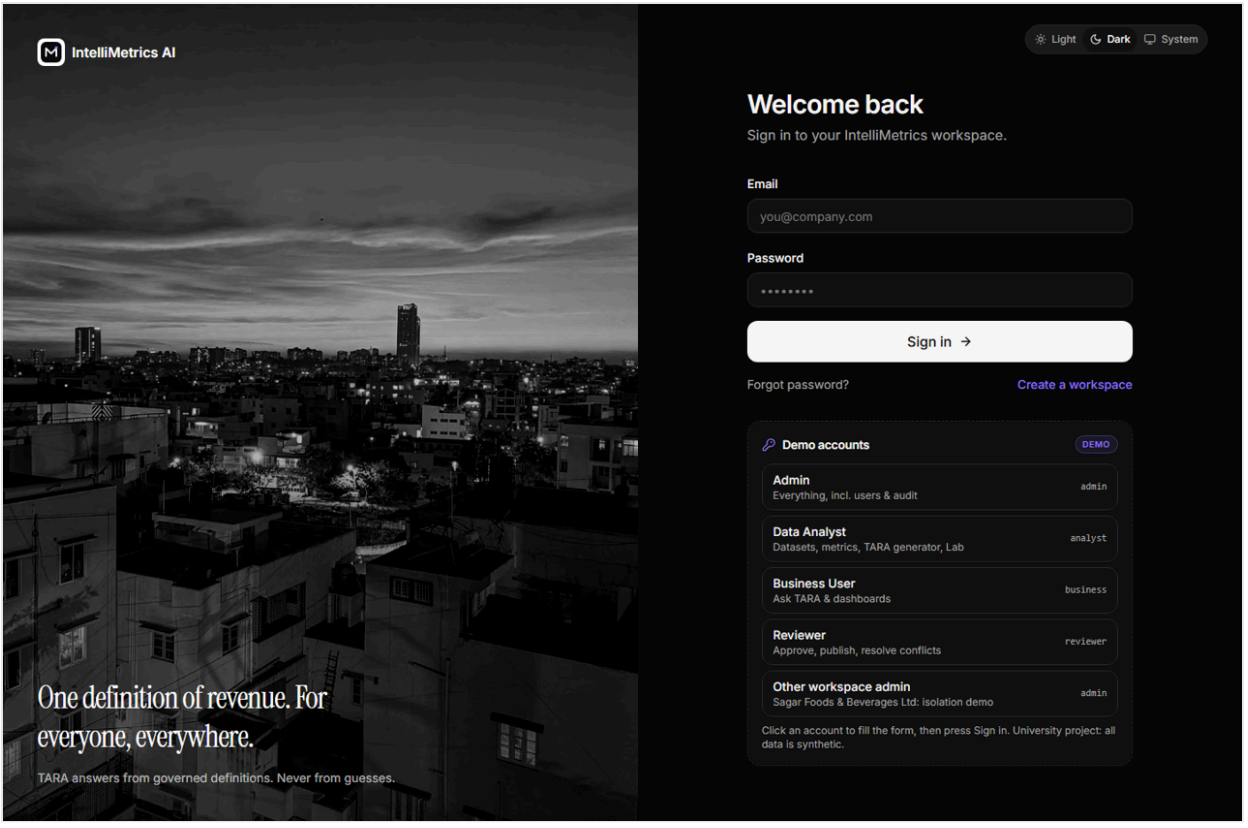


About / product tour, dark mode

Login with demo accounts panel

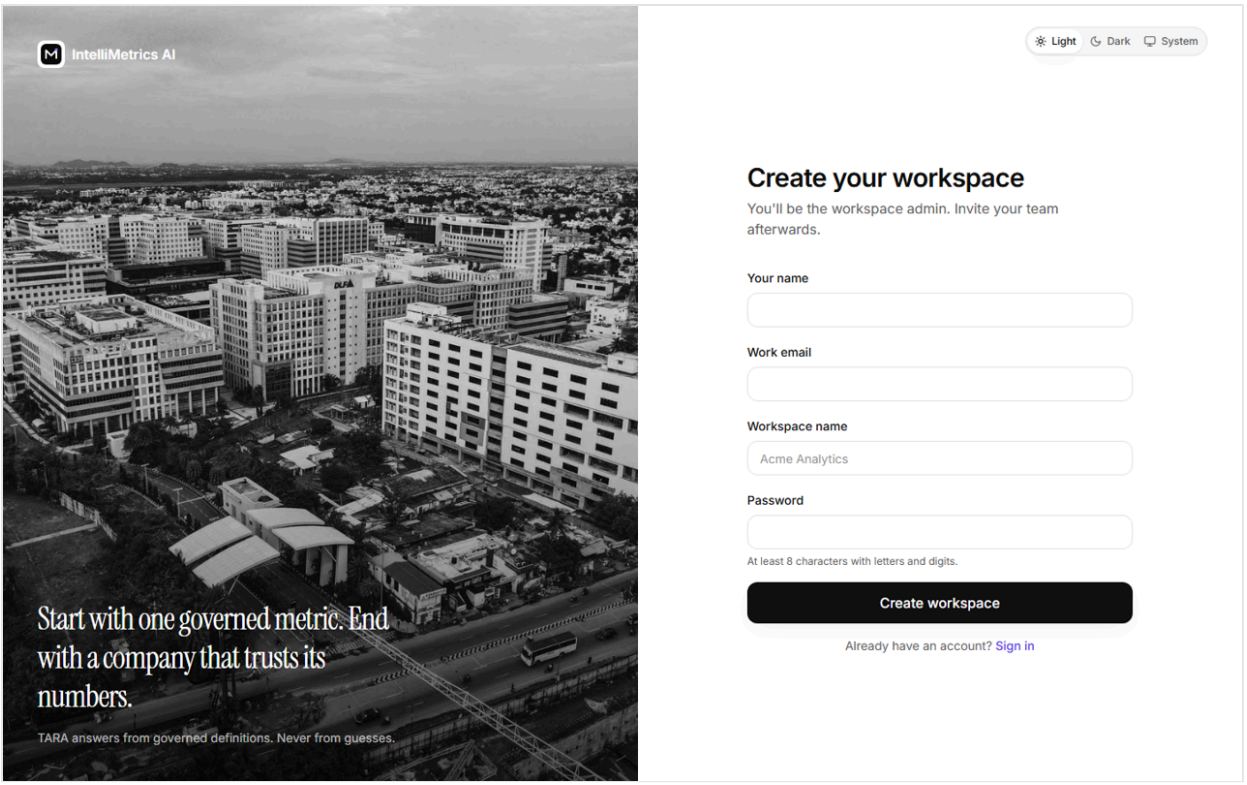


Login with demo accounts panel, light mode

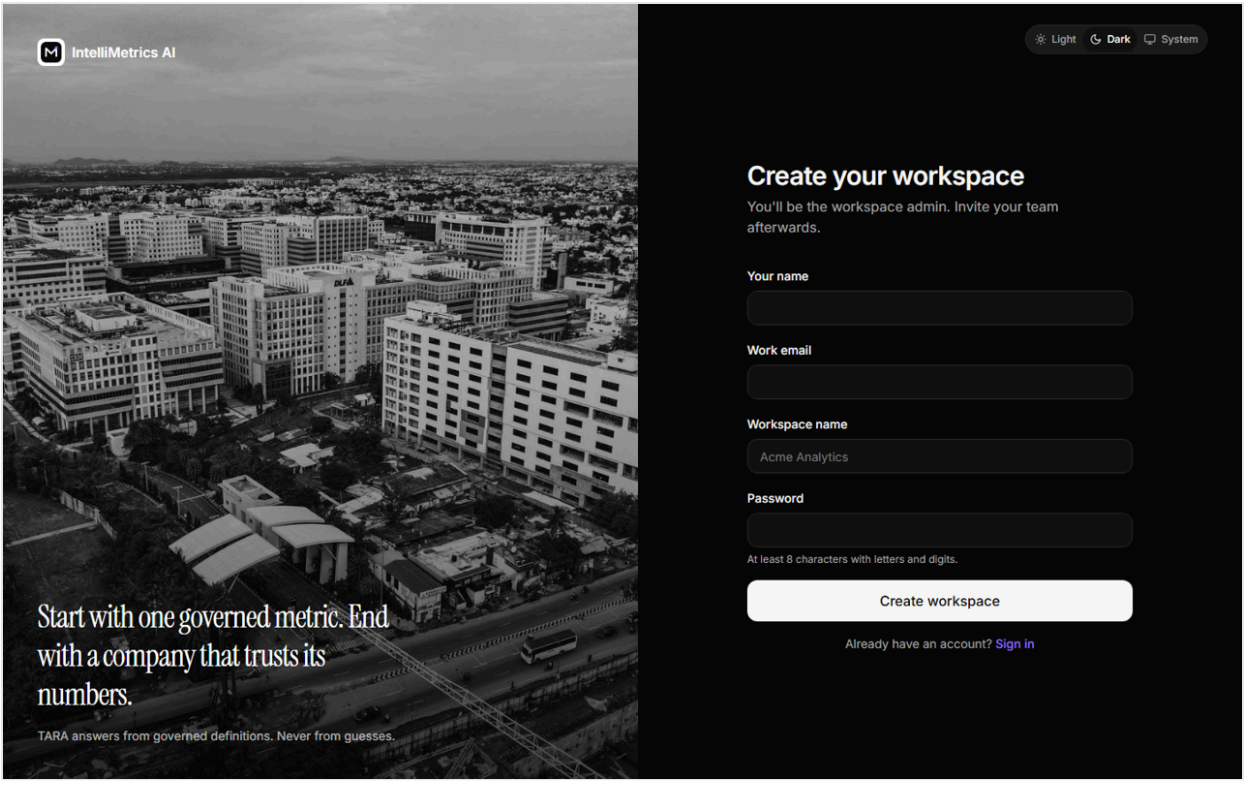


Login with demo accounts panel, dark mode

Register

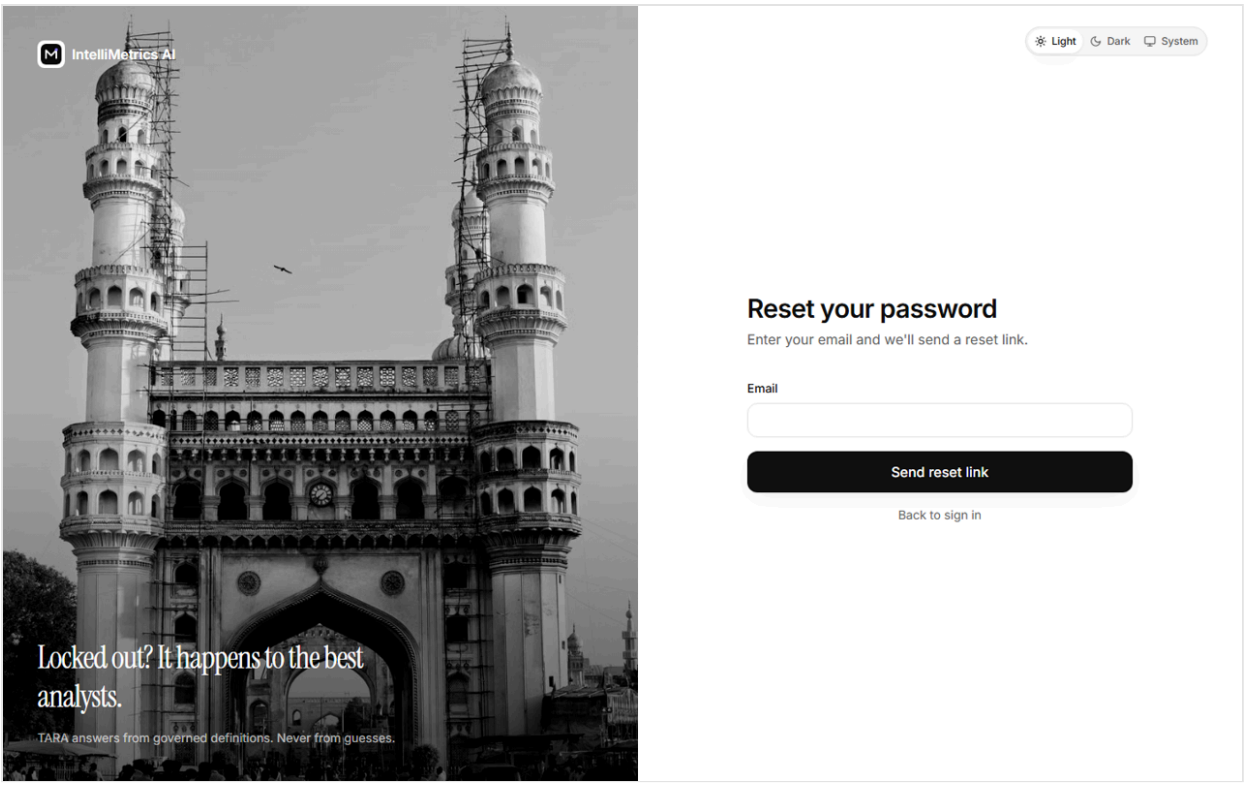


Register, light mode

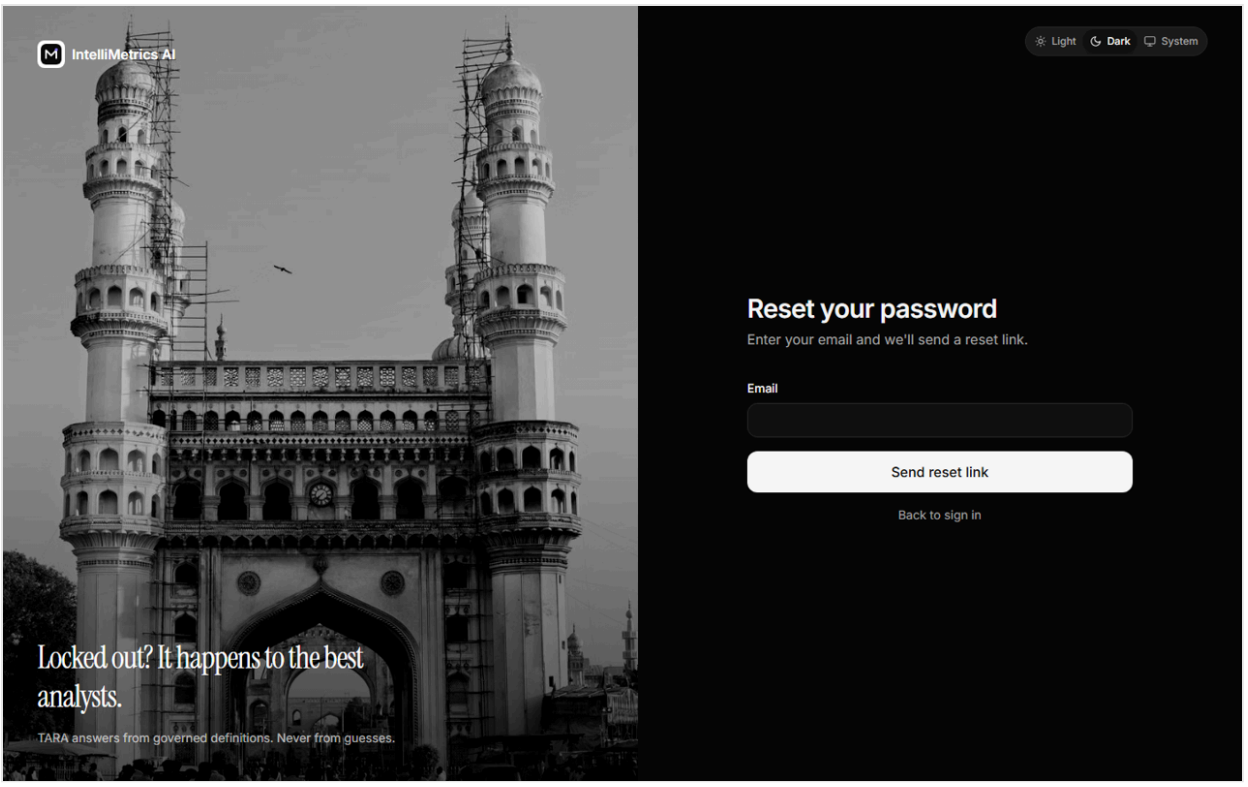


Register, dark mode

Forgot password

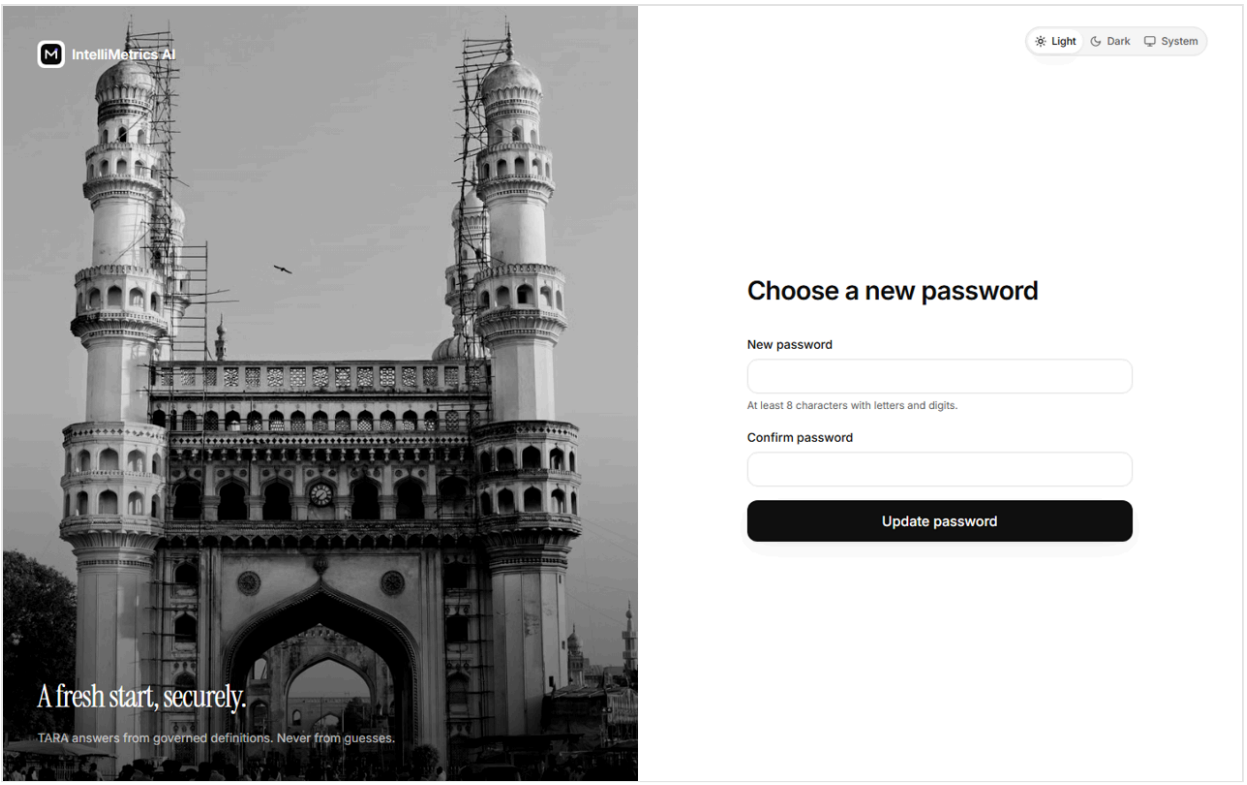


Forgot password, light mode

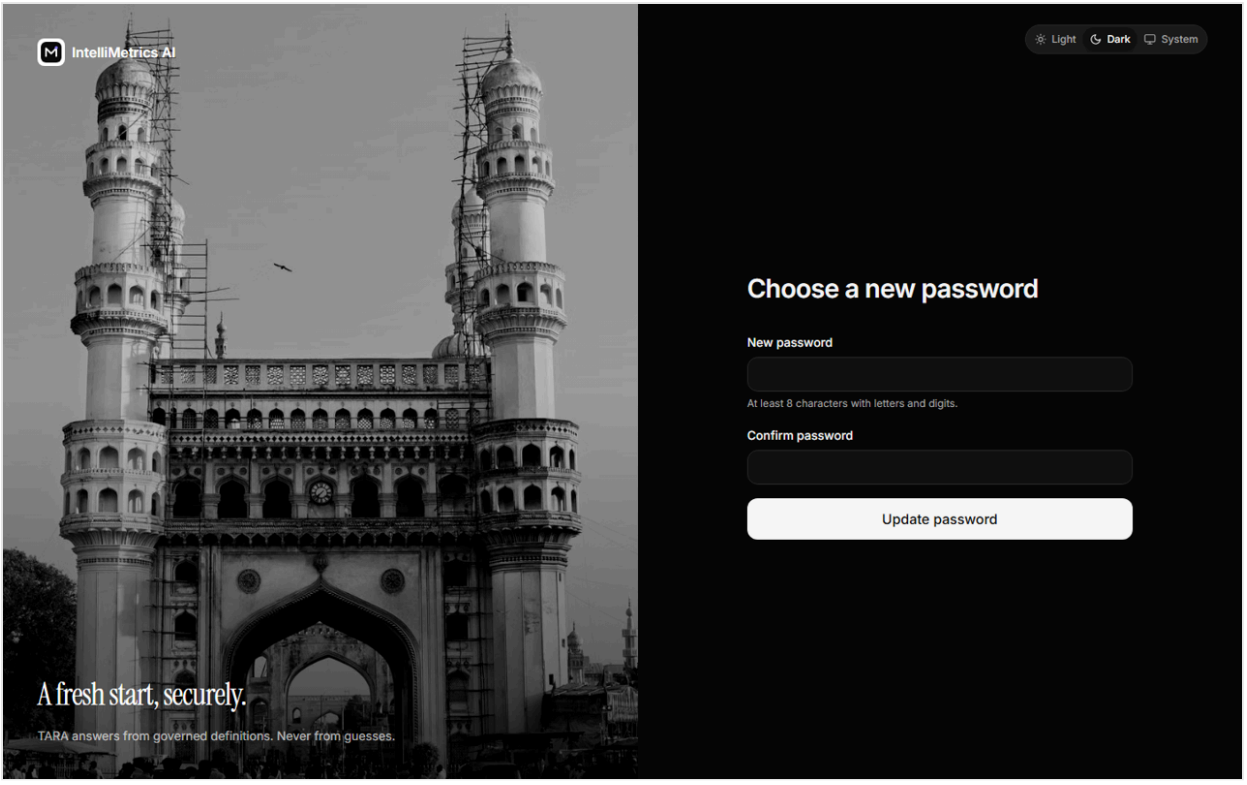


Forgot password, dark mode

Reset password

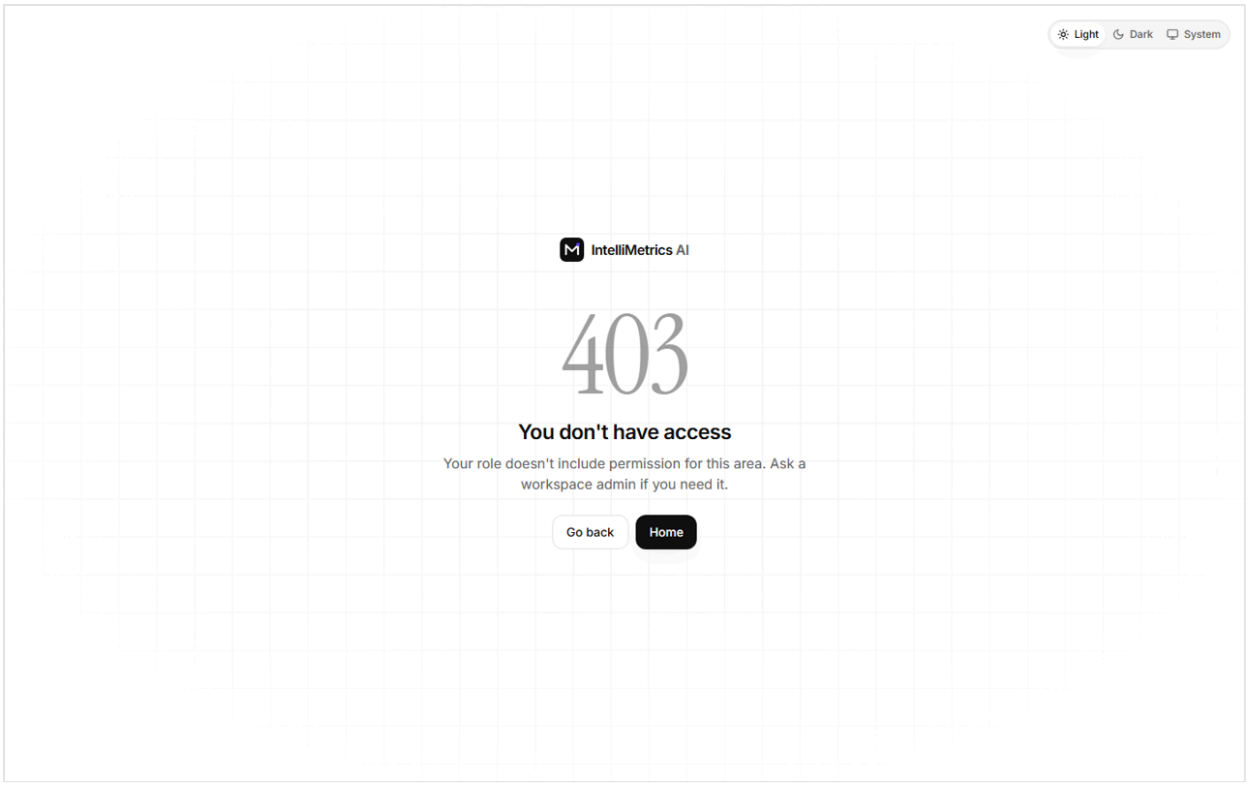


Reset password, light mode

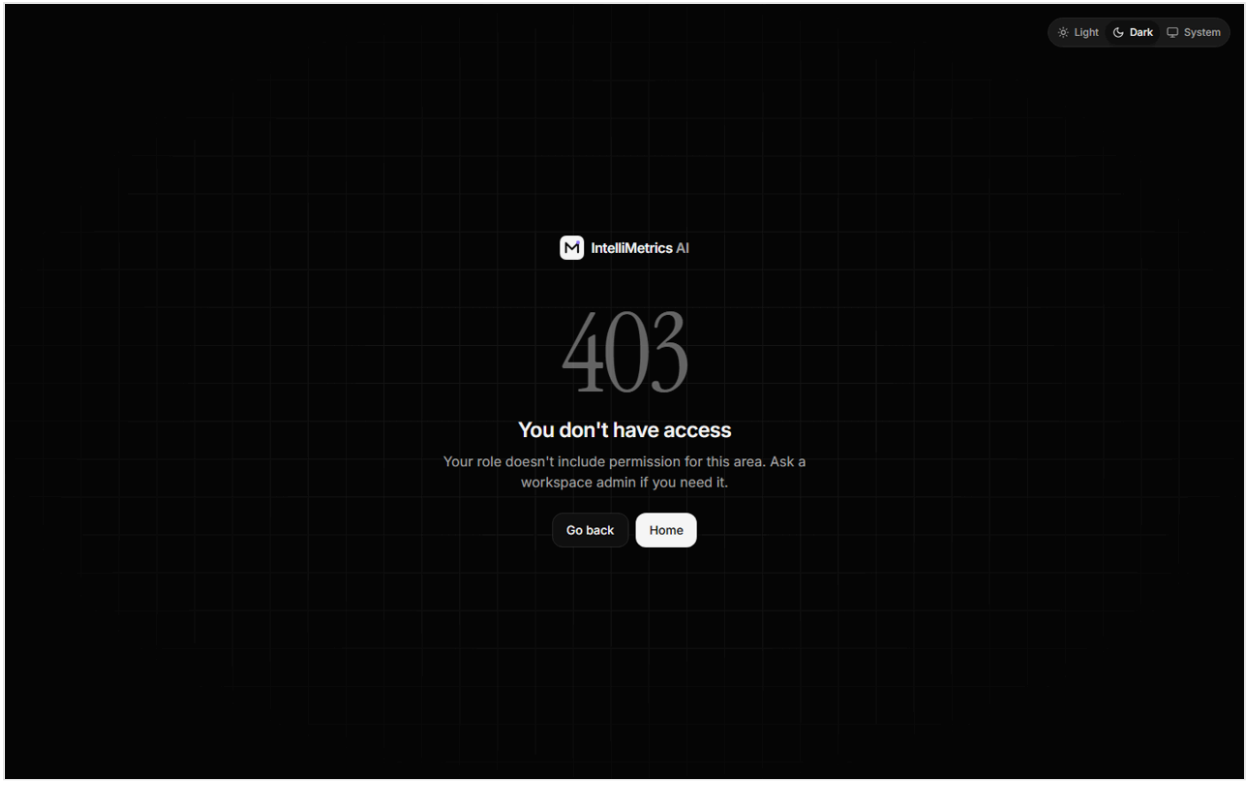


Reset password, dark mode

403 Forbidden

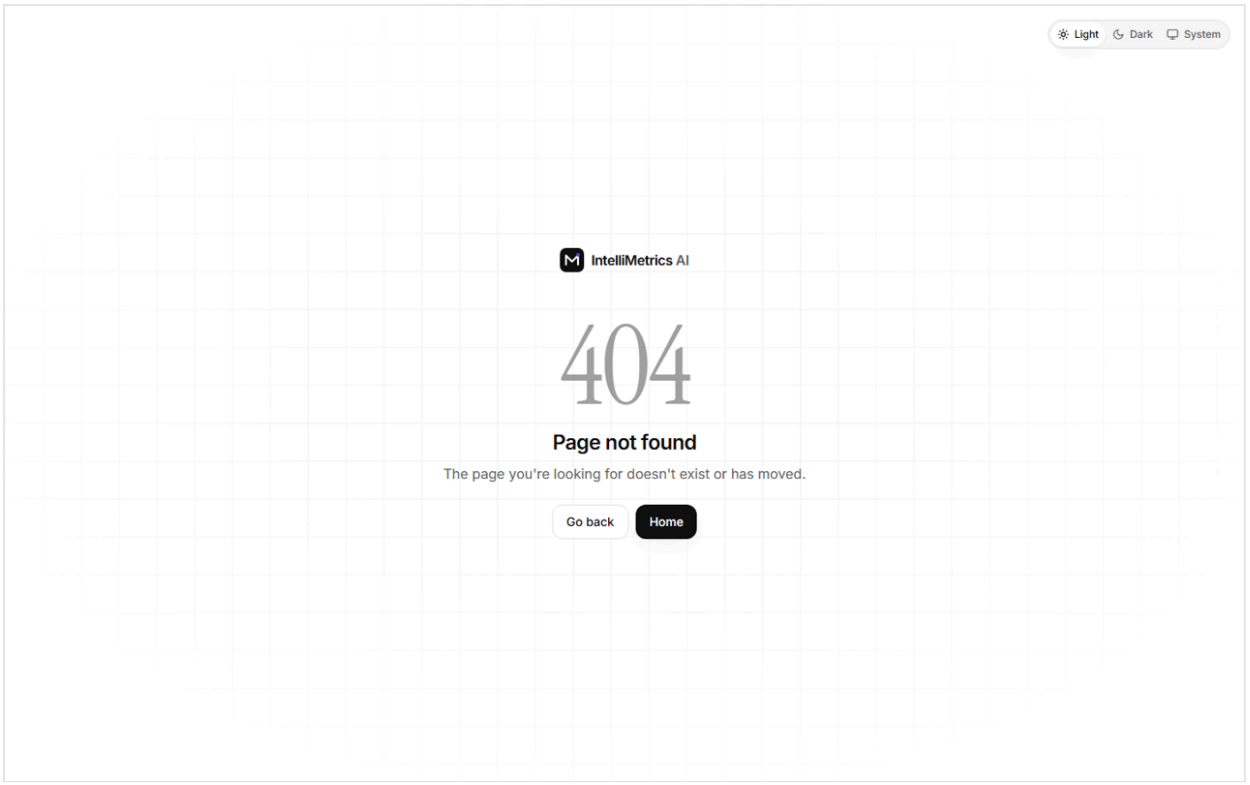


403 Forbidden, light mode

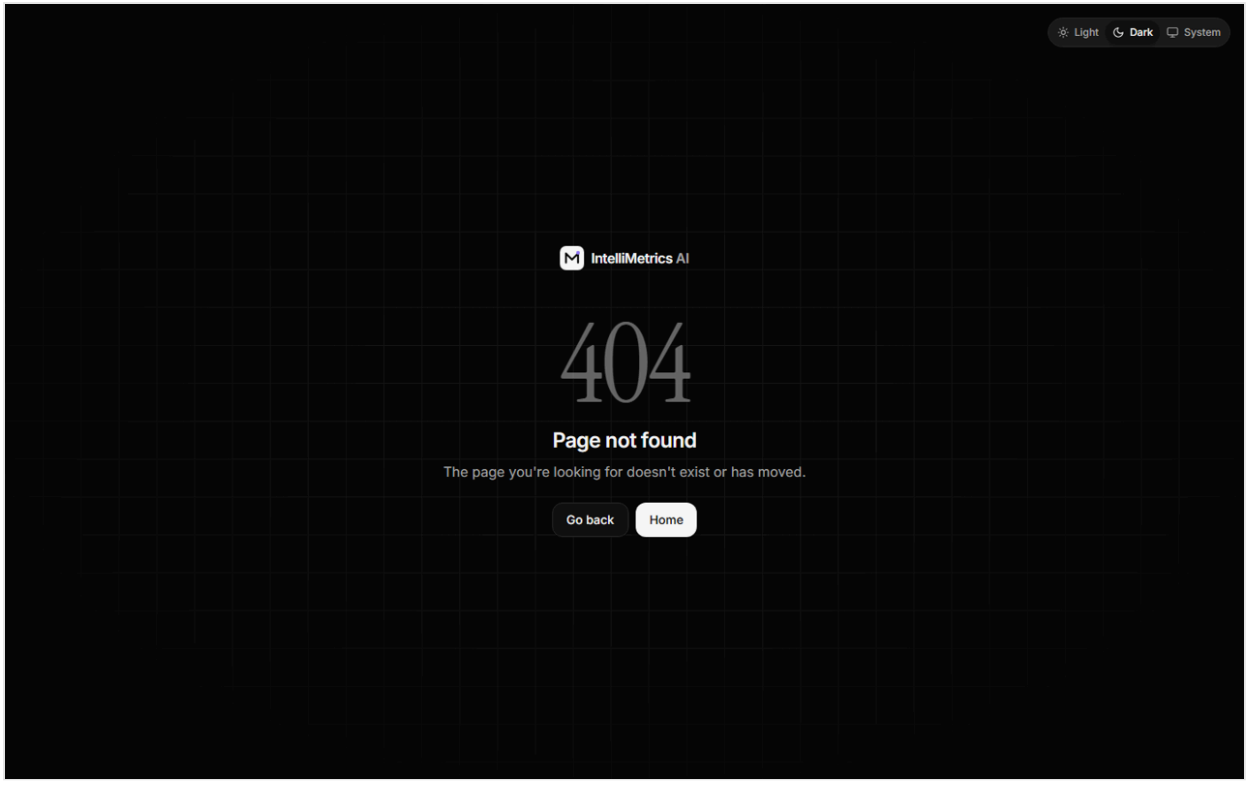


403 Forbidden, dark mode

404 Not found



404 Not found, light mode



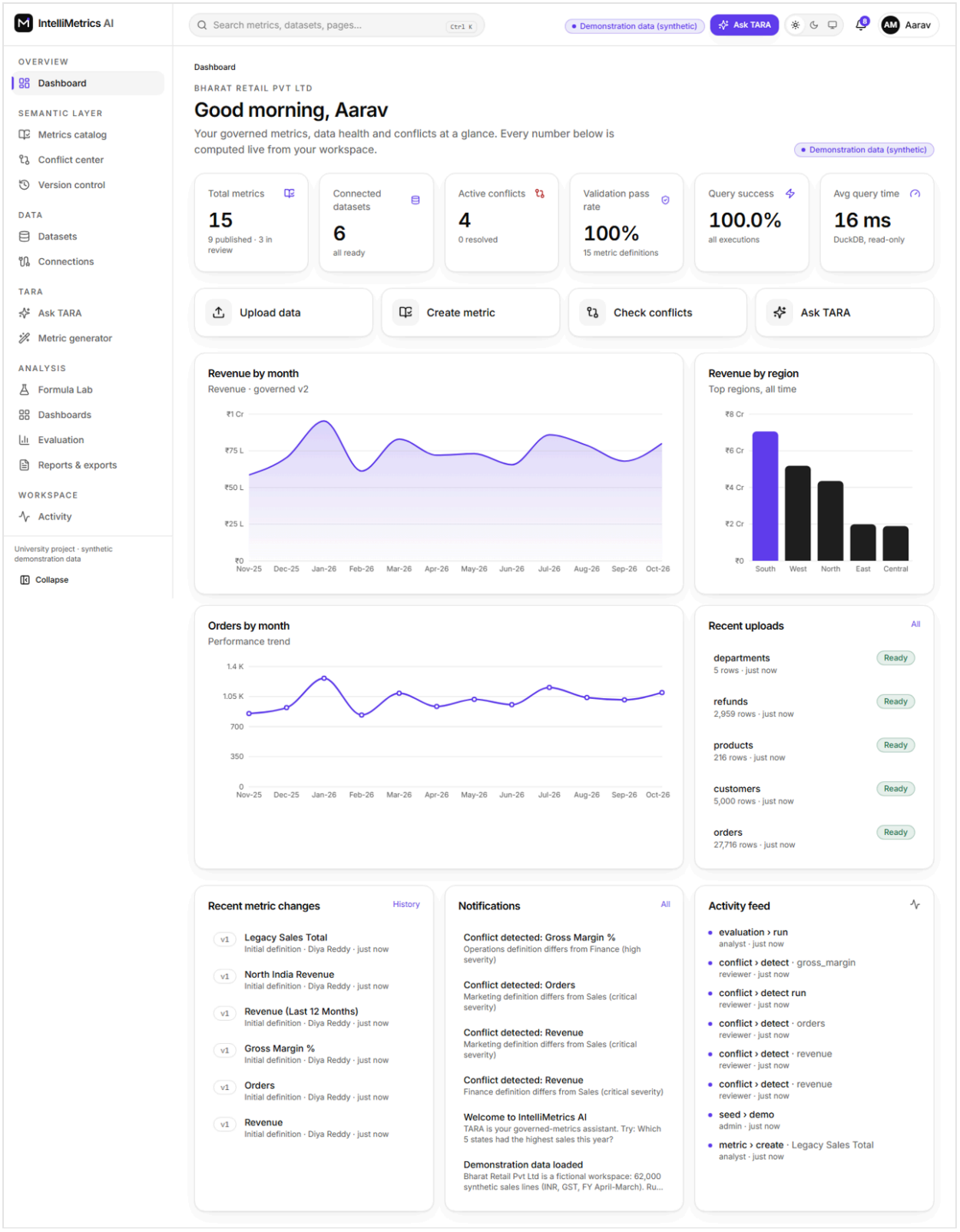
404 Not found, dark mode

7.2 Main application

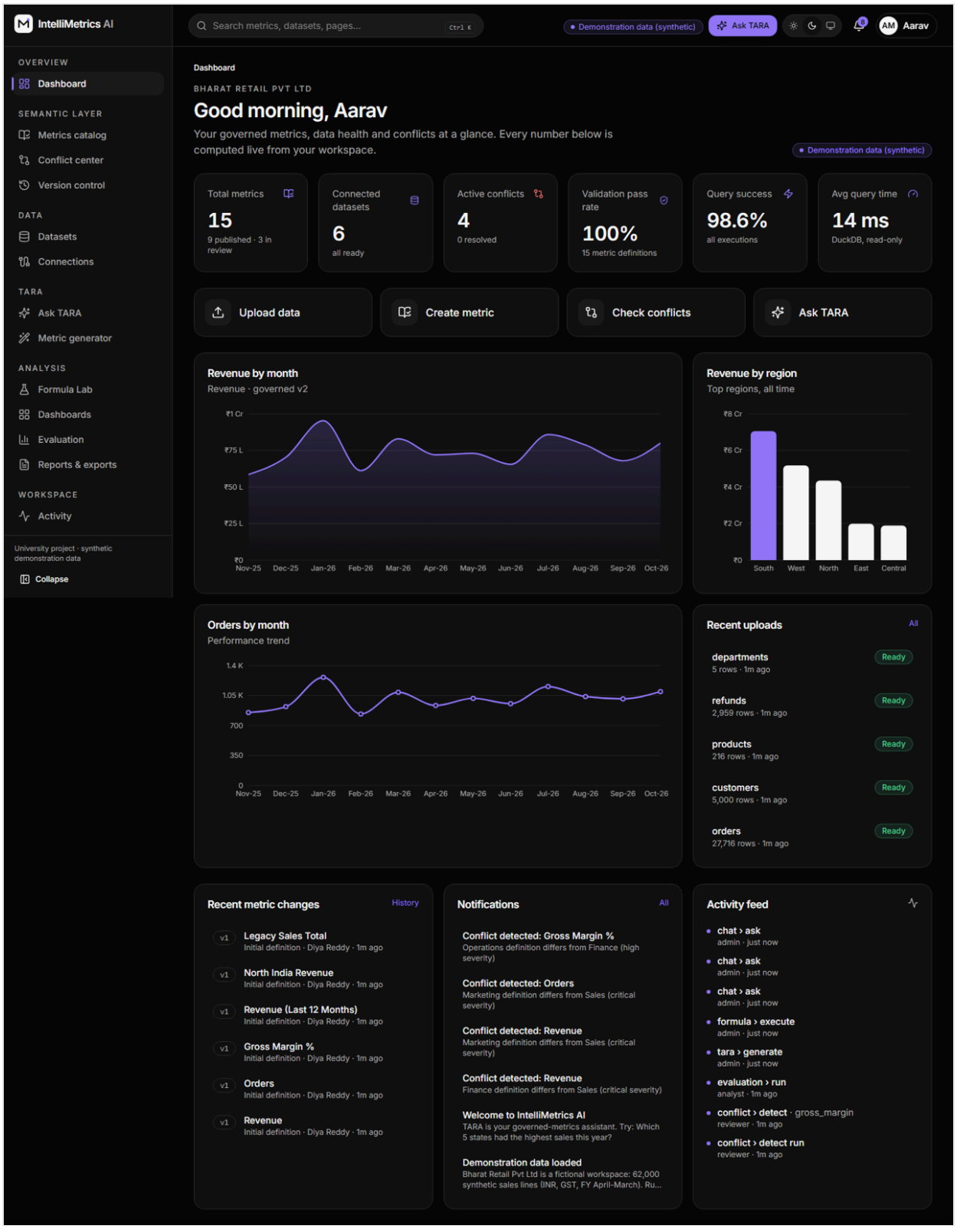
Pages are listed in the order of the sidebar. Screenshots are shown in light and dark mode; the sun/moon button in the top bar switches theme (Light, Dark or System) and the choice is remembered.

7.2.1 Dashboard

The first page after login: four key figures (all computed from real data), recent conflicts and activity, and the dataset status. Amounts are in rupees with lakh/crore grouping (for example ₹12,45,300) and compact forms (₹1.2 Cr). Dates are DD-MMM-YYYY and times are in IST.



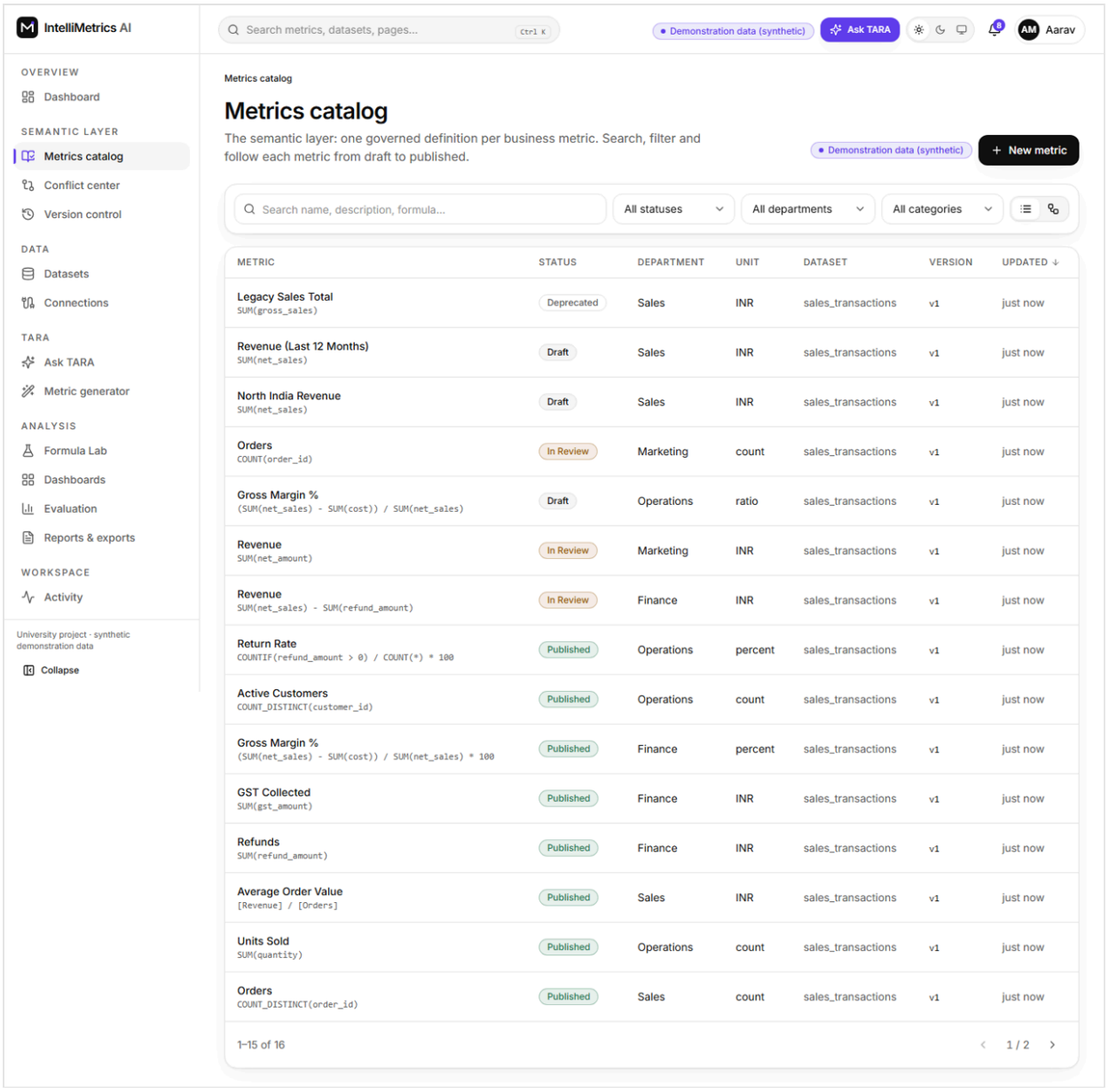
Main dashboard, light mode



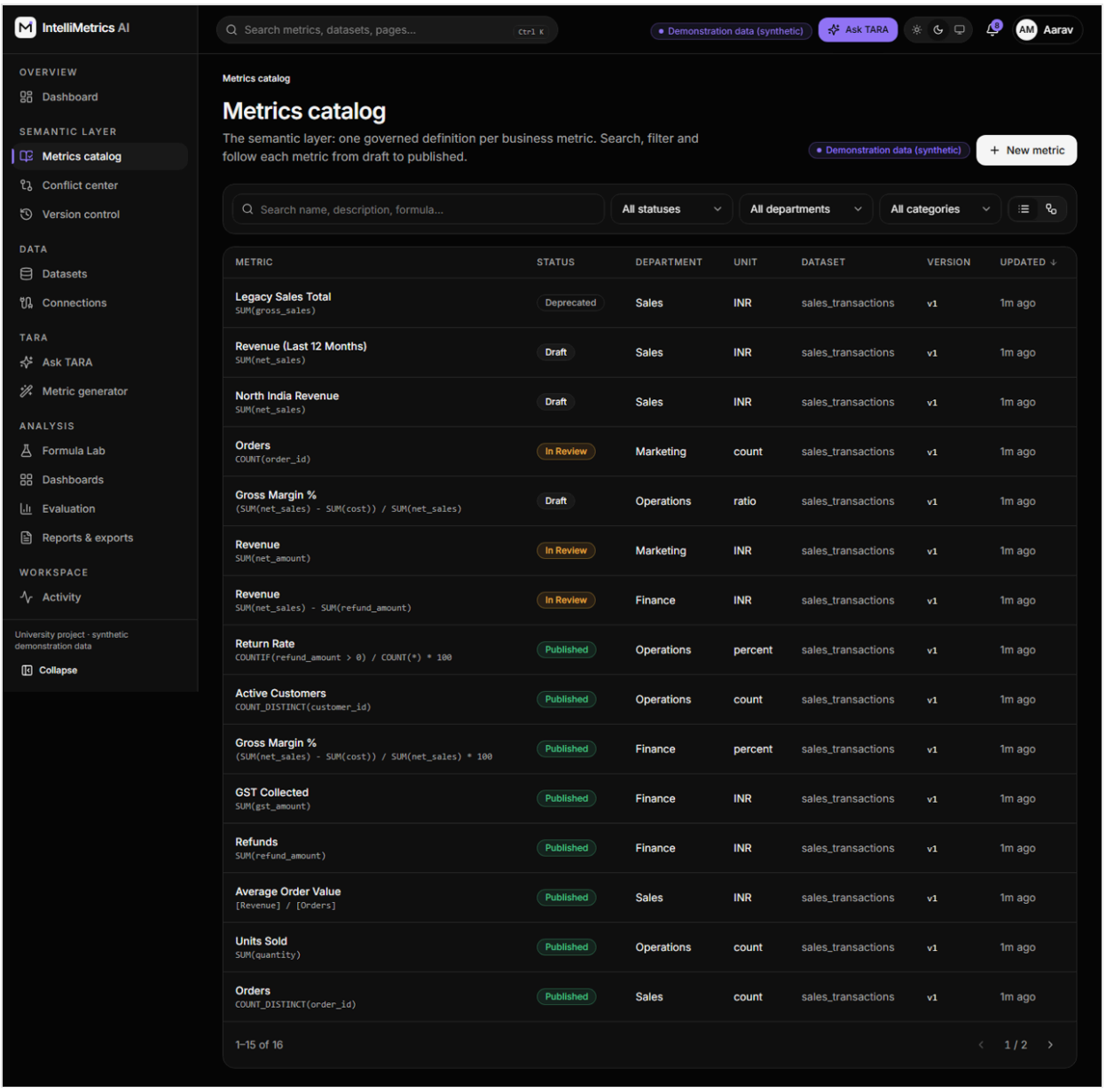
Main dashboard, dark mode

7.2.2 Metrics catalog and metric detail

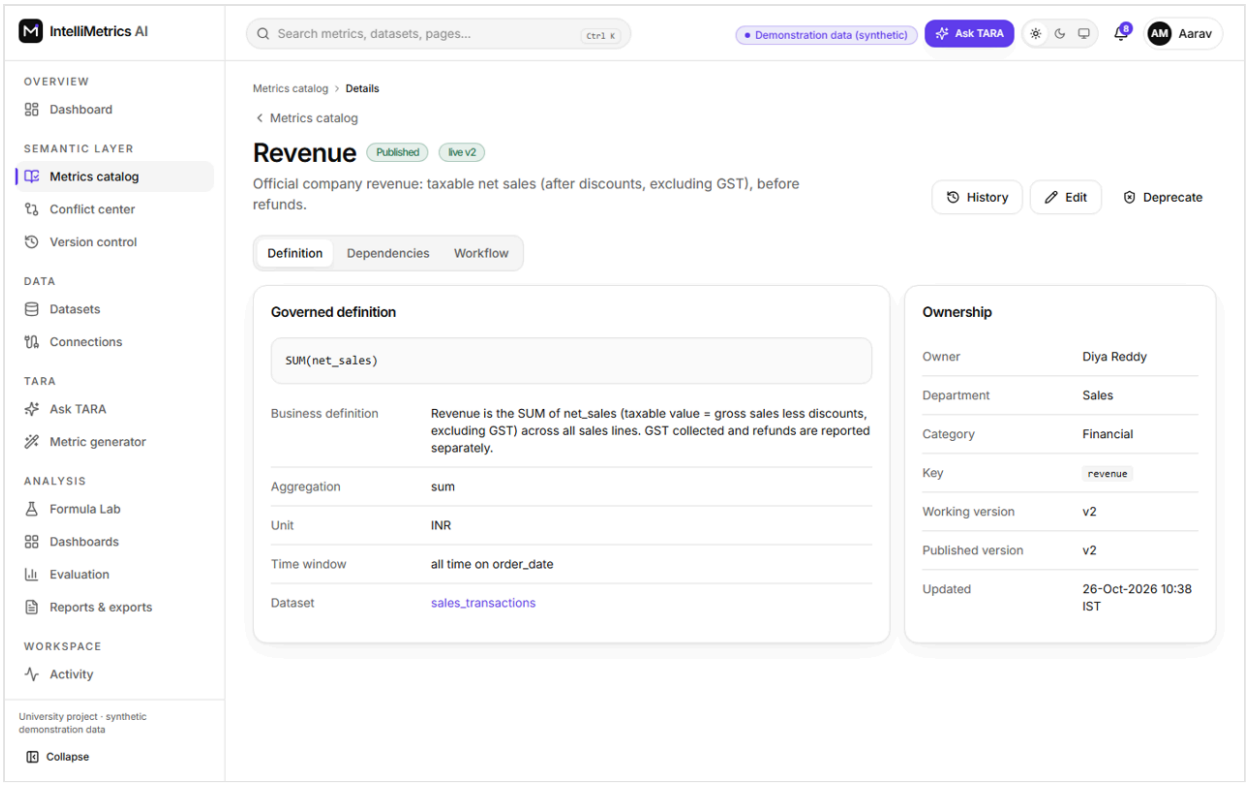
Search, filter and sort all metrics. A metric has a formula, a unit, a department, a status (Draft, In review, Approved, Published, Deprecated) and an owner. The detail page shows the live SQL preview, dependencies on other metrics, the latest value, conflicts and a link to the version history.



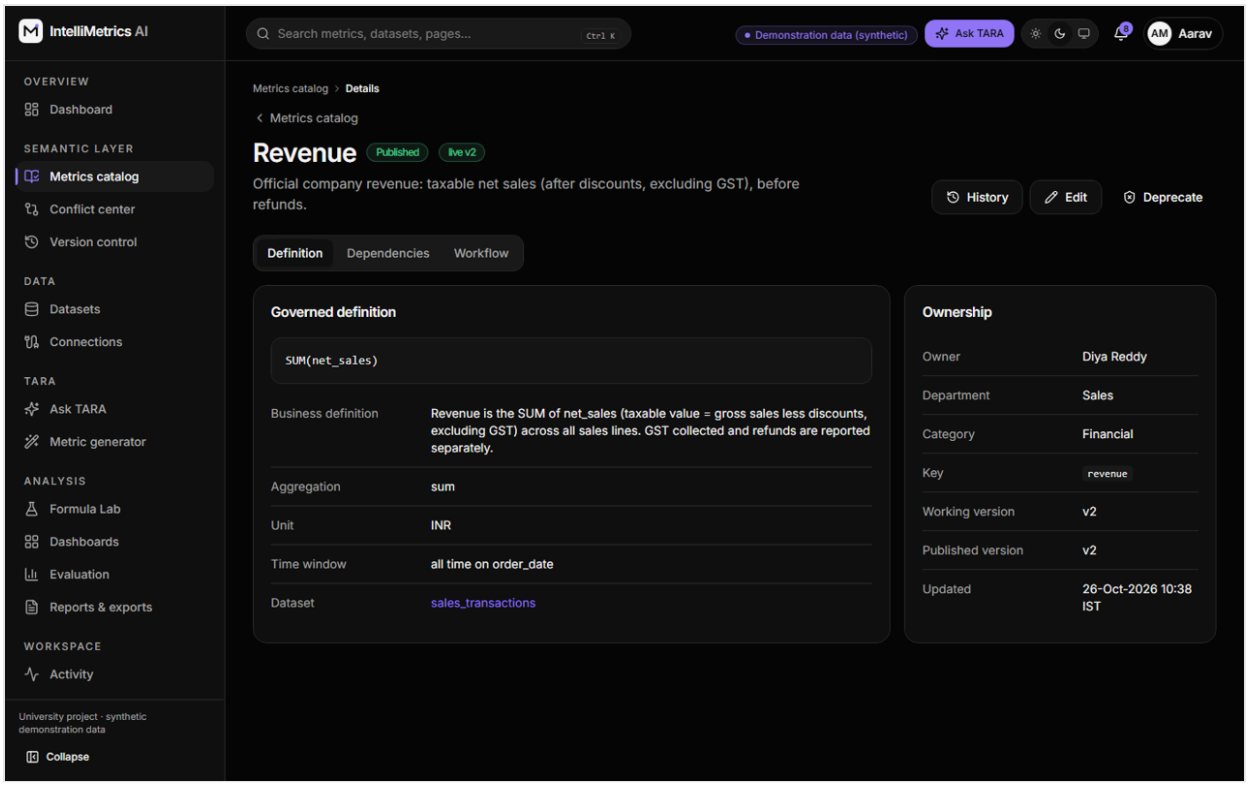
Metrics catalog, light mode



Metrics catalog, dark mode



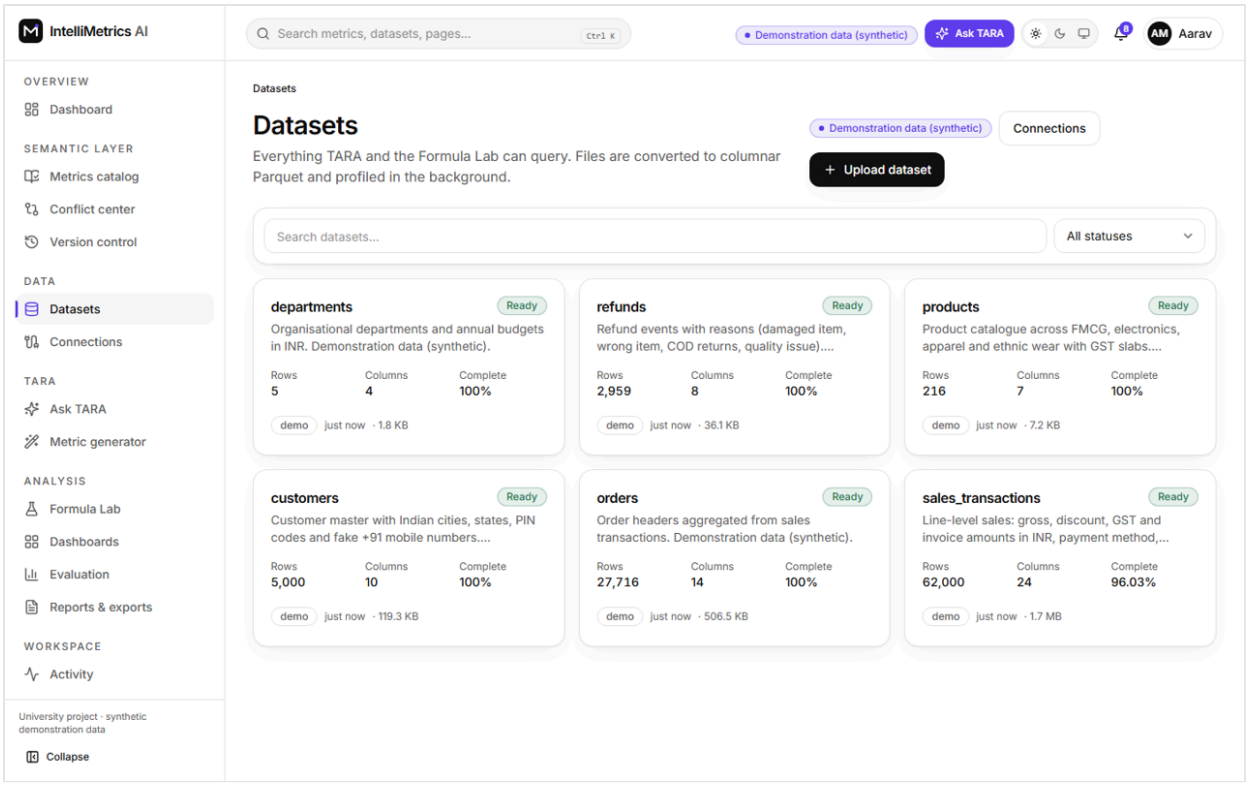
Metric detail (Revenue), light mode



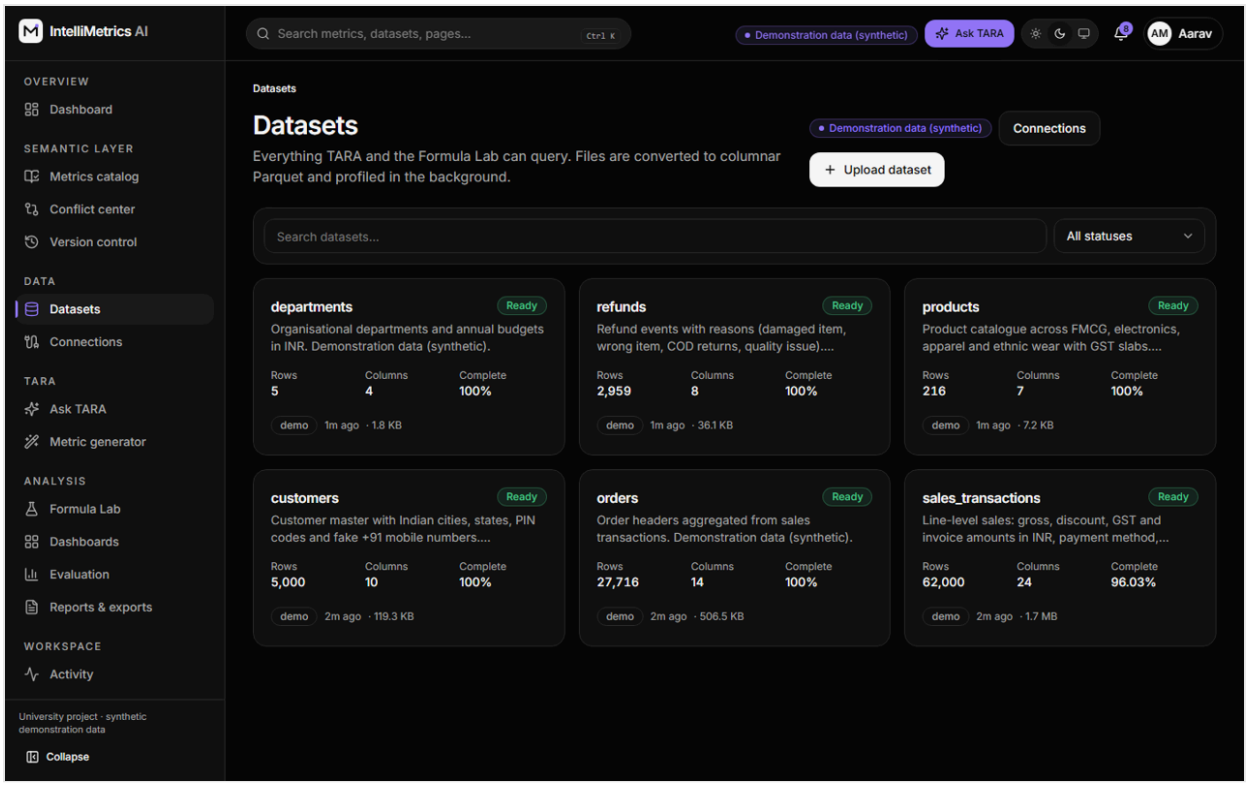
Metric detail (Revenue), dark mode

7.2.3 Datasets and connections

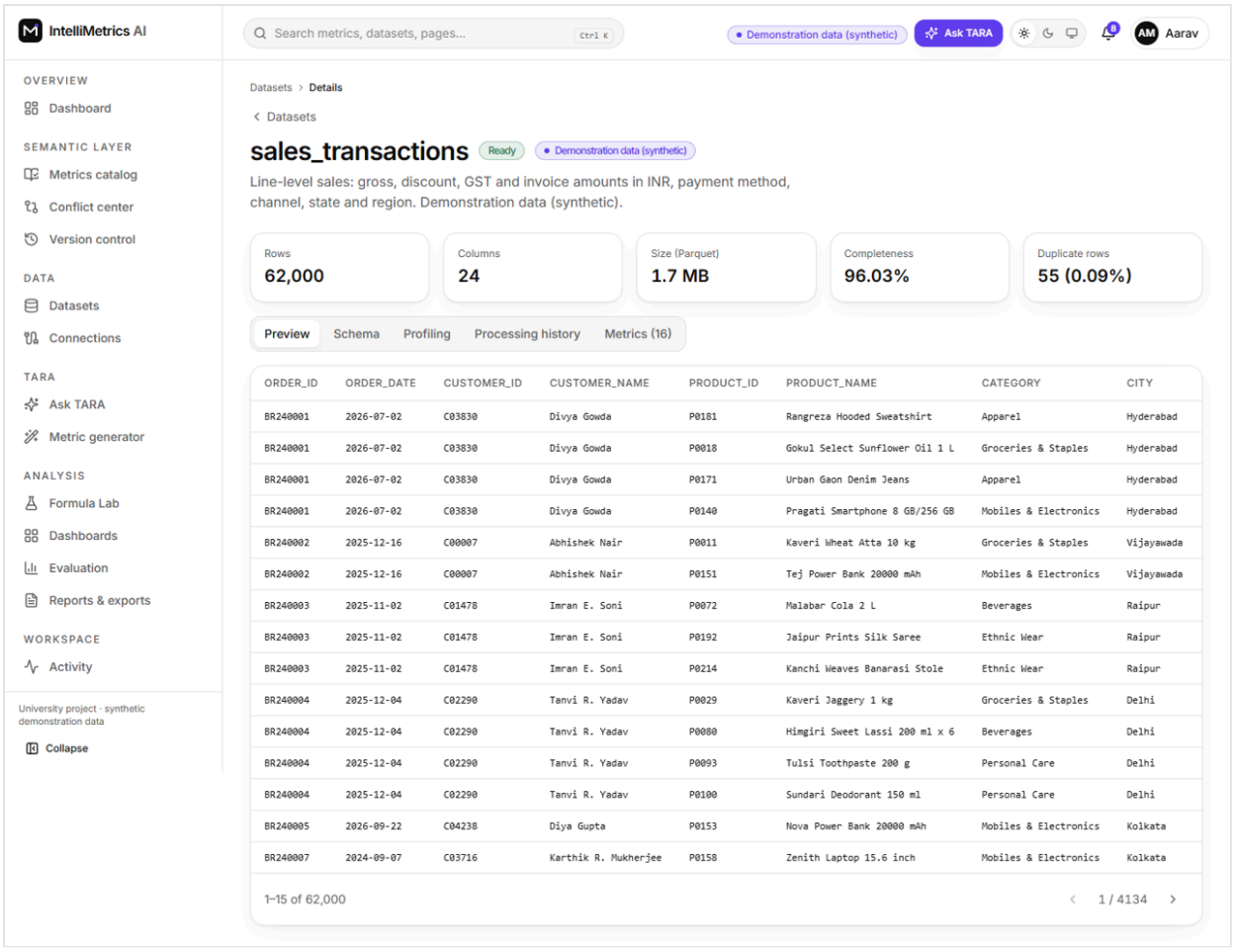
Upload a CSV, TSV, Excel, JSON, JSON-Lines or Parquet file, or connect to a PostgreSQL or MySQL database. Processing runs in the background: *pending*, *processing*, *ready*. The dataset page then shows a preview, the detected schema (with the date column) and a per-column profile (nulls, distinct values, minimum, maximum, mean, most frequent values, duplicates).



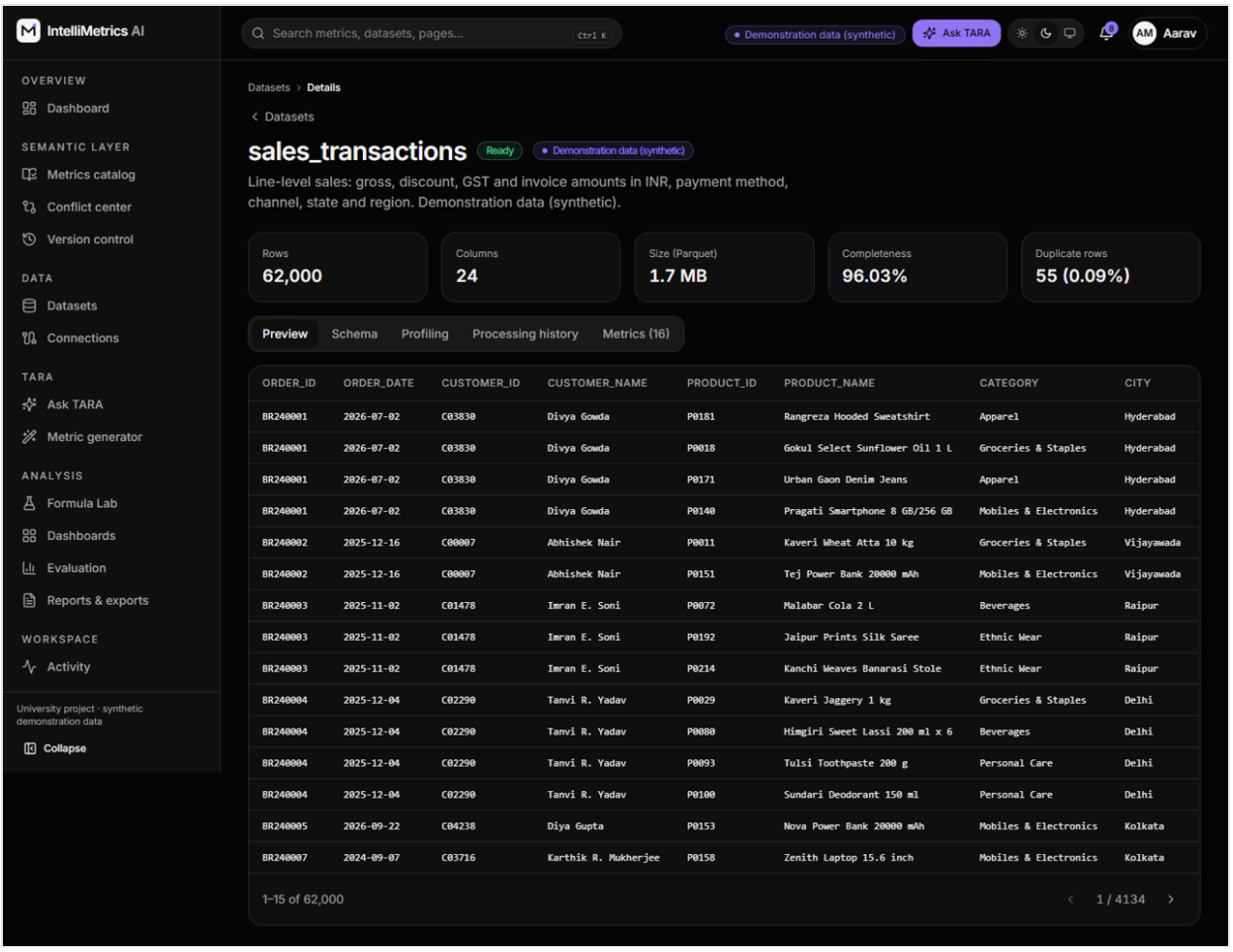
Datasets catalog, light mode



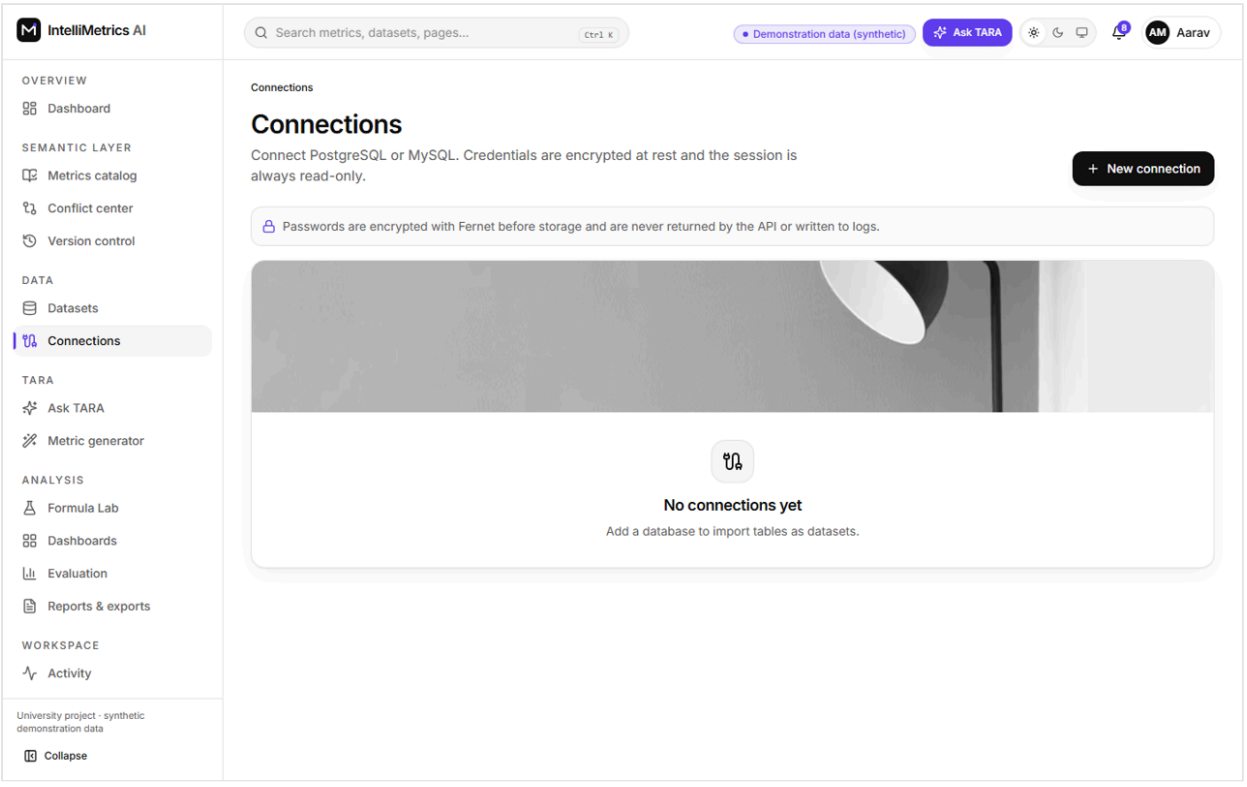
Datasets catalog, dark mode



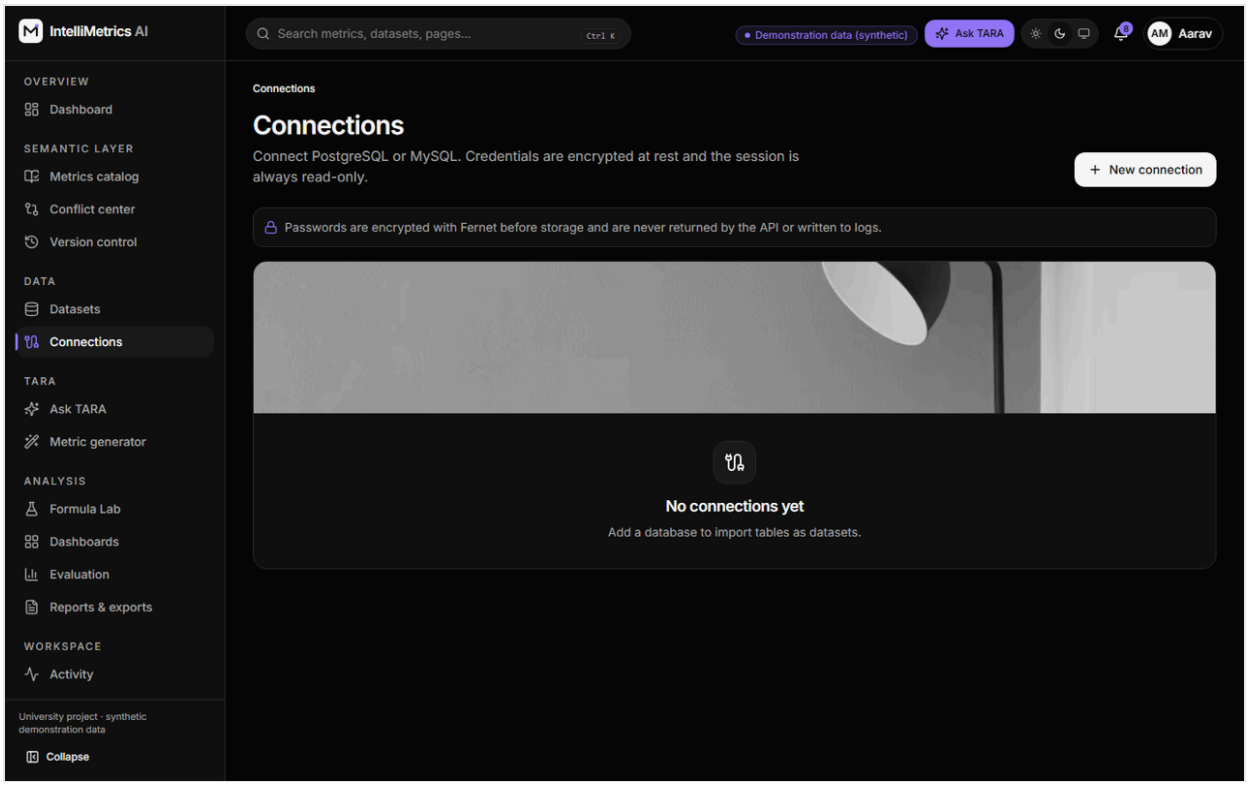
Dataset detail (preview), light mode



Dataset detail (preview), dark mode



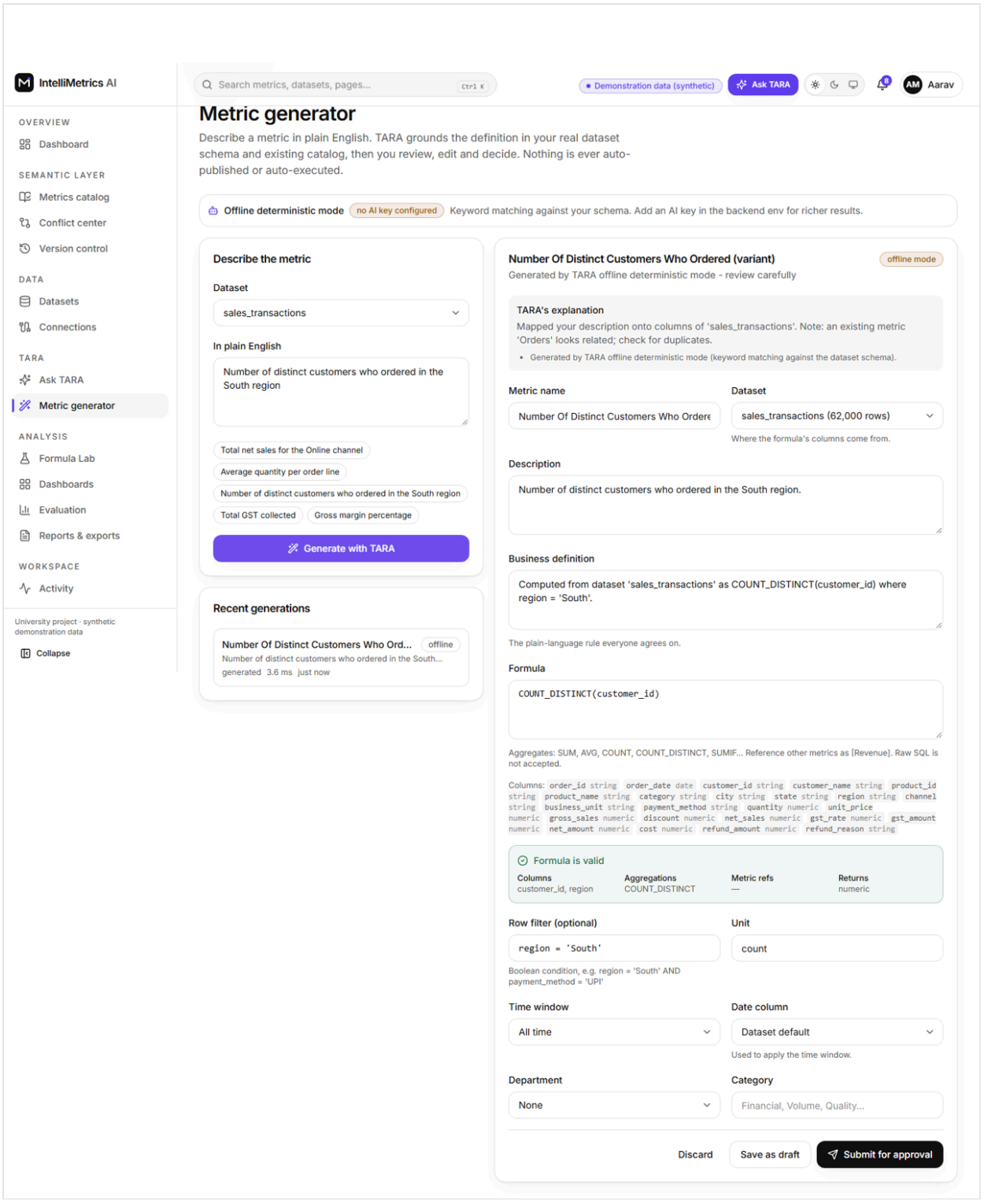
Database connections, light mode



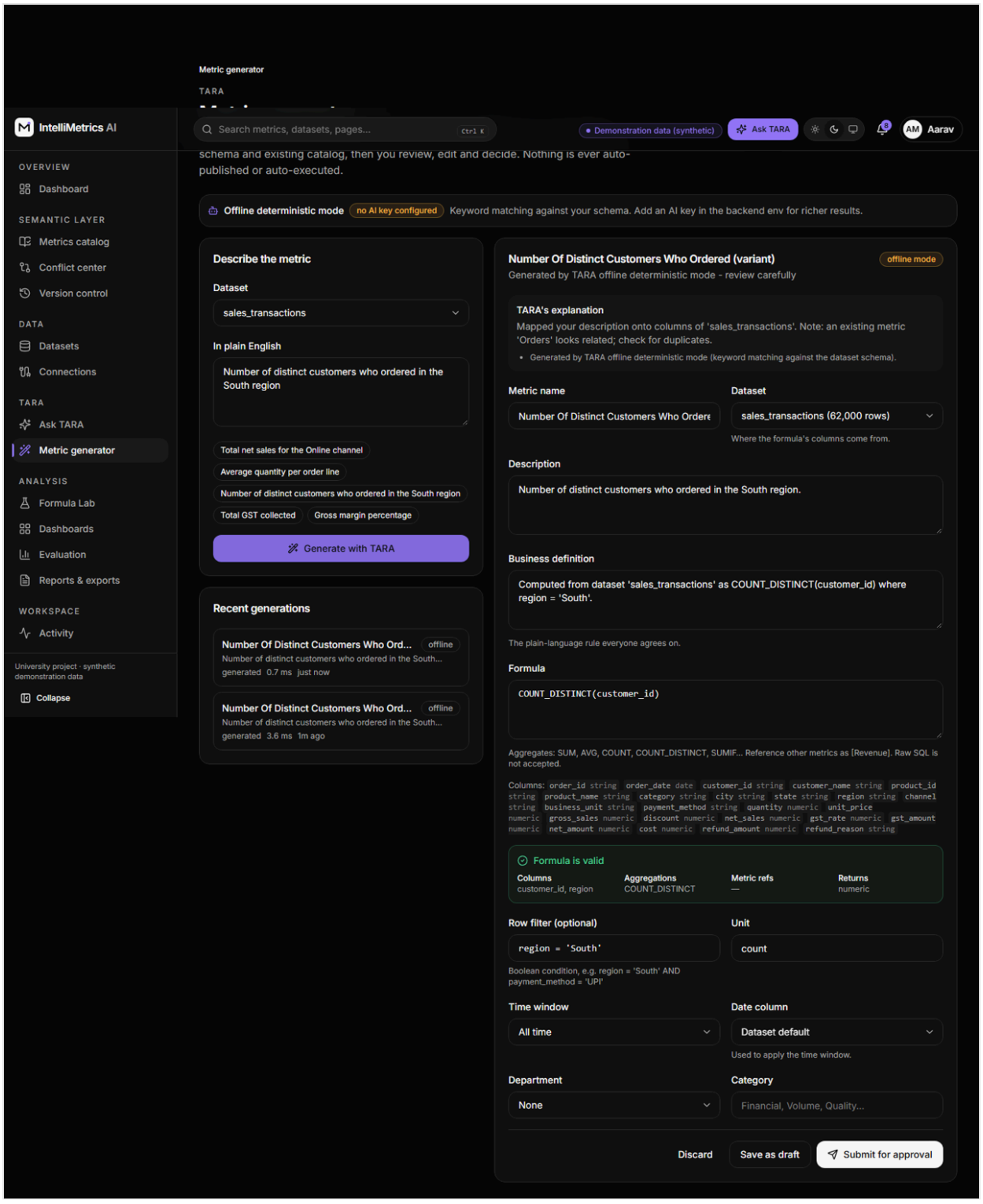
Database connections, dark mode

7.2.4 TARA metric generator

Choose a dataset and describe a metric in English. TARA returns an editable formula, explanation and warnings. **Nothing is saved or published automatically**; the analyst saves a draft or submits for approval.



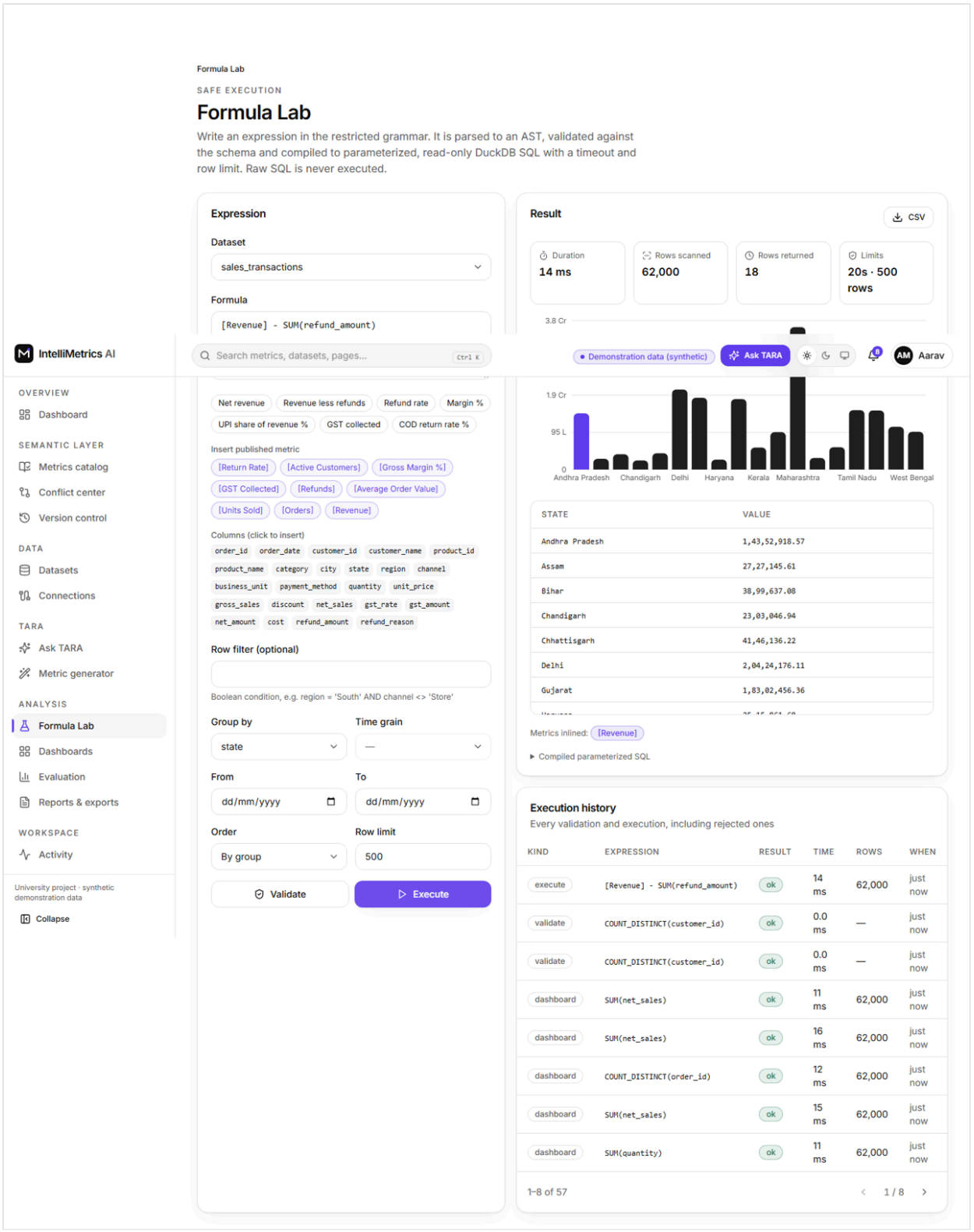
TARA metric generator (offline deterministic mode, editable suggestion), light mode



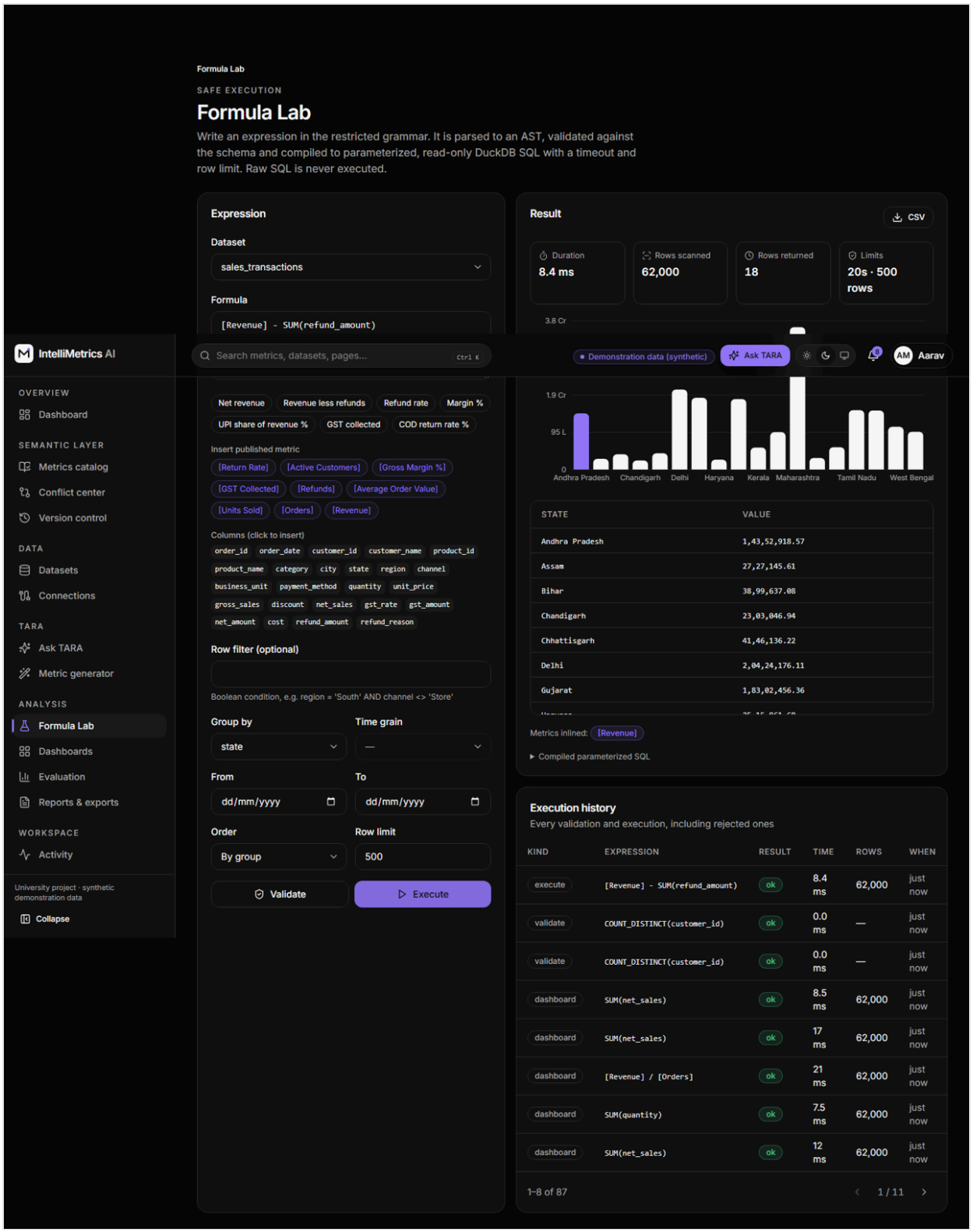
TARA metric generator (offline deterministic mode, editable suggestion), dark mode

7.2.5 Formula Lab

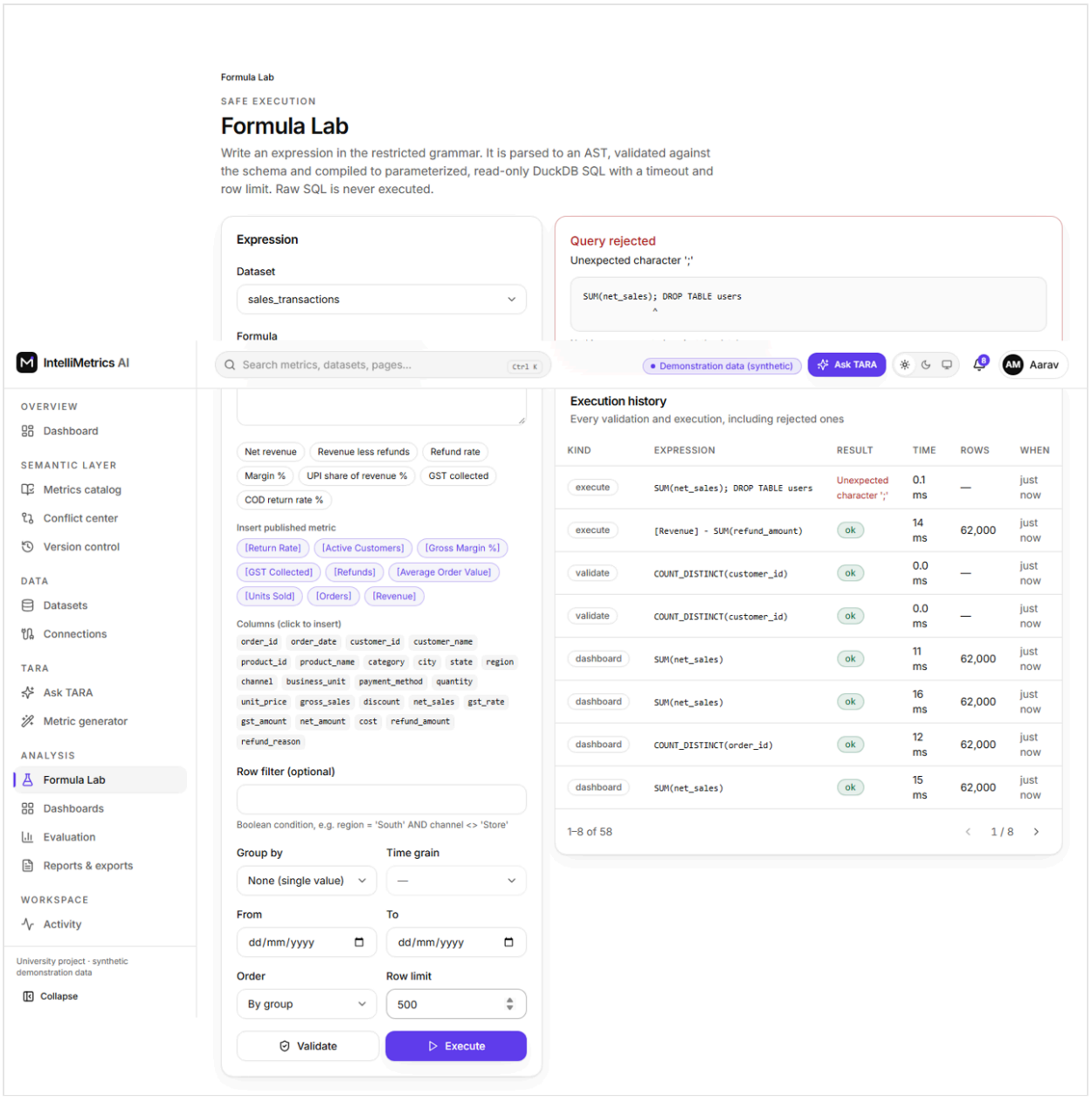
A safe place to try formulas. Metrics are written as `[Revenue]`, columns by name, and the grammar allows only whitelisted functions. The Lab validates syntax, columns and types, shows the generated parameterised SQL, executes it on the real data with a time limit, and reports rows scanned and duration. Hostile input is rejected without running.



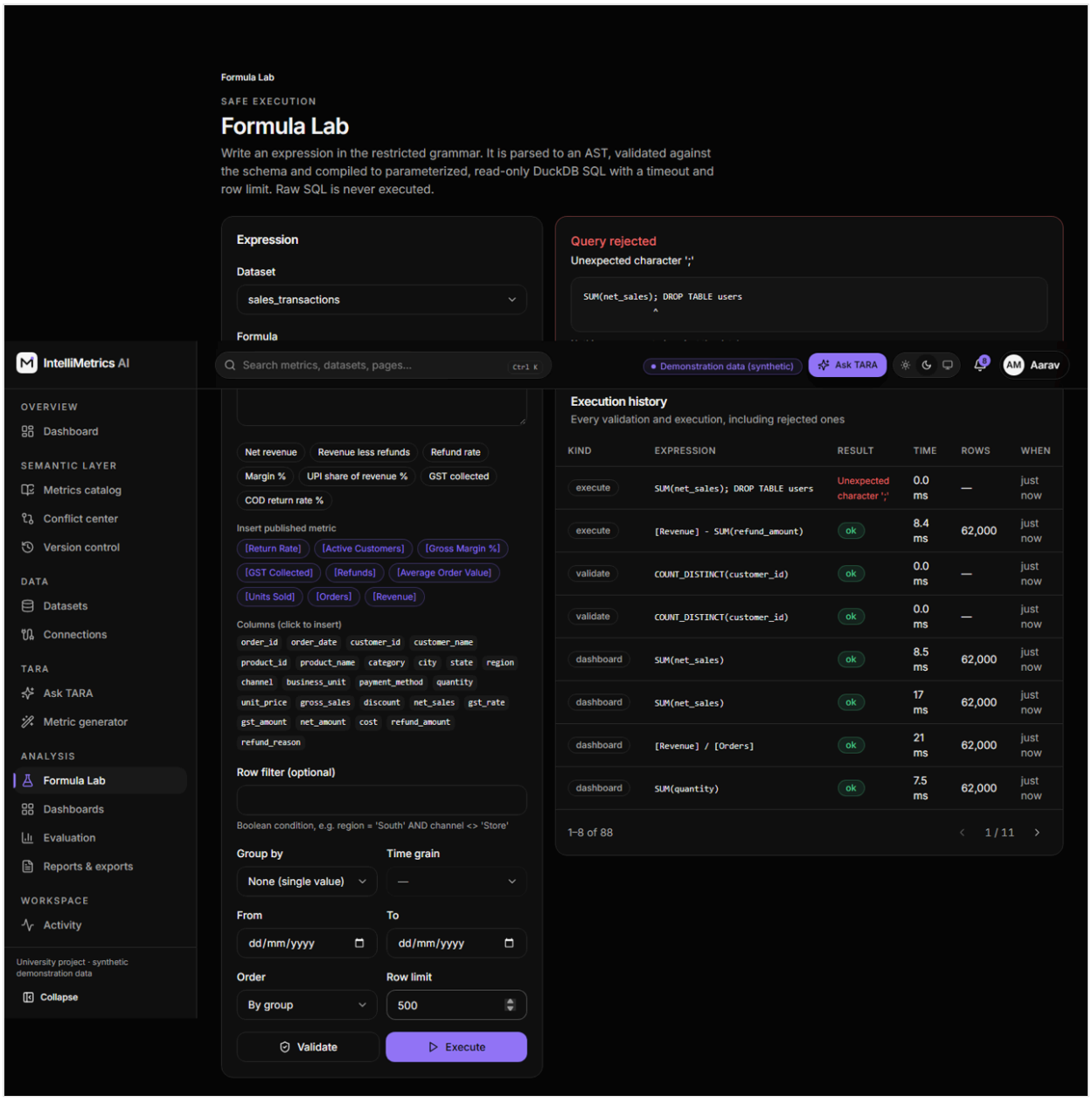
Formula Lab: validate and execute (Revenue less refunds by state), light mode



Formula Lab: validate and execute (Revenue less refunds by state), dark mode



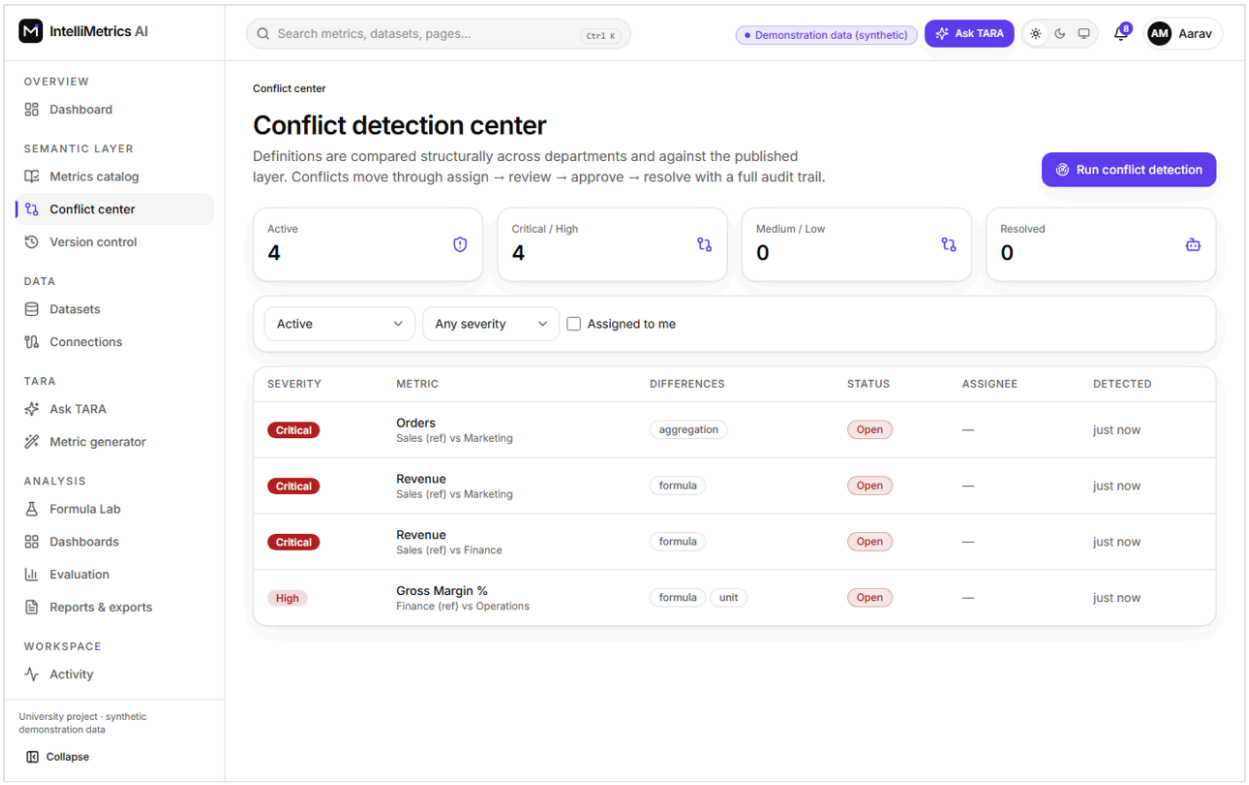
Formula Lab rejecting an injection attempt, light mode



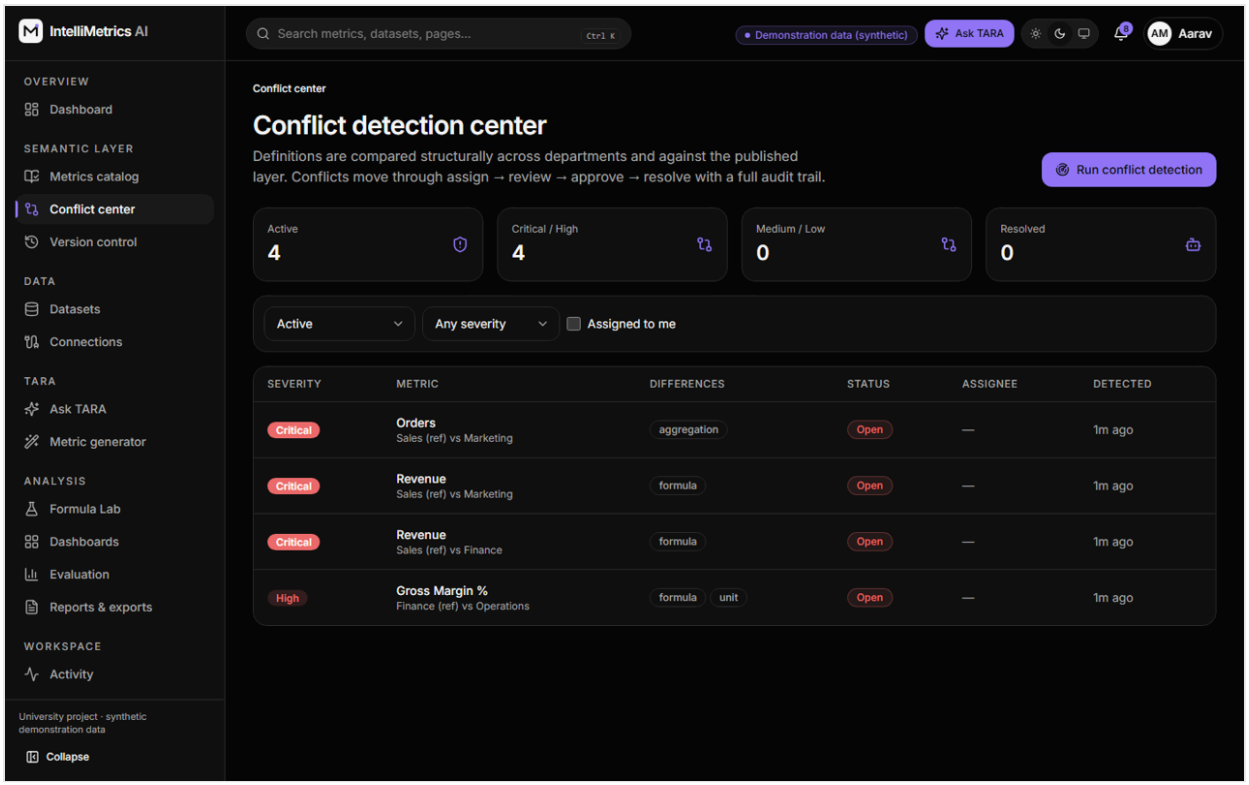
Formula Lab rejecting an injection attempt, dark mode

7.2.6 Conflict detection center

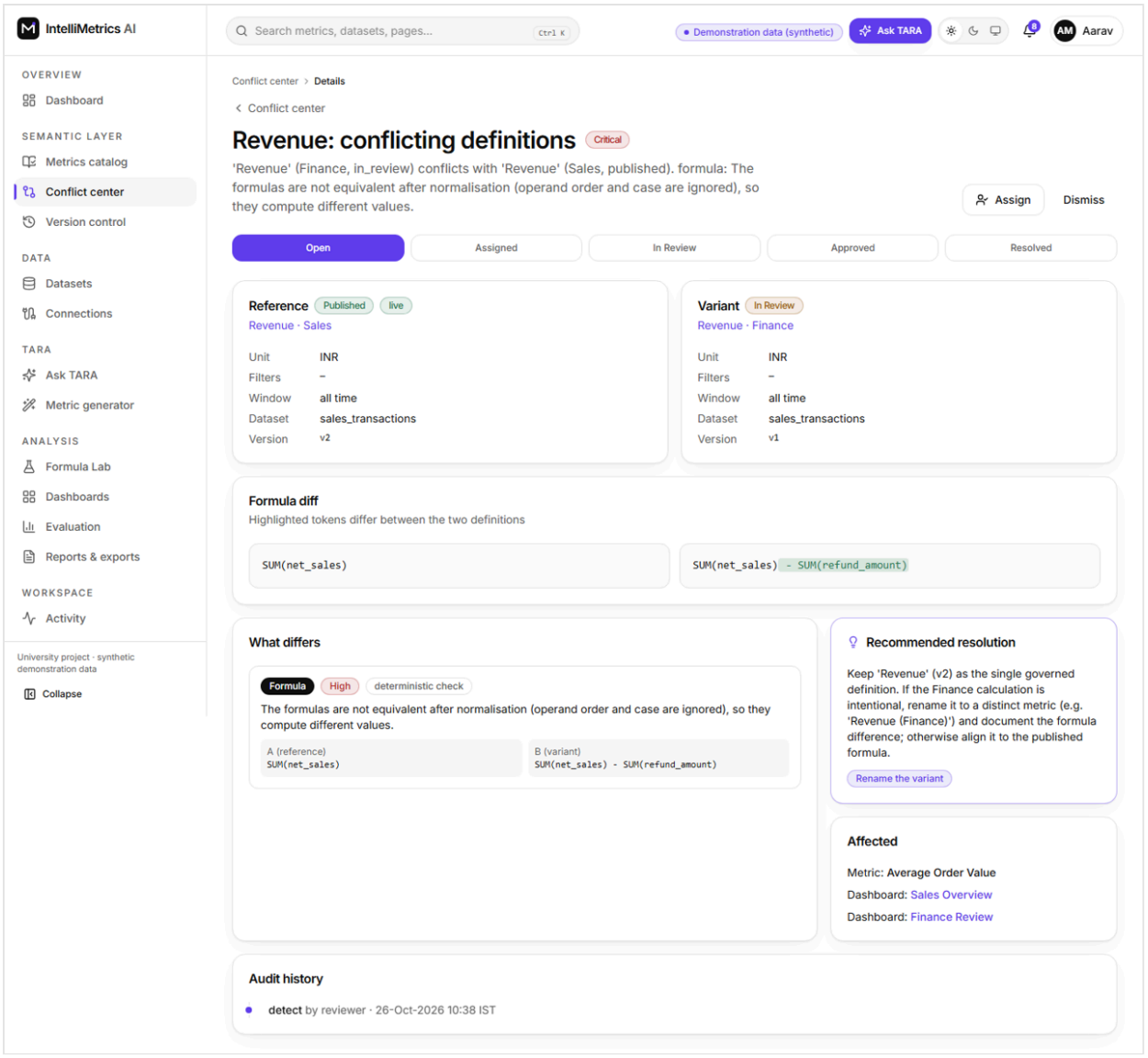
Run conflict detection compares every pair of definitions of the same business idea, and lists conflicts with a type (formula, aggregation, filter, unit, time window, dataset), a severity and a recommendation. The detail page shows the two definitions side by side with the differences highlighted, then the workflow: **assign, review, approve, resolve**, each step recorded.



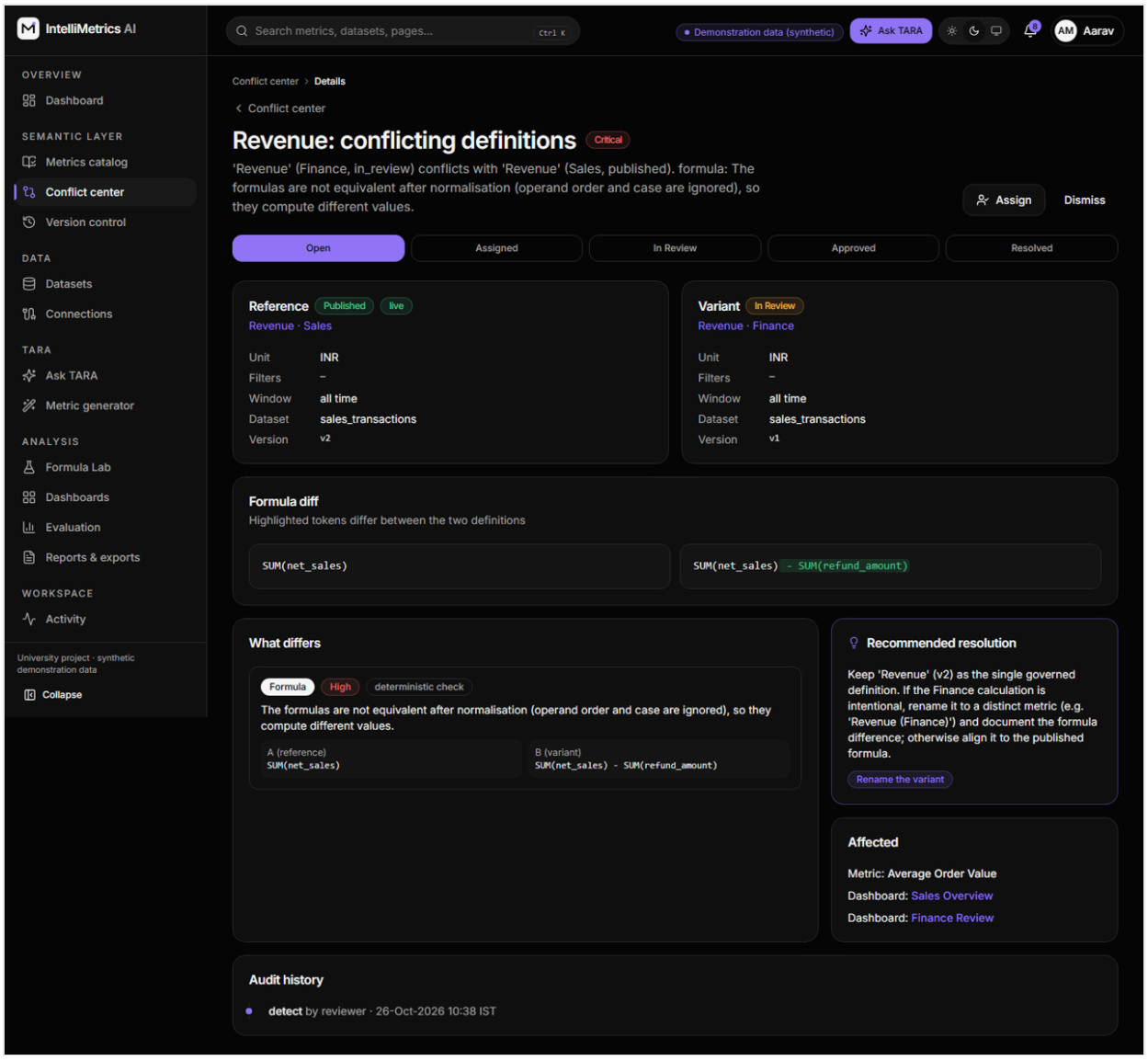
Conflict detection center, light mode



Conflict detection center, dark mode



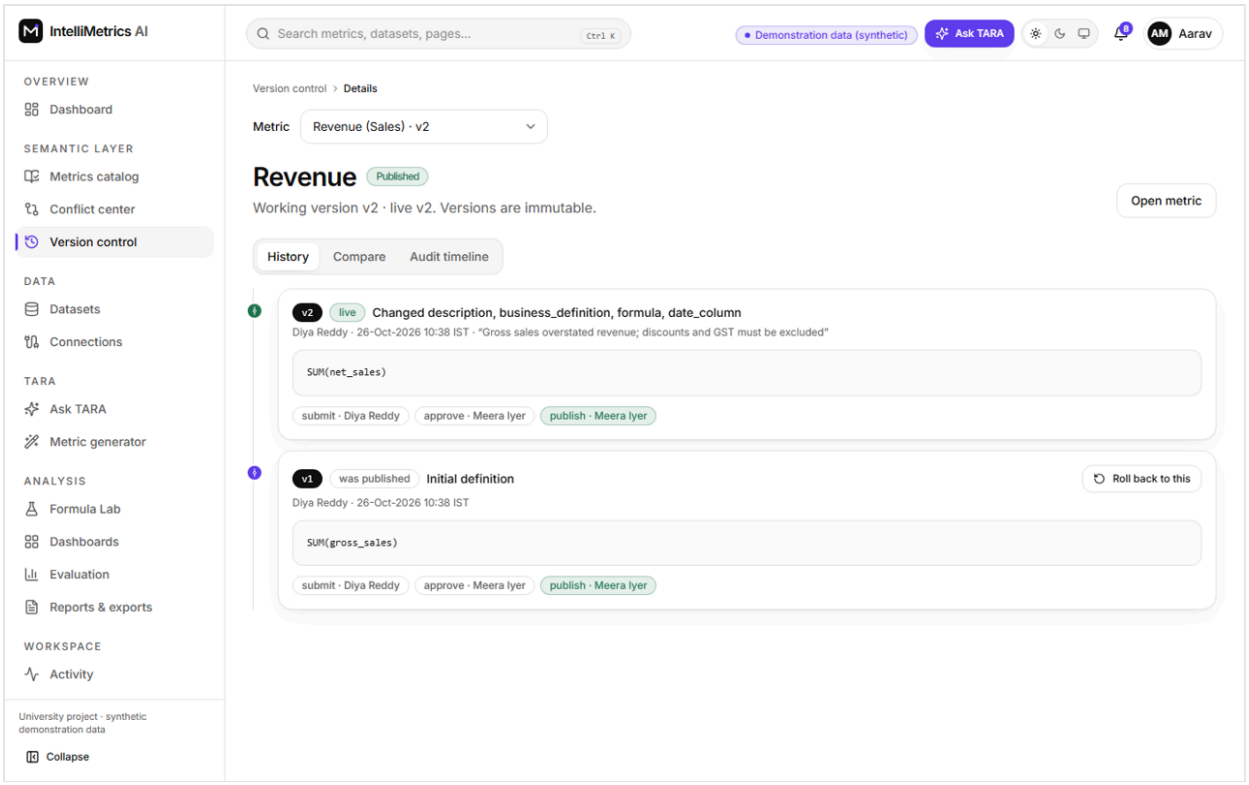
Conflict detail (side-by-side), light mode



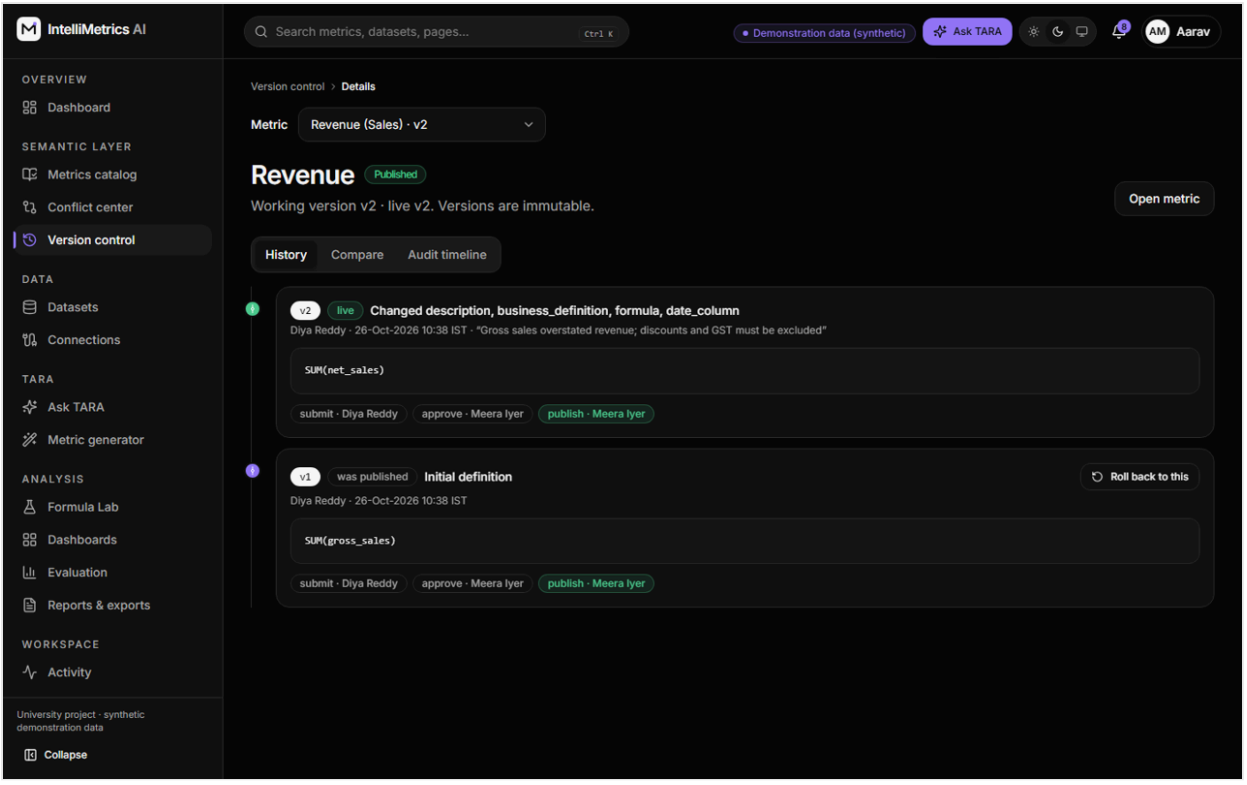
Conflict detail (side-by-side), dark mode

7.2.7 Version control

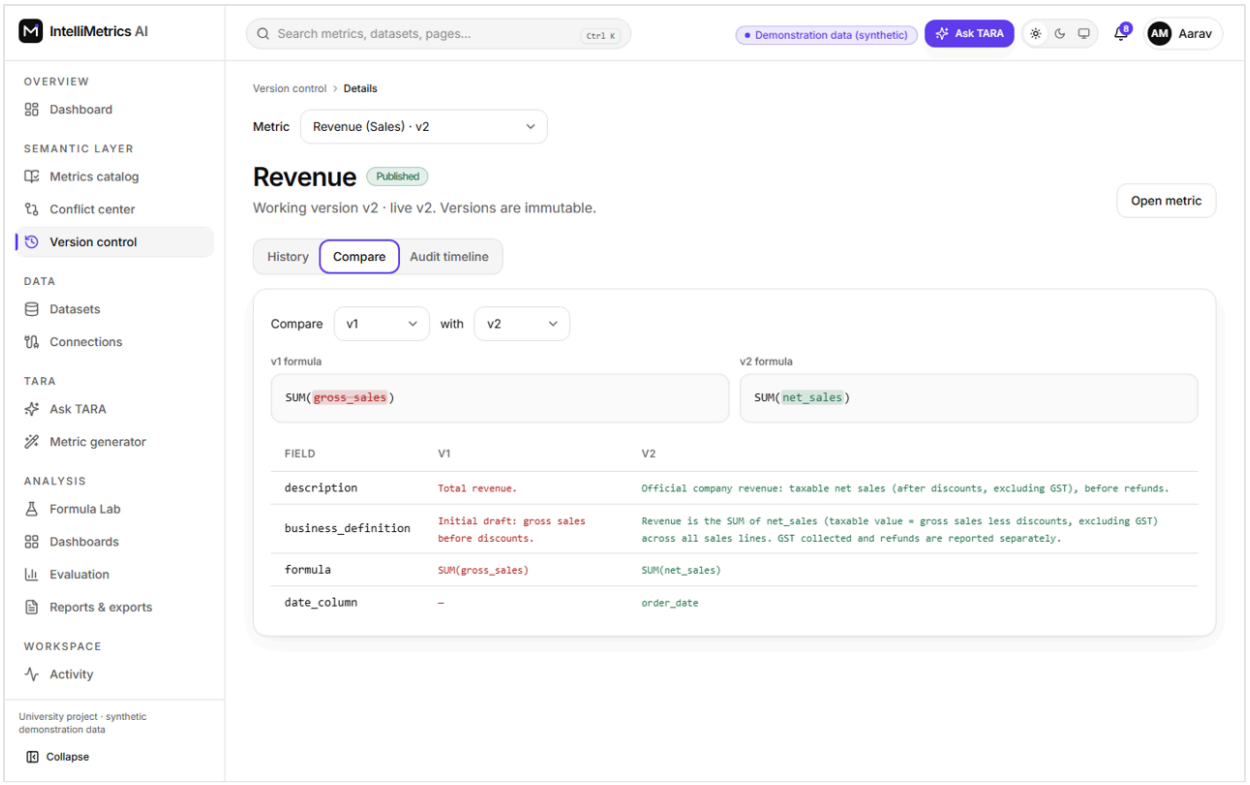
Every change creates an immutable version. The history page lists versions, compares any two (word-level formula diff) and rolls back. Rolling back never rewrites history: it creates a **new** version that copies the old definition.



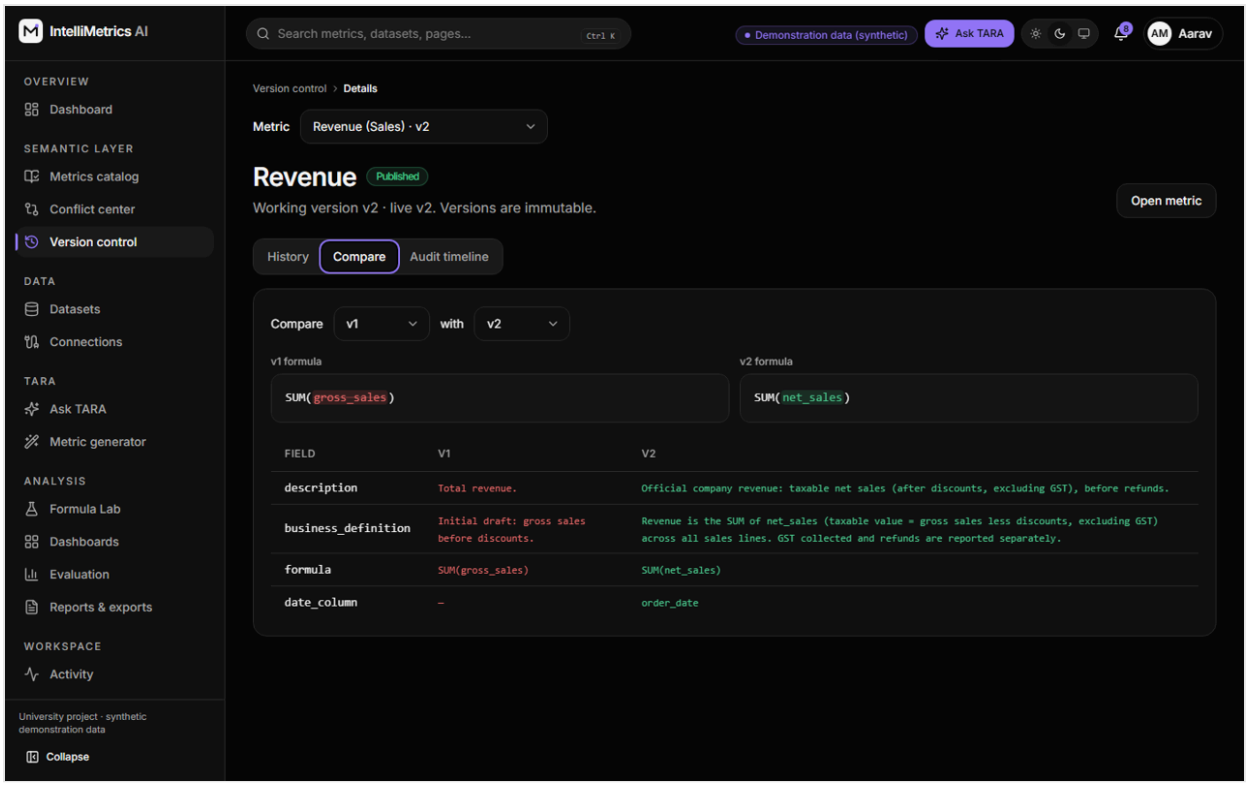
Version control (Revenue history), light mode



Version control (Revenue history), dark mode



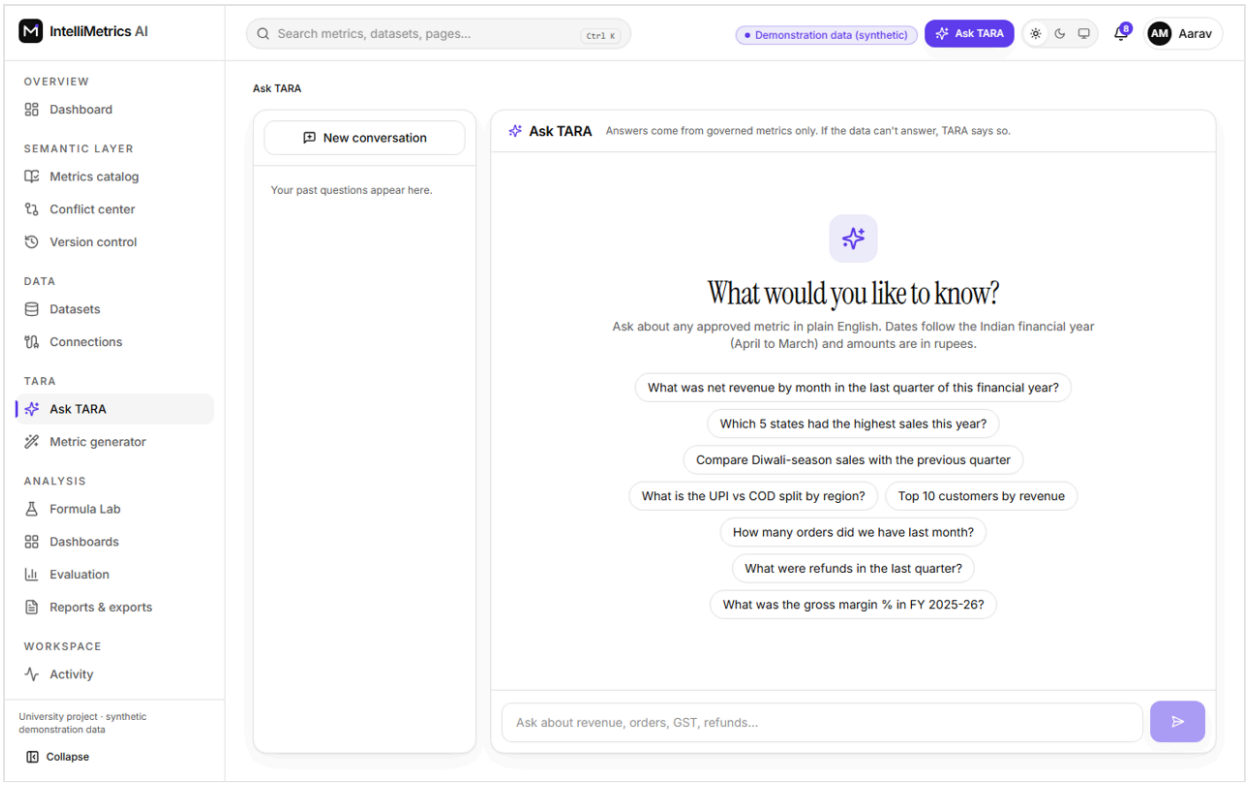
Version compare (diff view), light mode



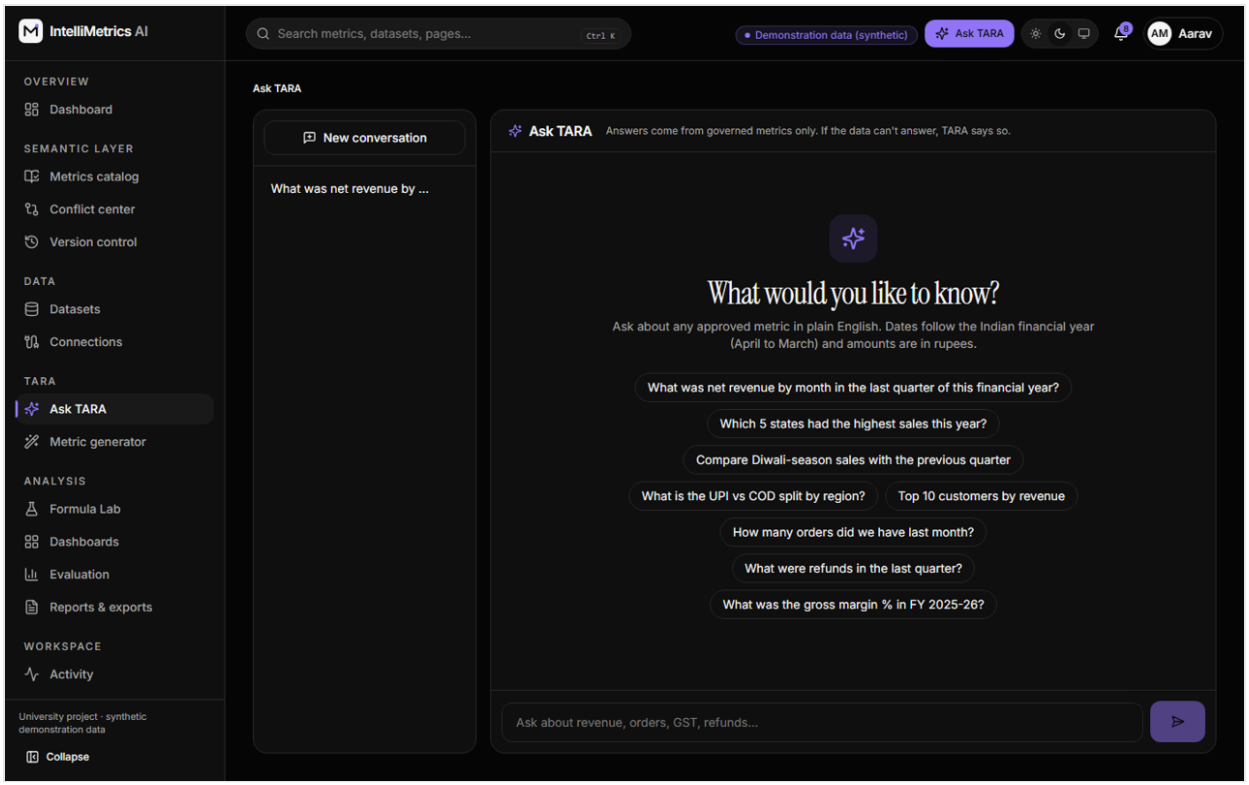
Version compare (diff view), dark mode

7.2.8 Ask TARA

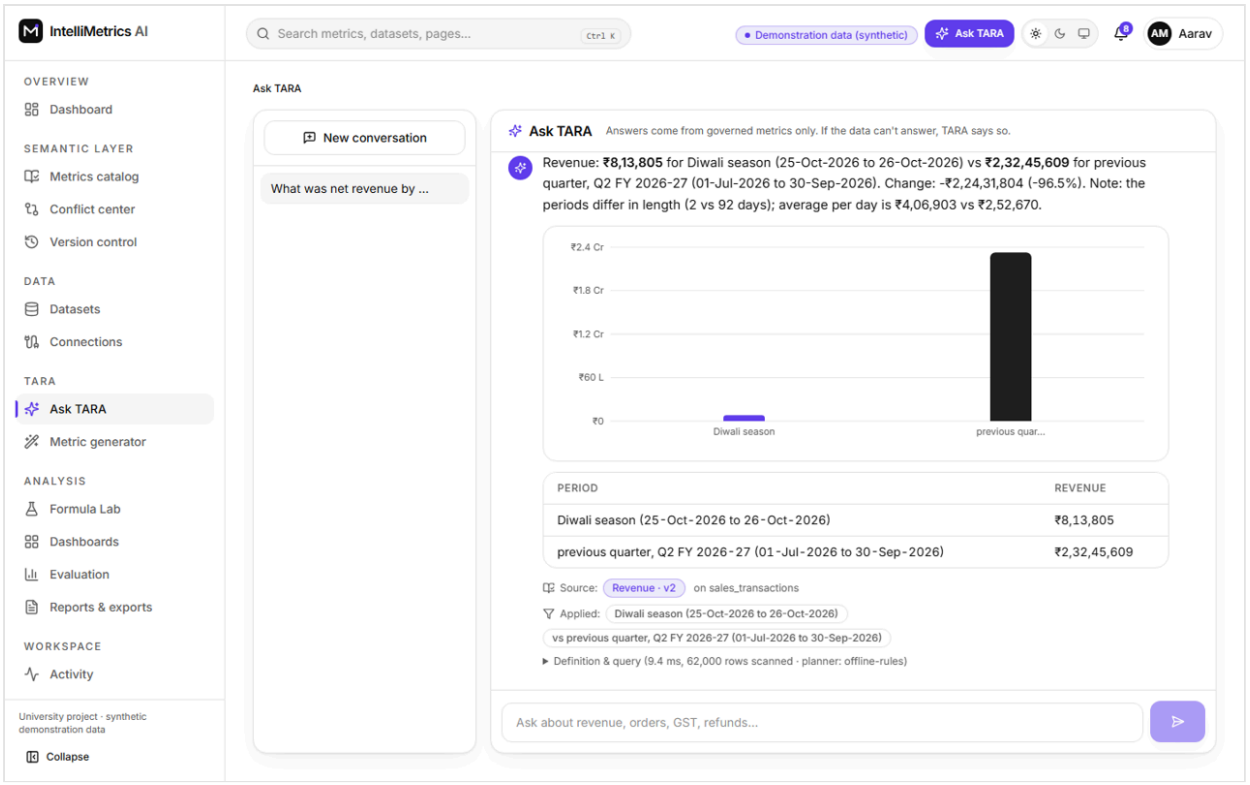
Type a question in plain English. TARA turns it into a checked query plan, runs it on approved metrics only, and shows the answer as text, table and chart, with the definition used, the date range and the filters. If the question cannot be answered from the data it says so.



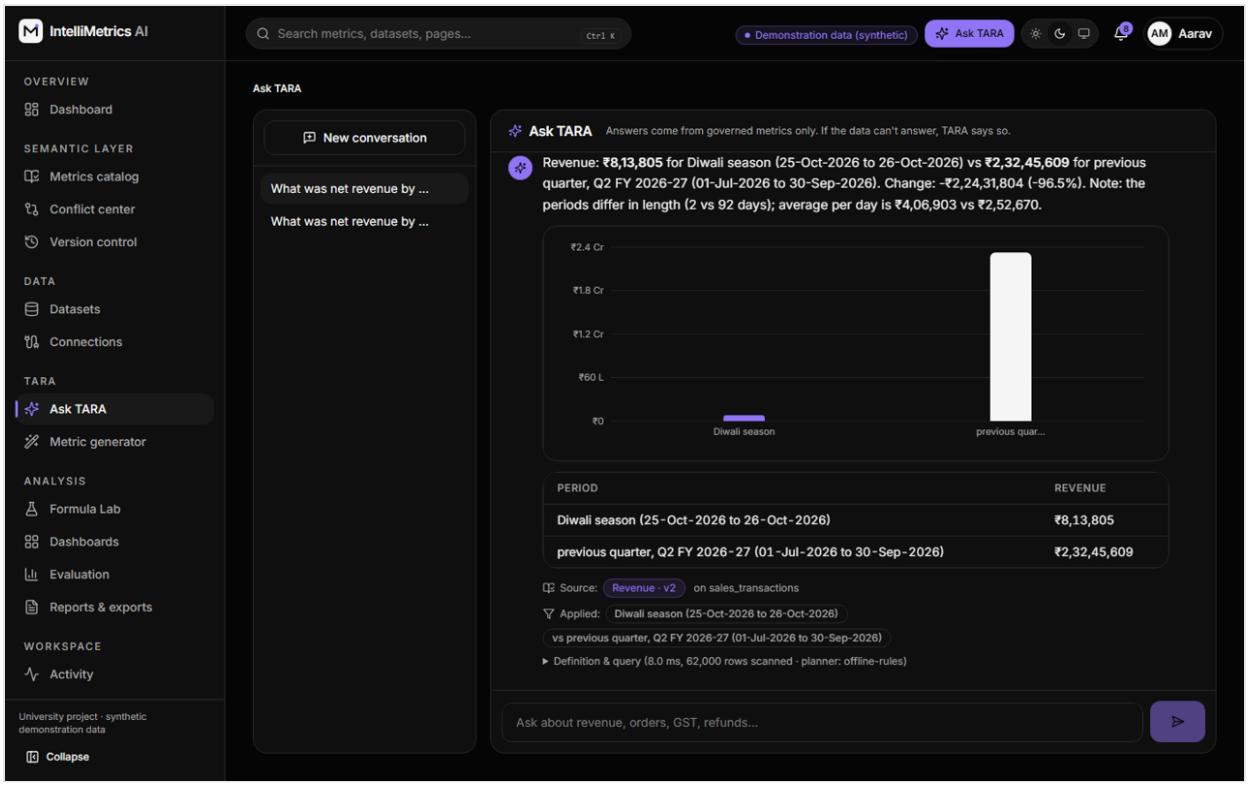
Ask TARA: suggested Indian questions, light mode



Ask TARA: suggested Indian questions, dark mode



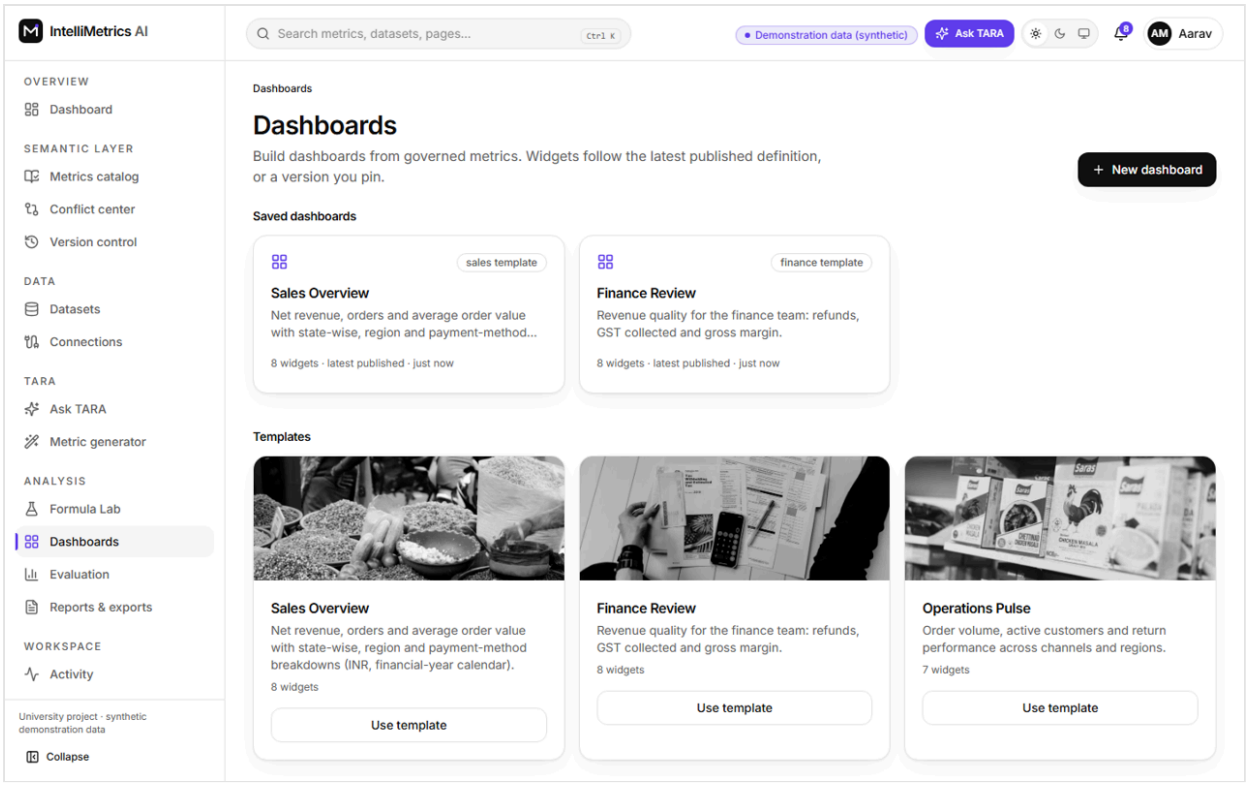
Ask TARA: answers with chart, table, sources and applied filters, light mode



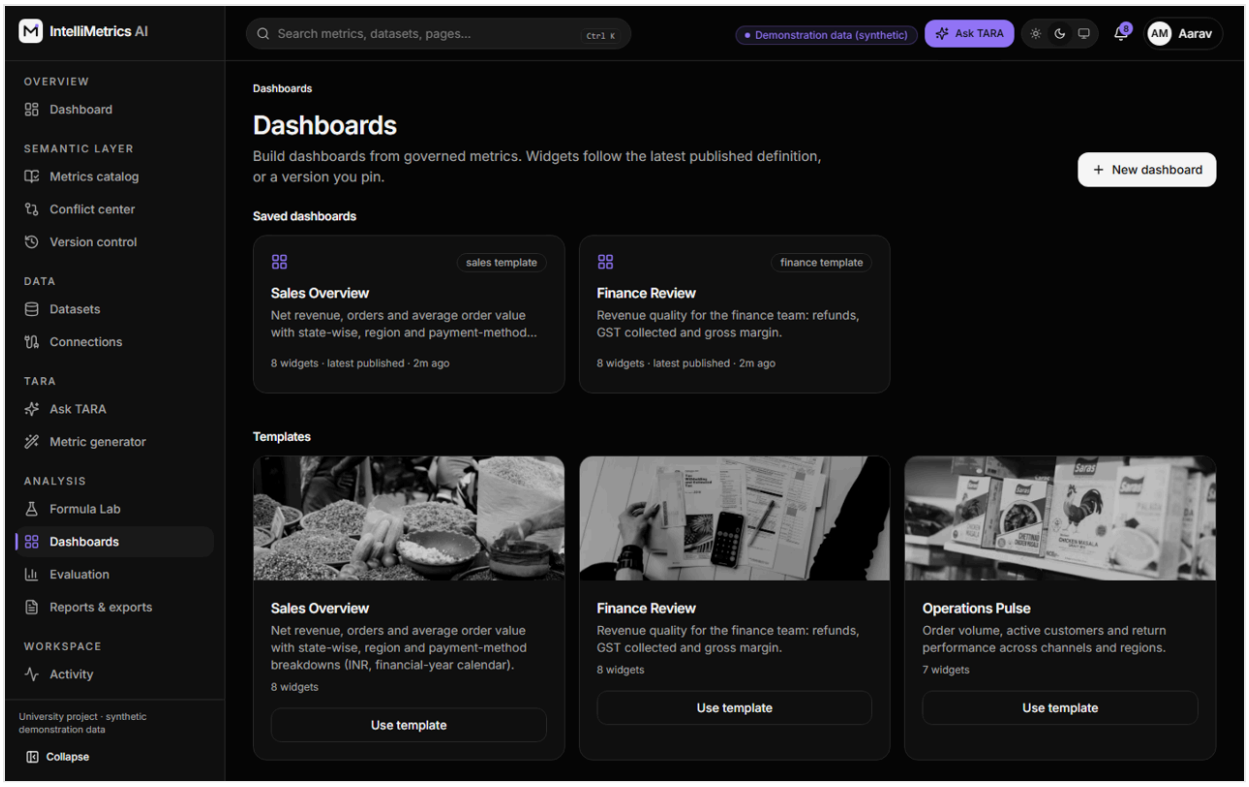
Ask TARA: answers with chart, table, sources and applied filters, dark mode

7.2.9 Dashboards

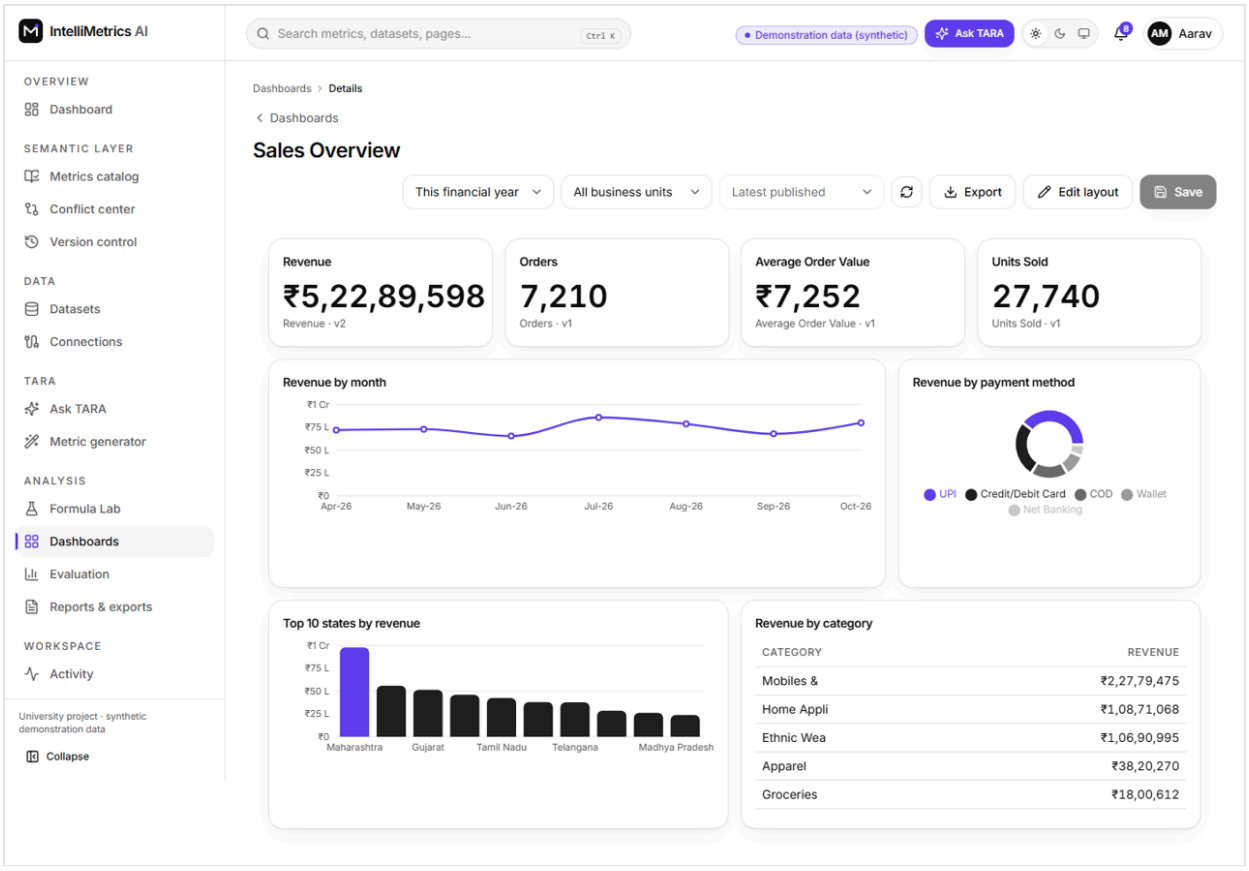
Saved dashboards with KPI, line, bar, pie and table widgets. *Edit layout* enables drag-and-resize. Dashboards can follow the latest published definition of each metric or be pinned to a version. Export as PNG, PDF or CSV.



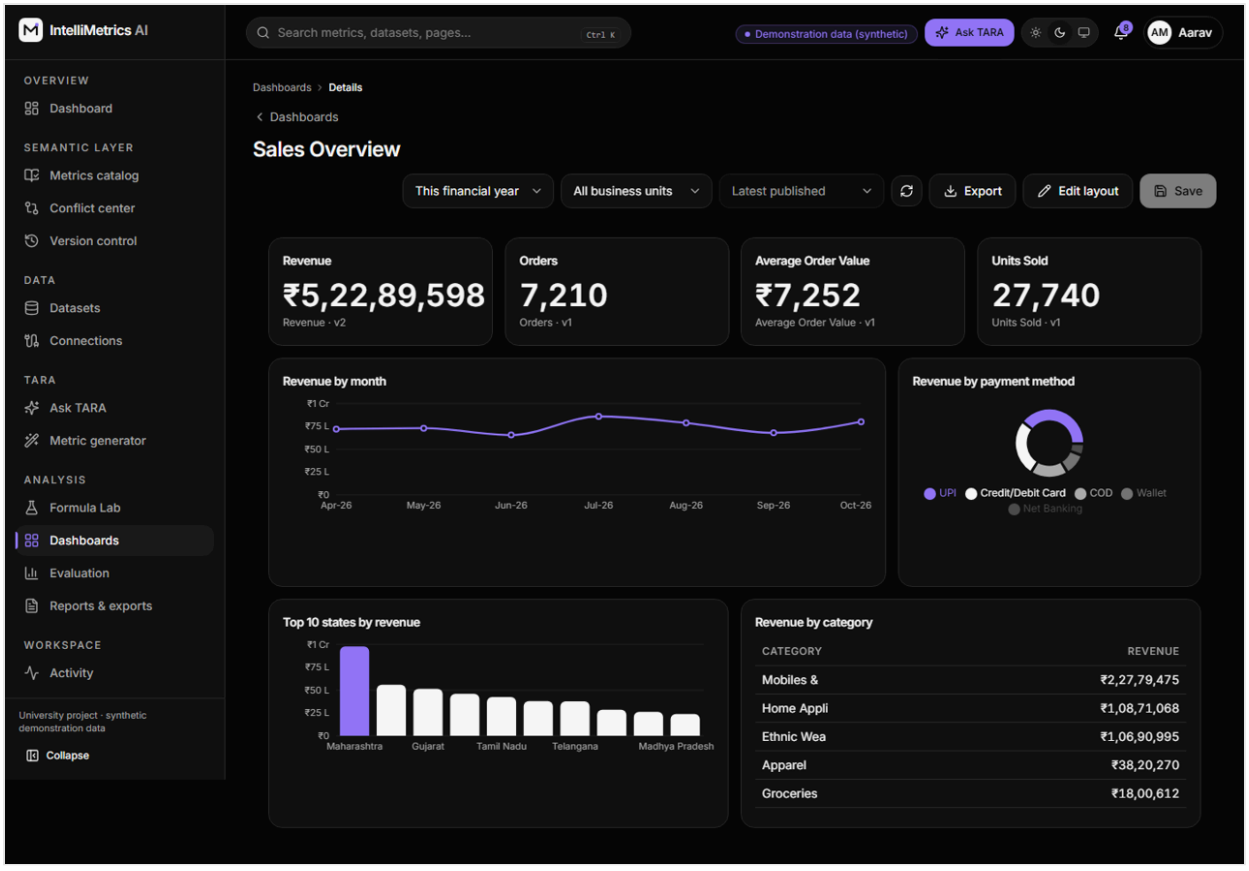
Saved dashboards and templates, light mode



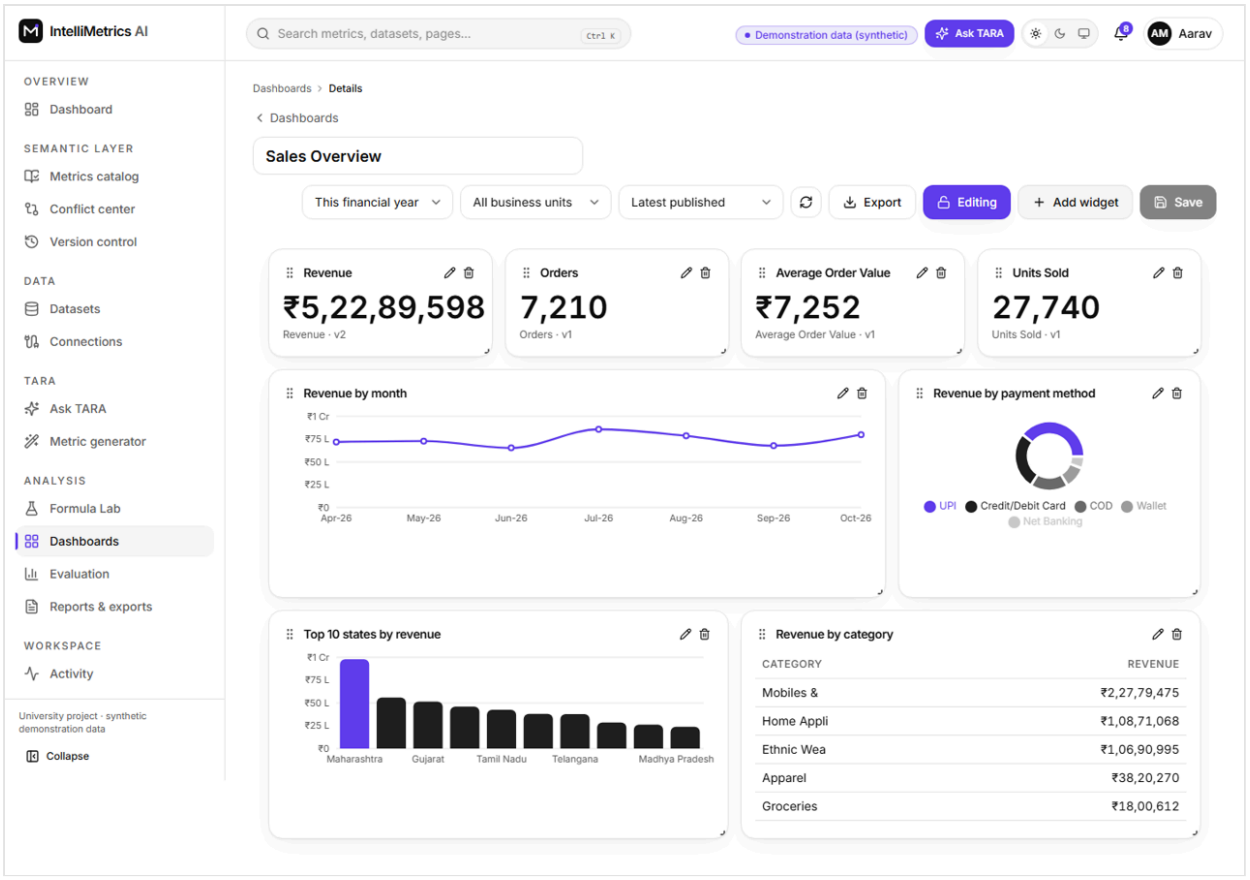
Saved dashboards and templates, dark mode



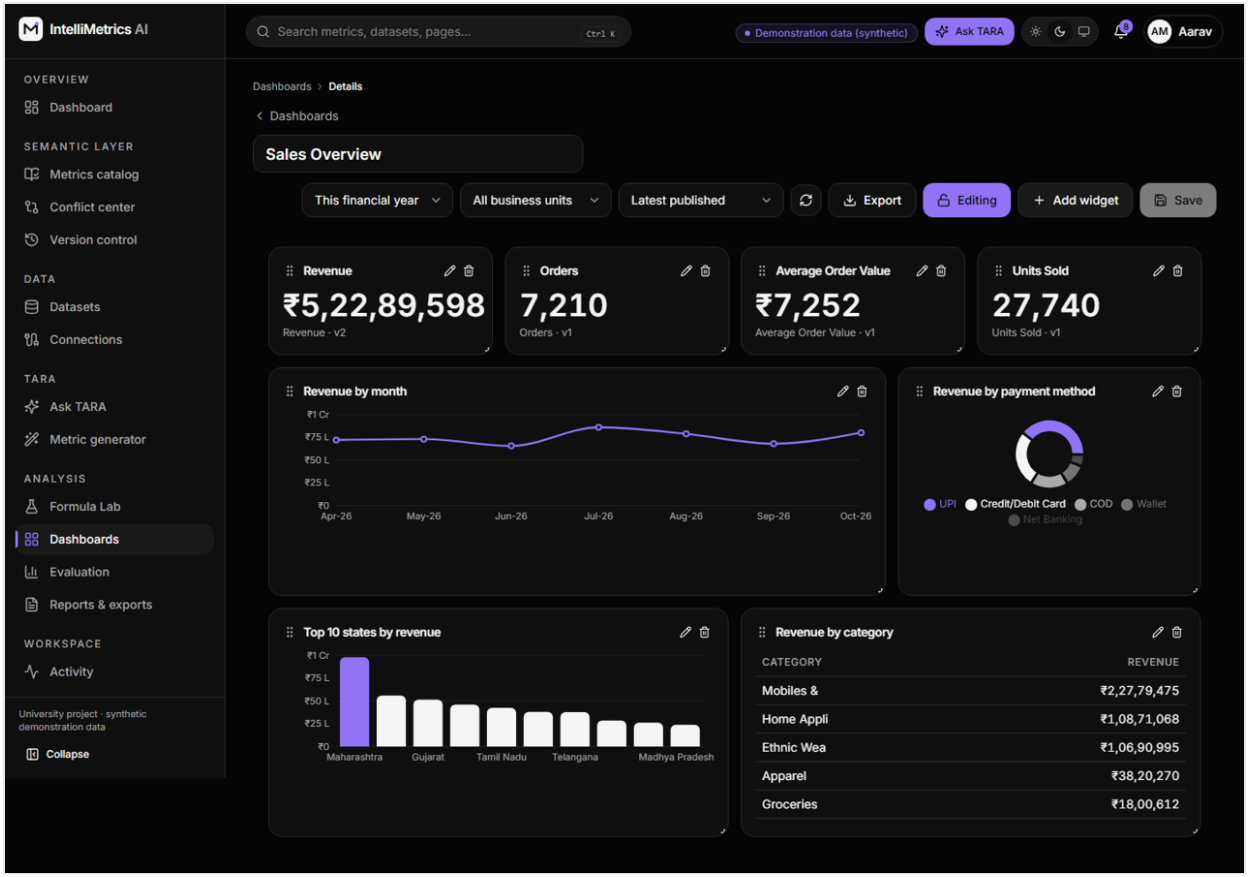
Dashboard builder (Sales Overview), light mode



Dashboard builder (Sales Overview), dark mode



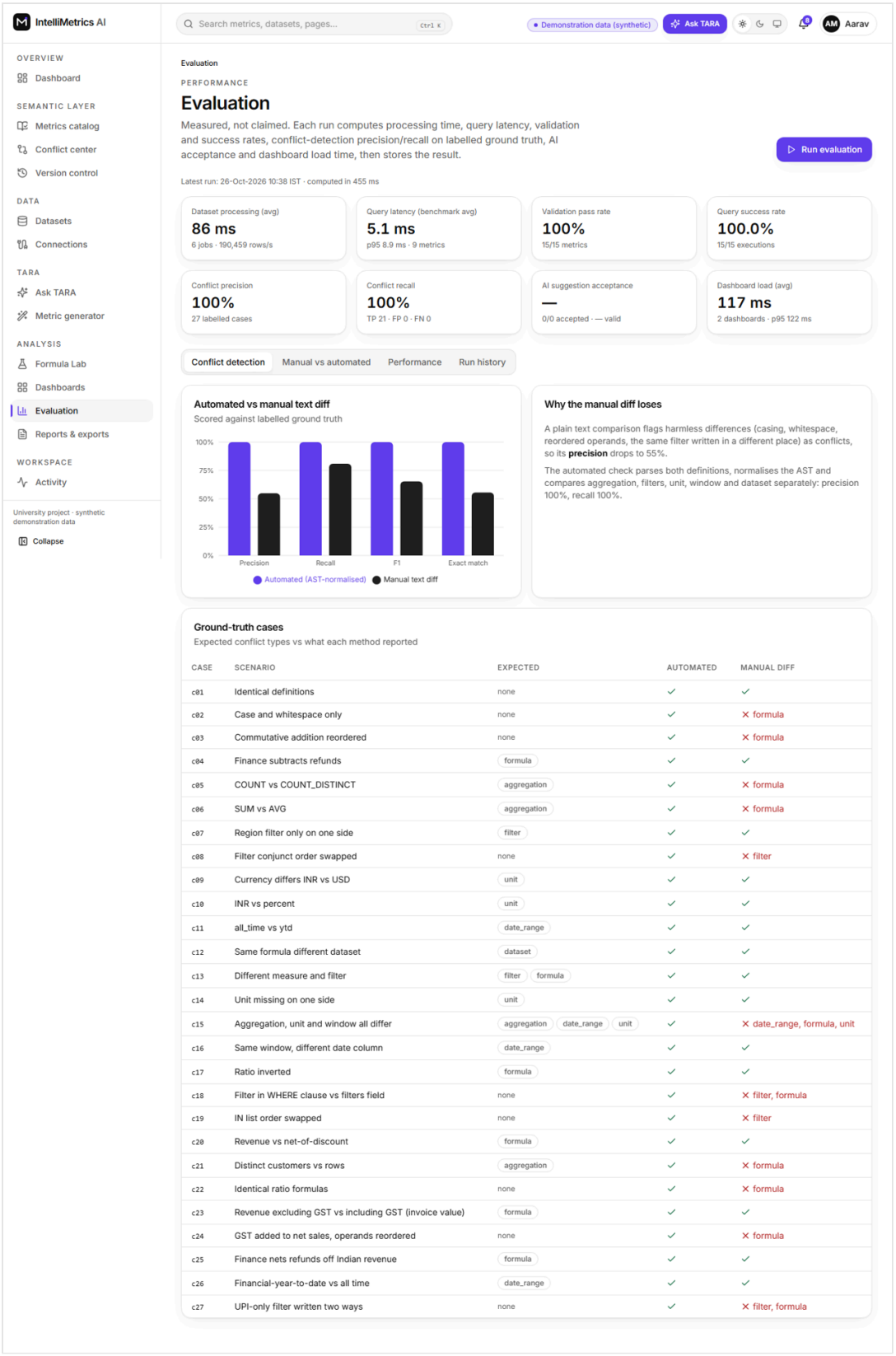
Dashboard builder in edit mode (drag handles), light mode



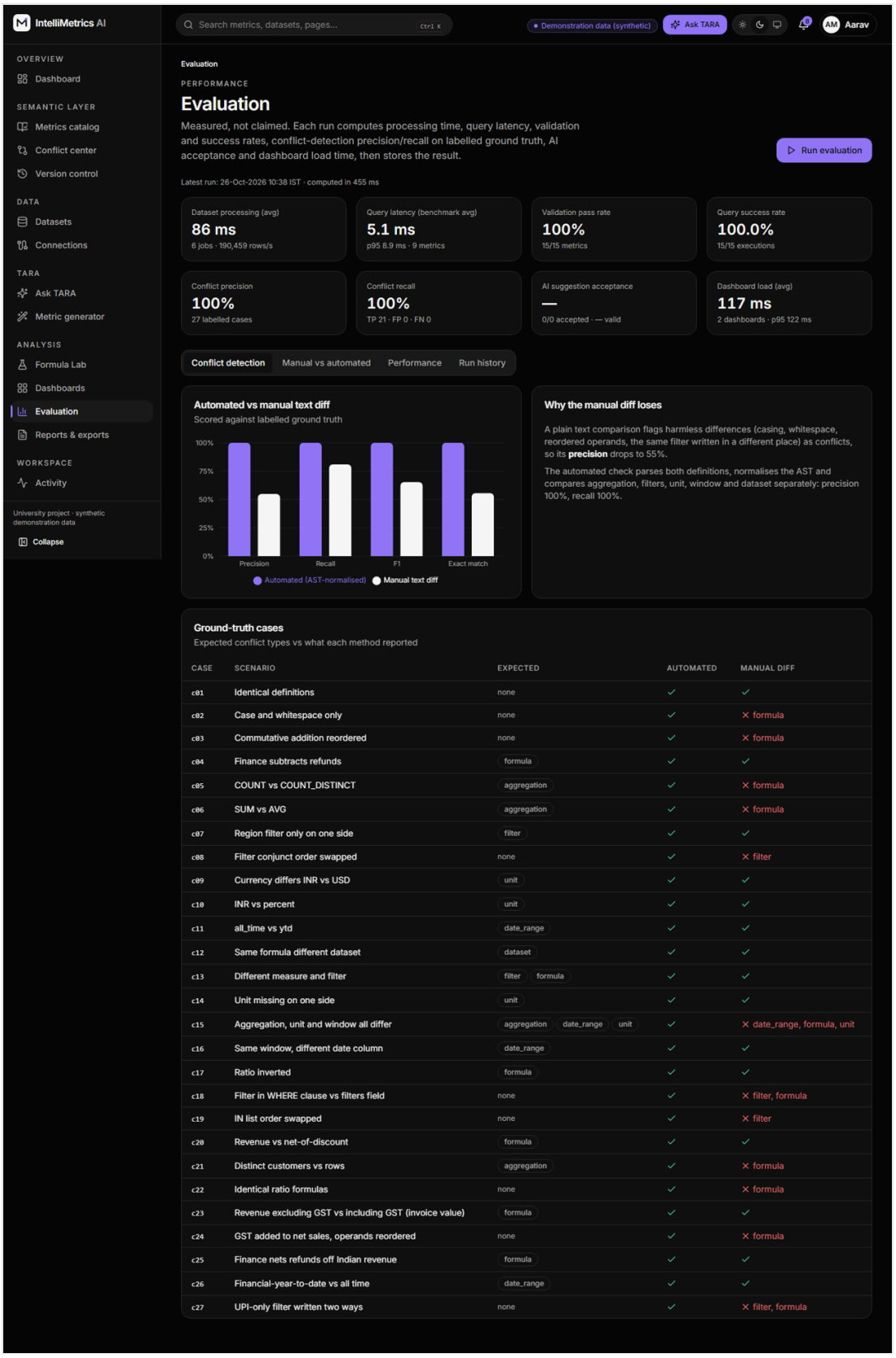
Dashboard builder in edit mode (drag handles), dark mode

7.2.10 Evaluation and reports

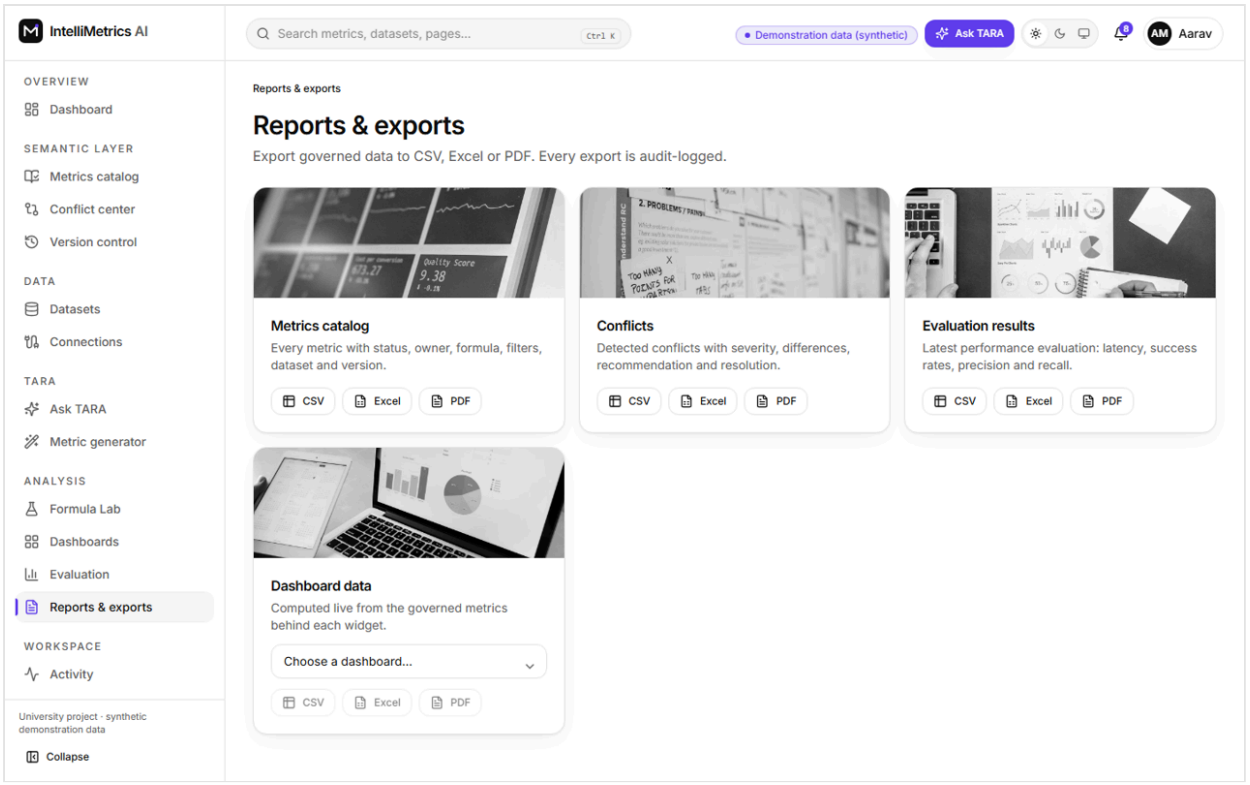
Run evaluation measures processing time, query latency, validation pass rate, query success rate, conflict-detection precision and recall on 27 labelled cases, and the AI suggestion acceptance rate. Reports can be exported in CSV, Excel or PDF with rupee formatting and IST timestamps.



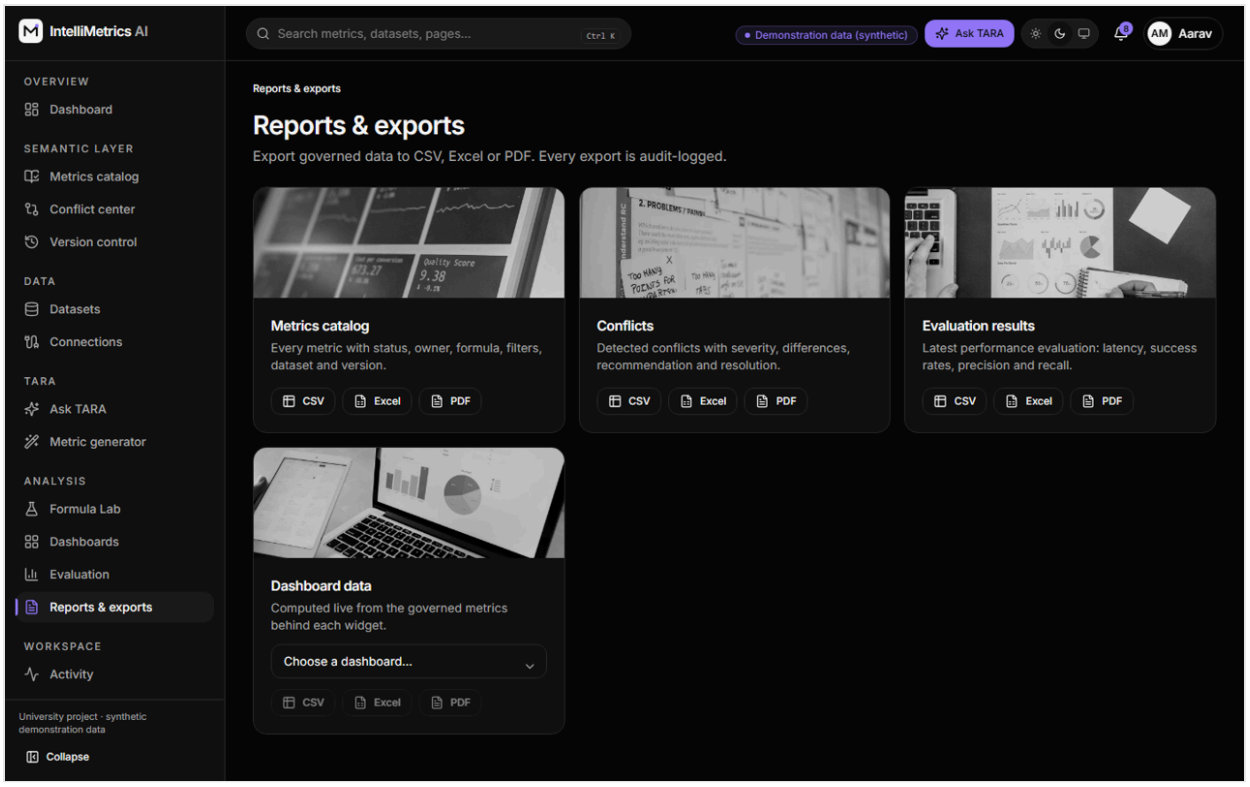
Performance evaluation, light mode



Performance evaluation, dark mode



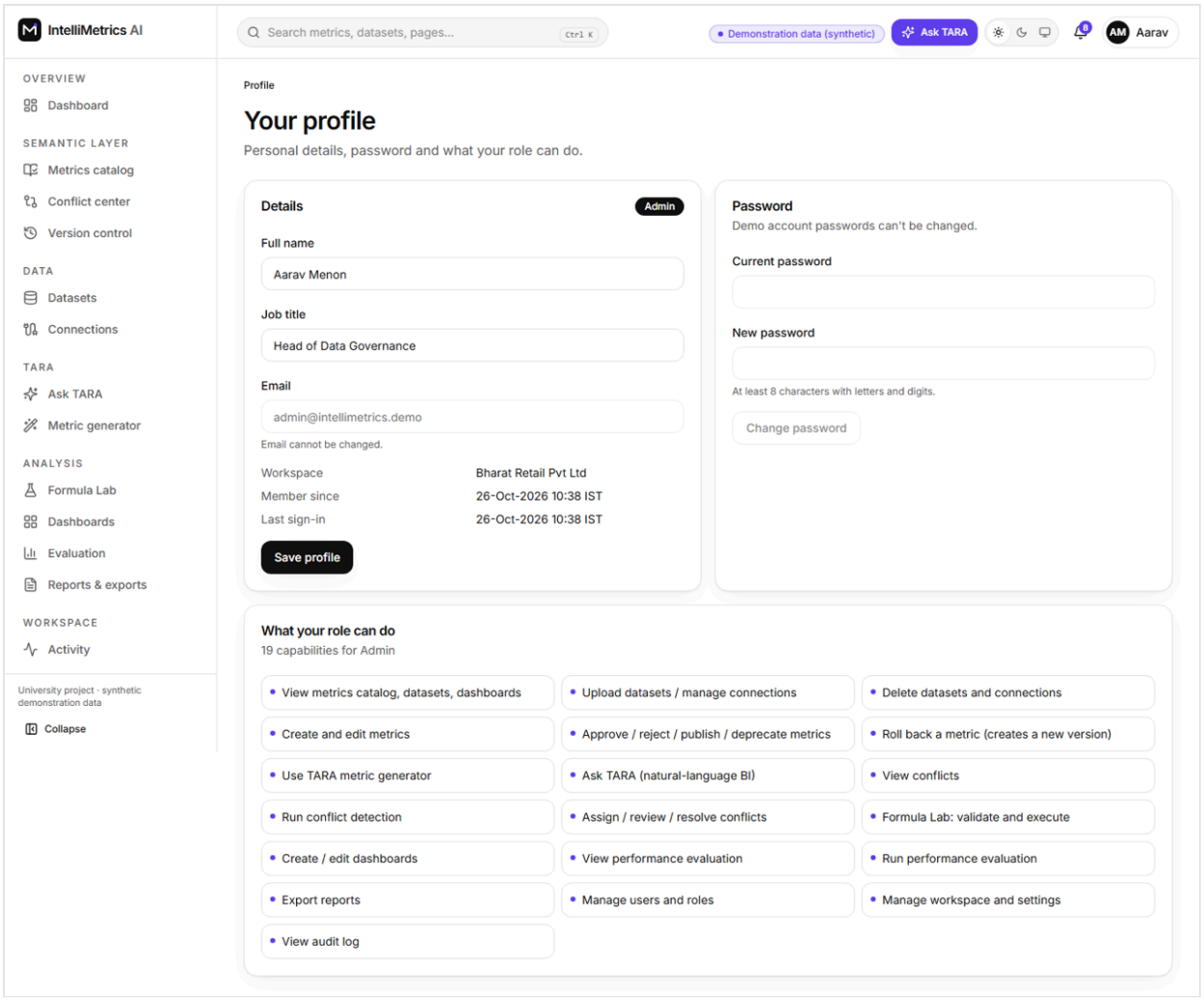
Reports and exports, light mode



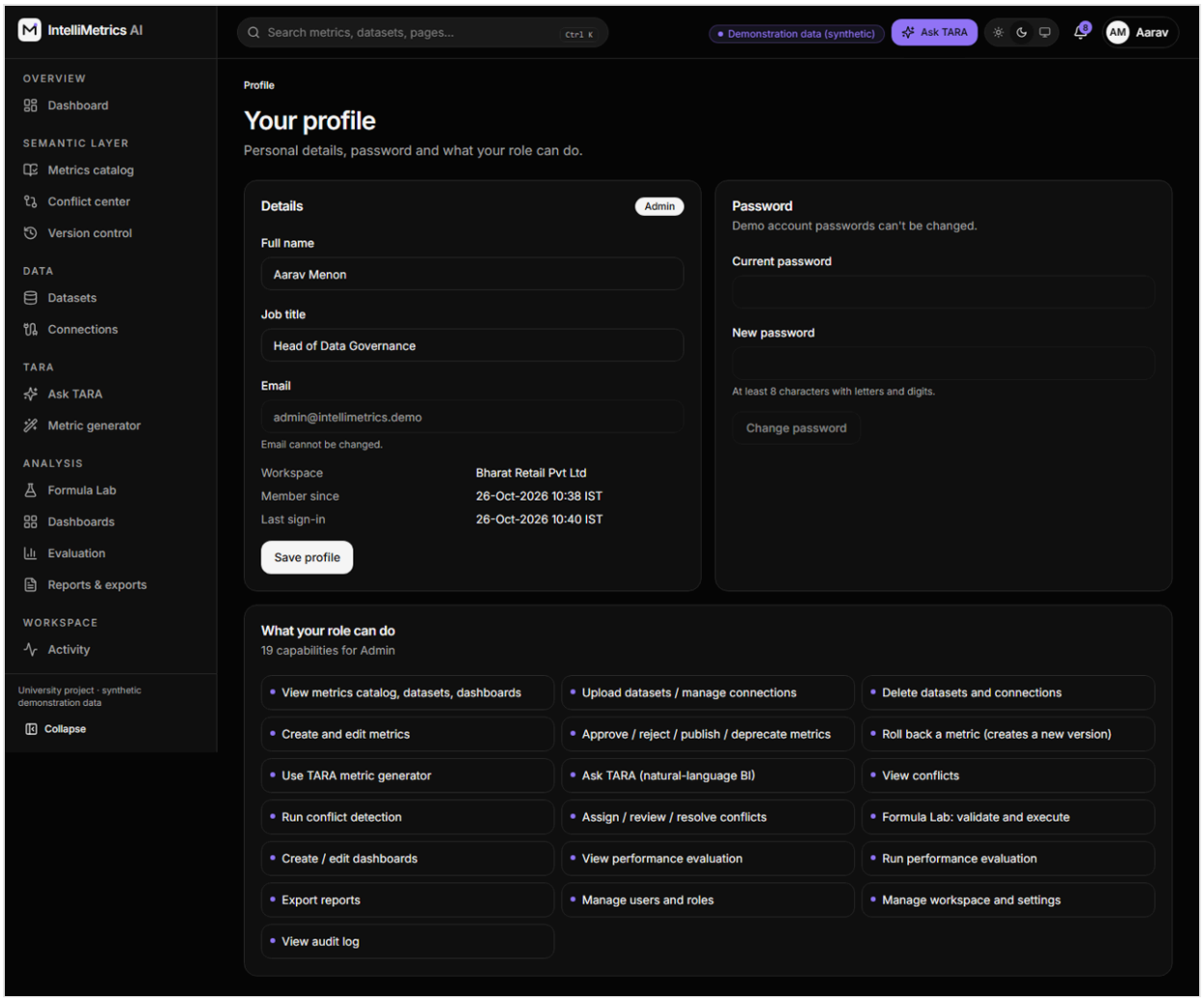
Reports and exports, dark mode

7.2.11 Profile, administration, settings and activity

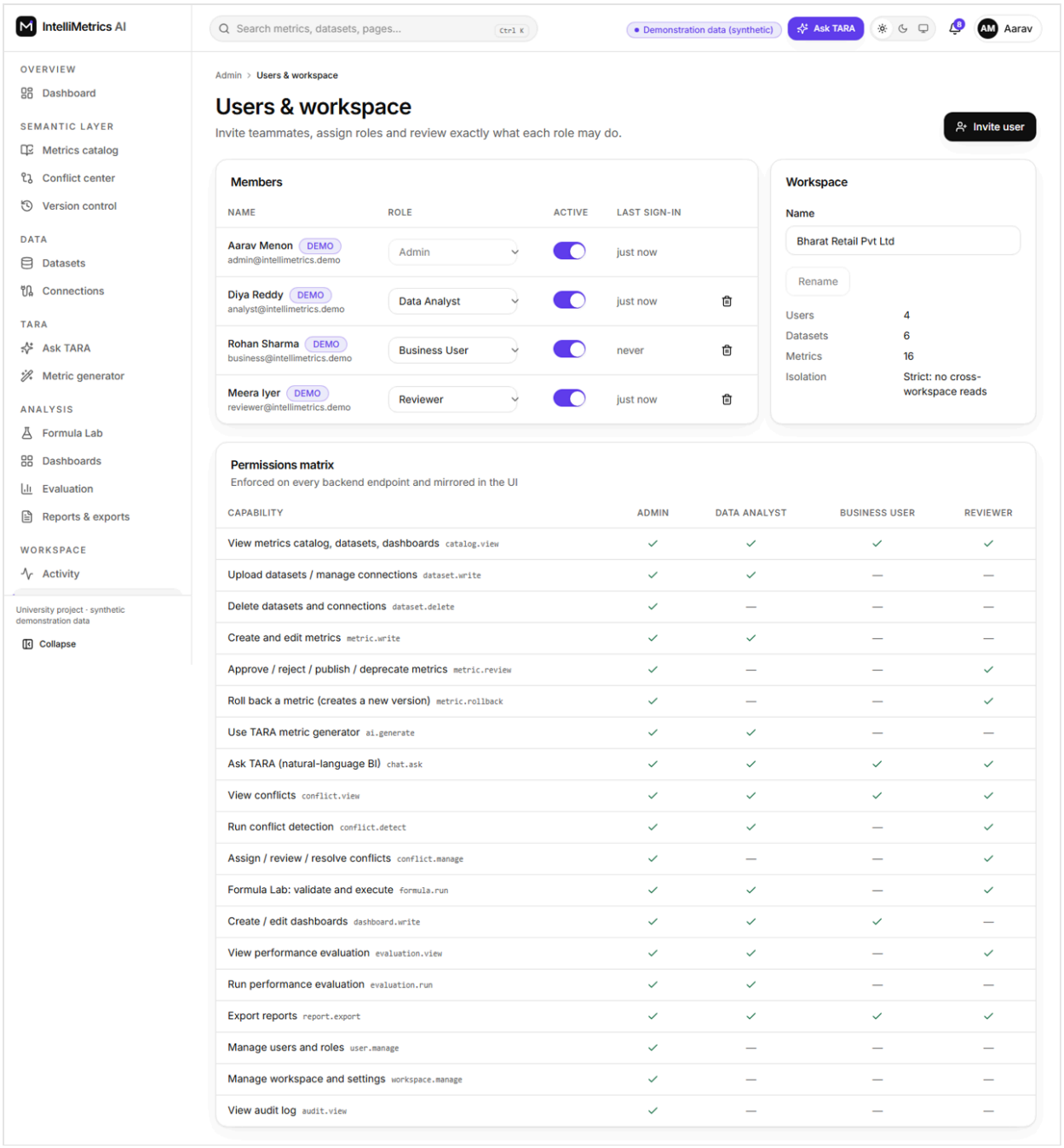
Admins manage users and roles, and read the audit log (who did what, when, from where). Settings show the AI mode, query limits and API usage.



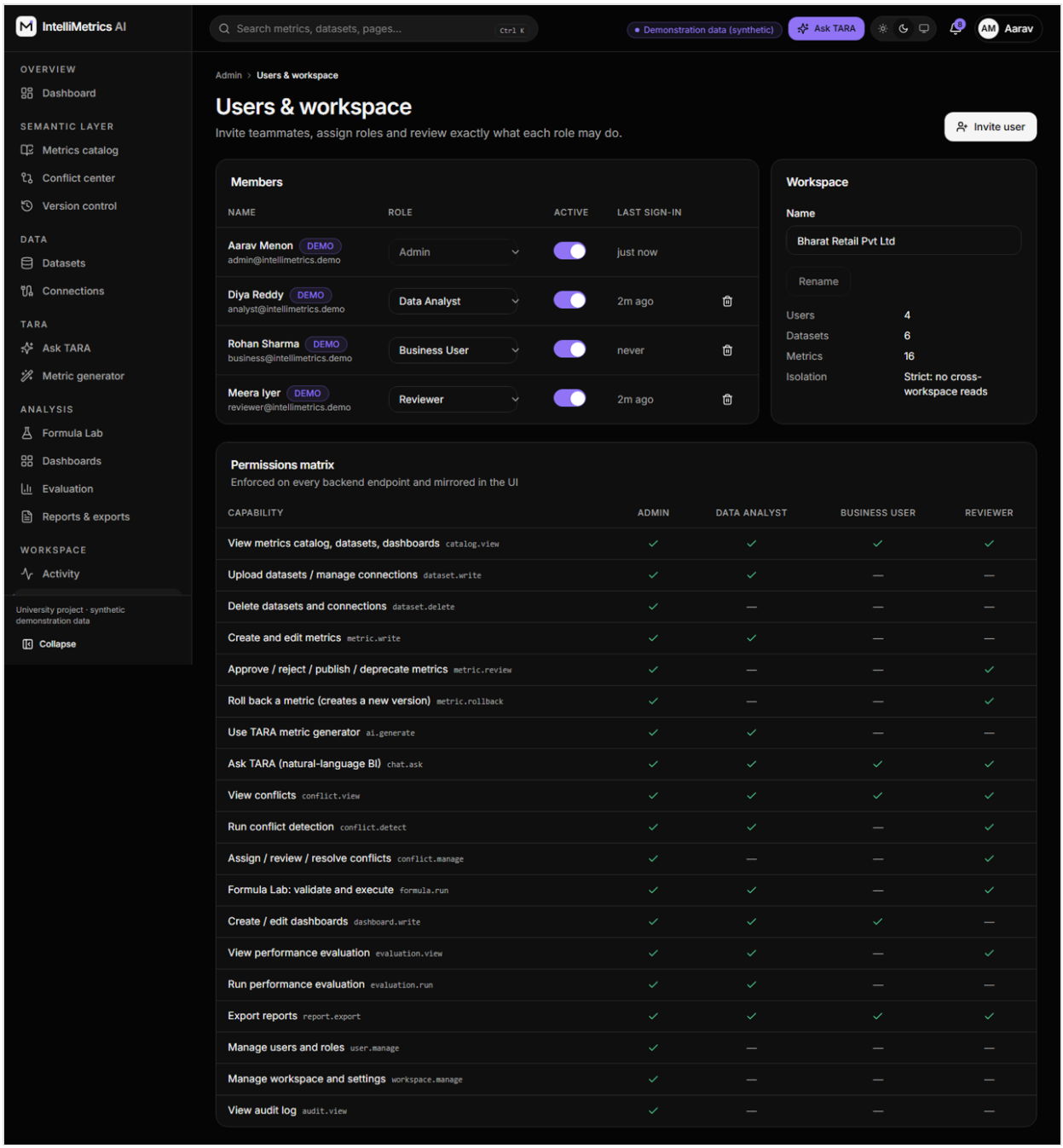
User profile, light mode



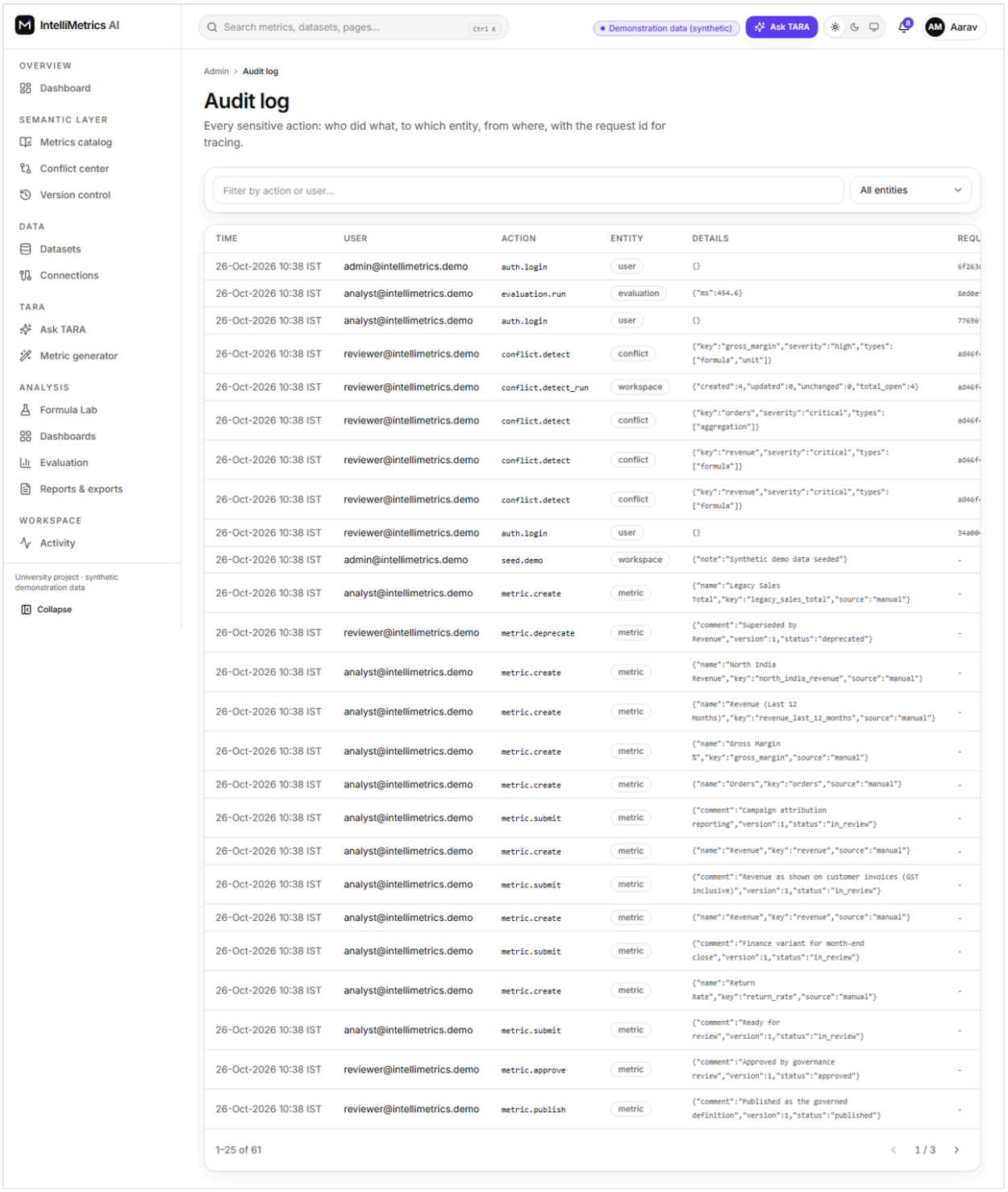
User profile, dark mode



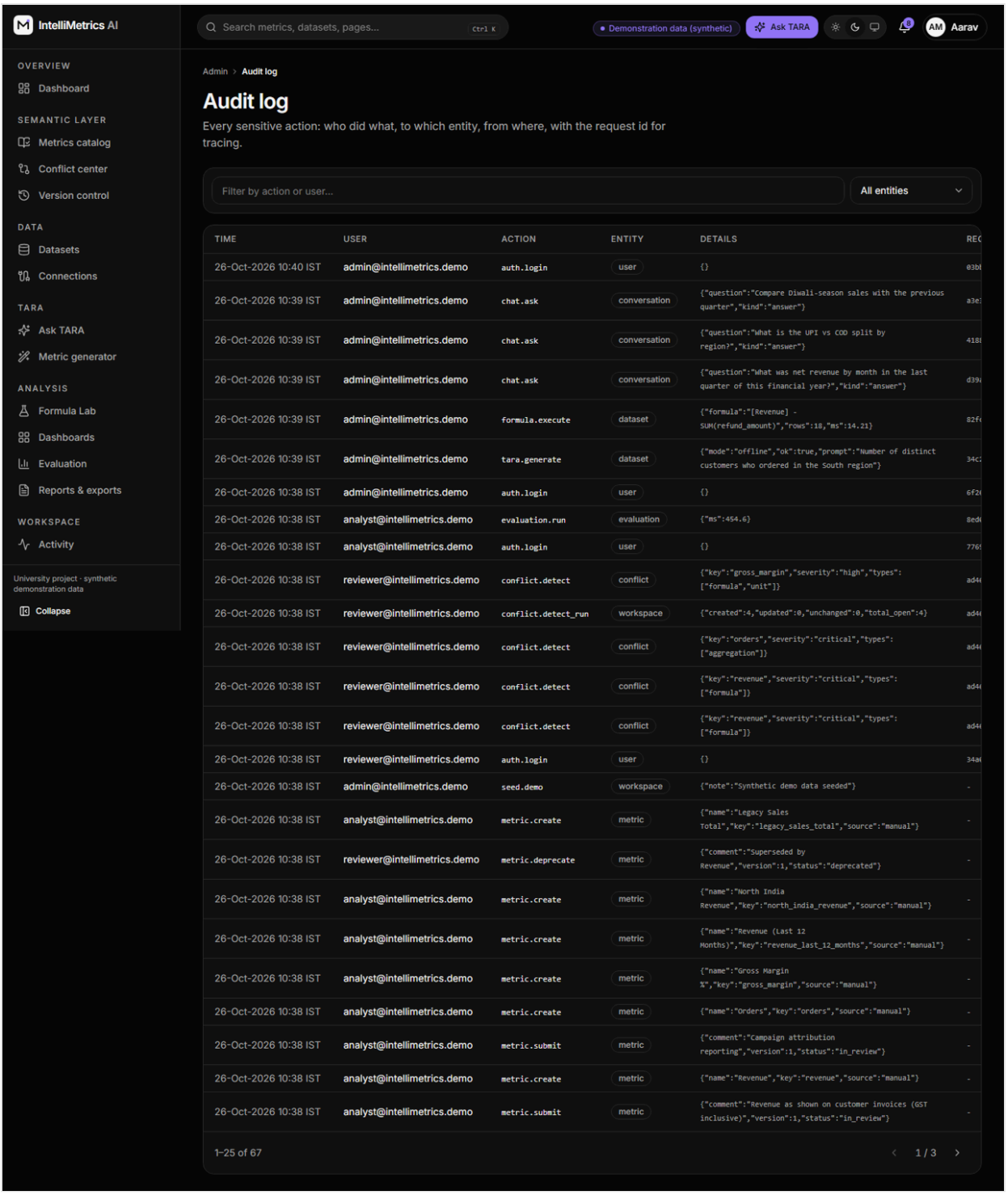
Workspace and user management, light mode



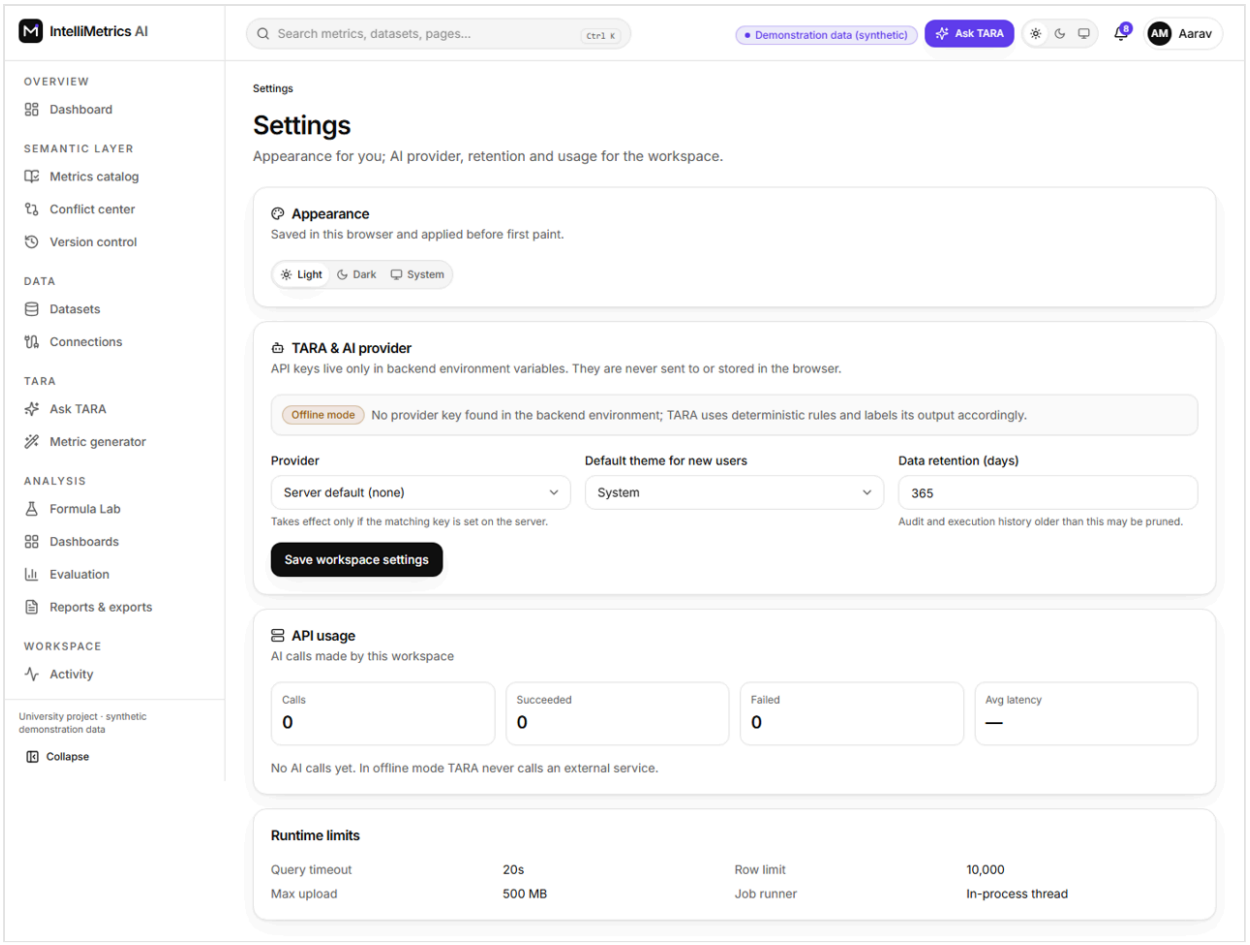
Workspace and user management, dark mode



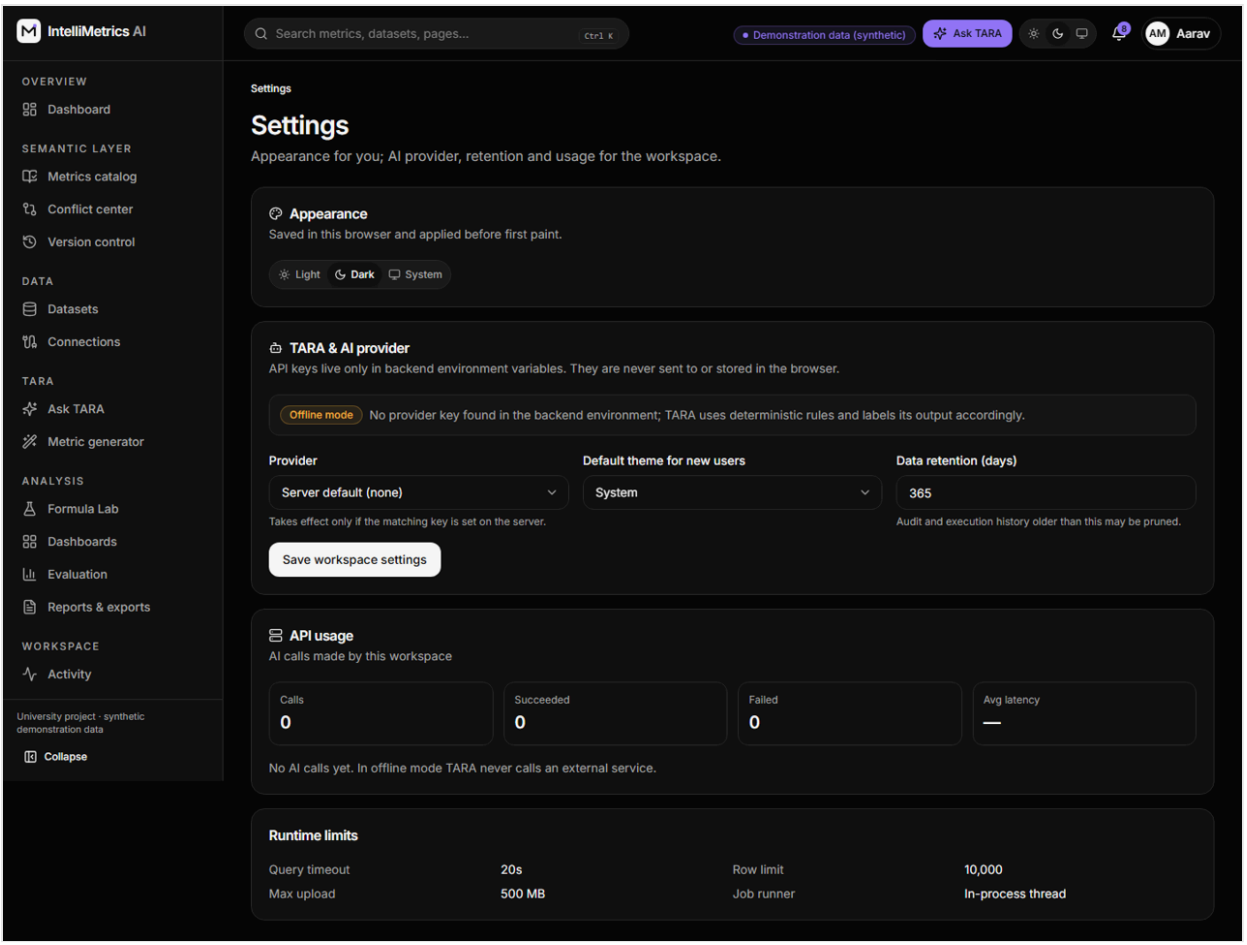
Audit log, light mode



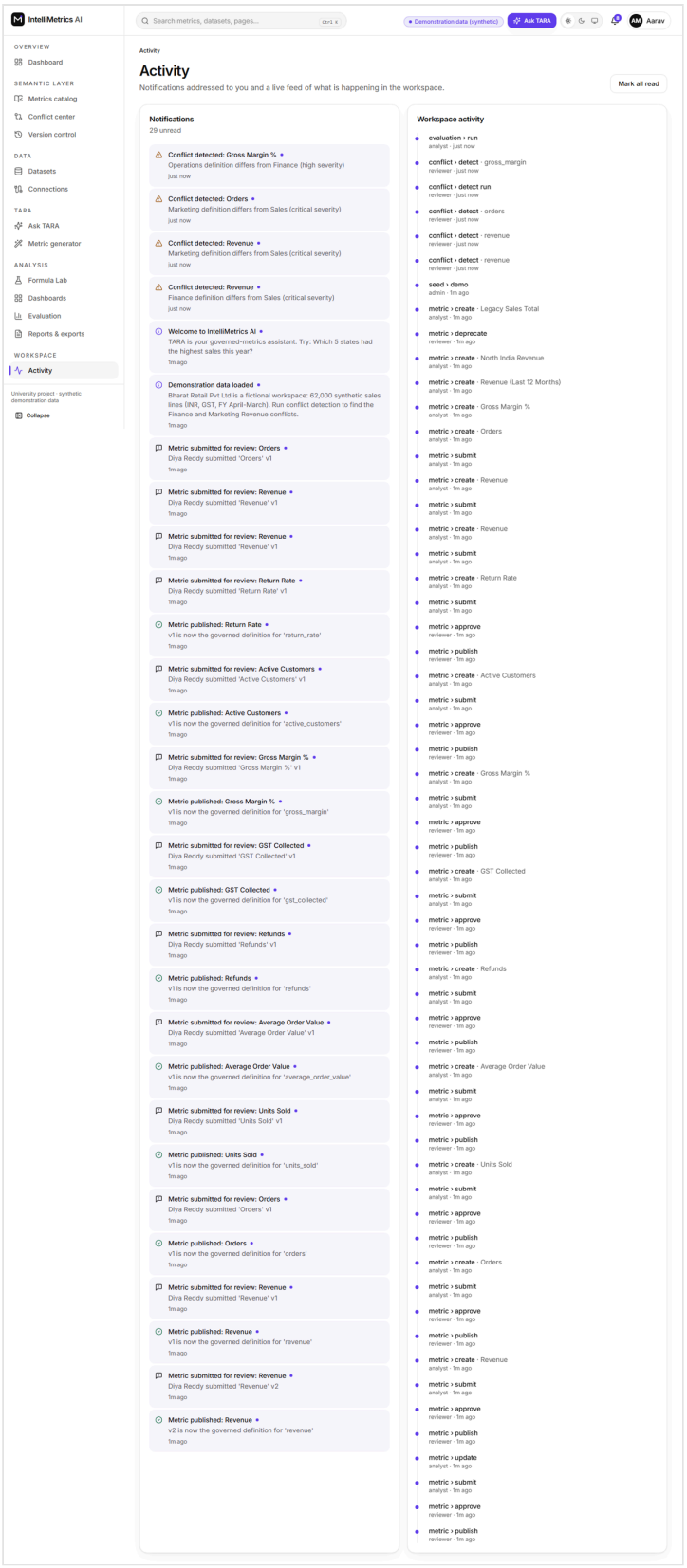
Audit log, dark mode



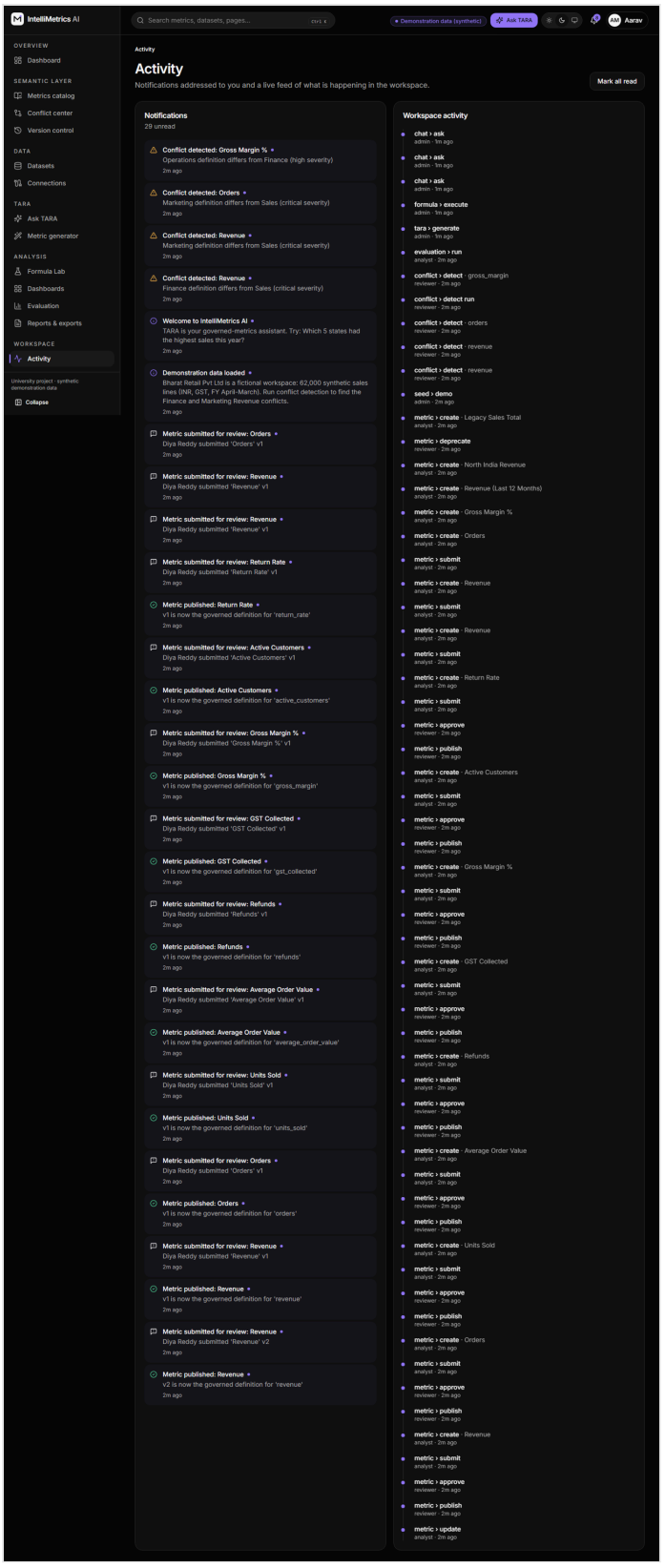
Application settings, light mode



Application settings, dark mode



Notifications and activity, light mode

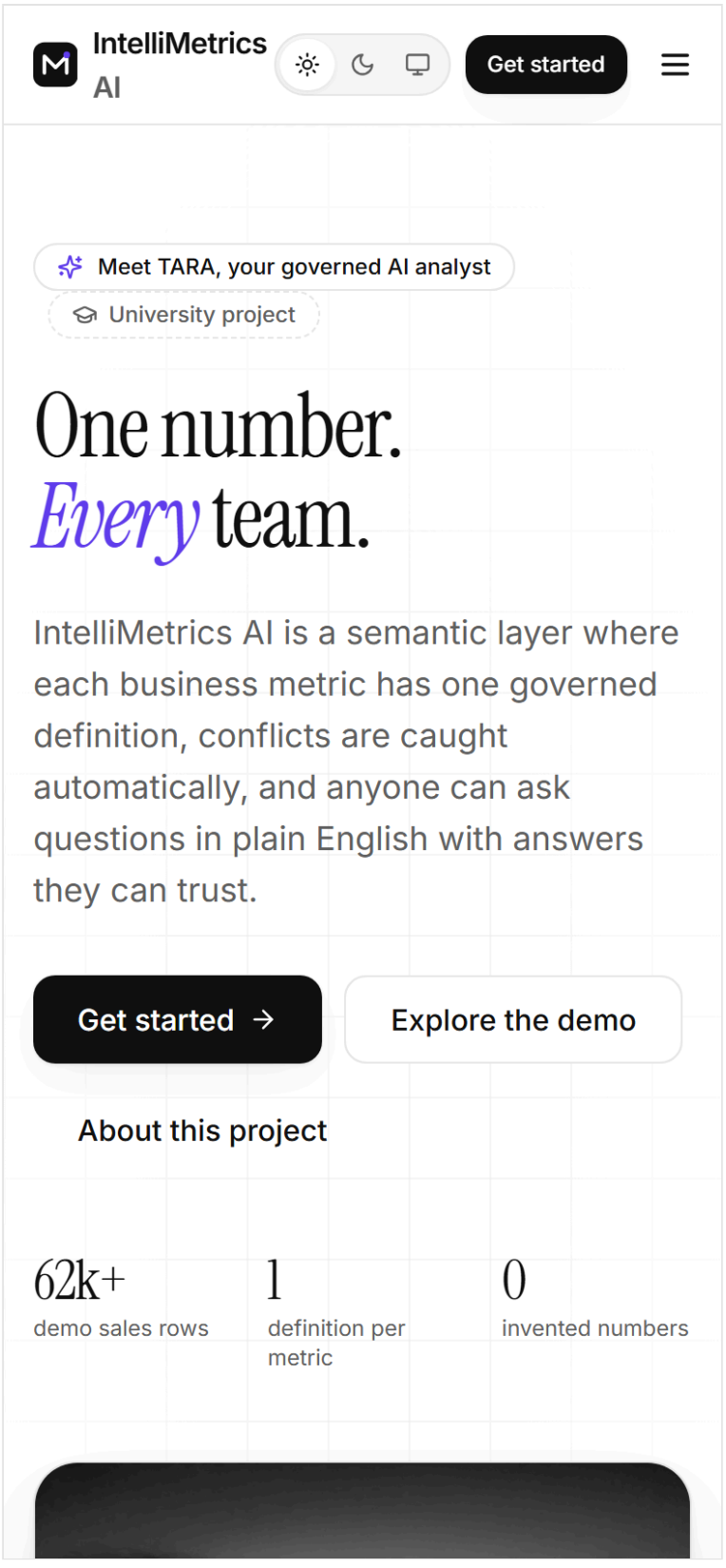


Notifications and activity, dark mode

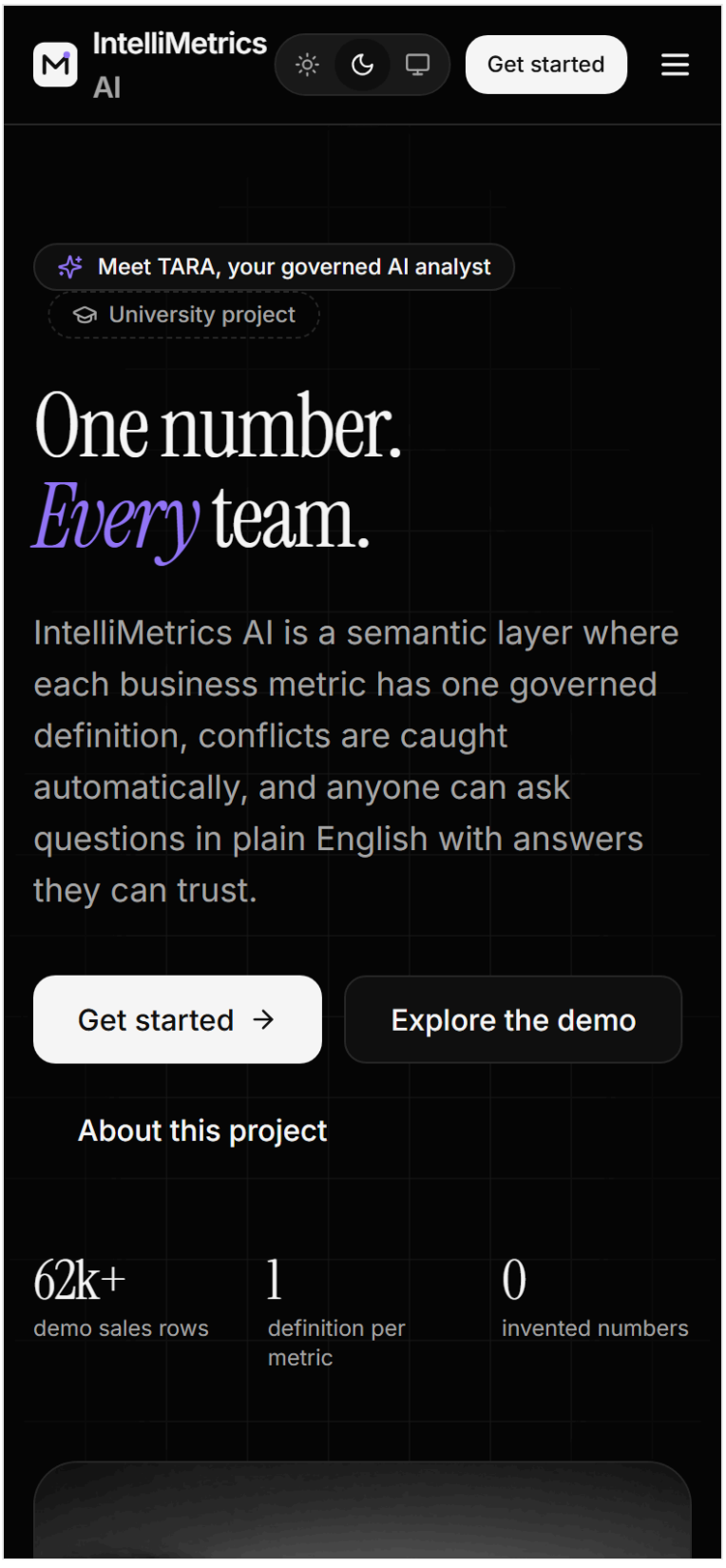
7.3 Phone-sized screens

The layout adapts to phone widths (390 px was checked).

Mobile: landing page

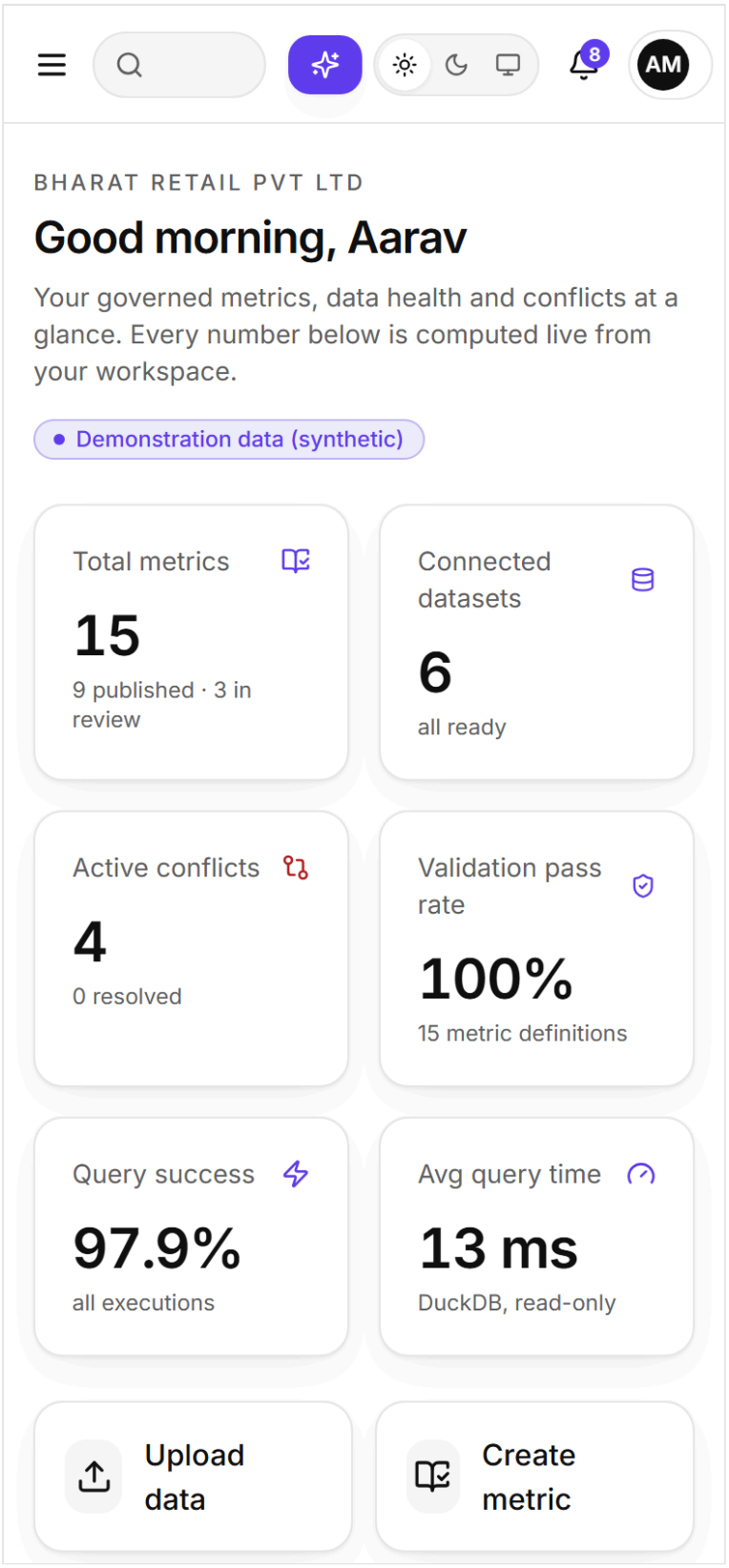


Mobile: landing page, light mode

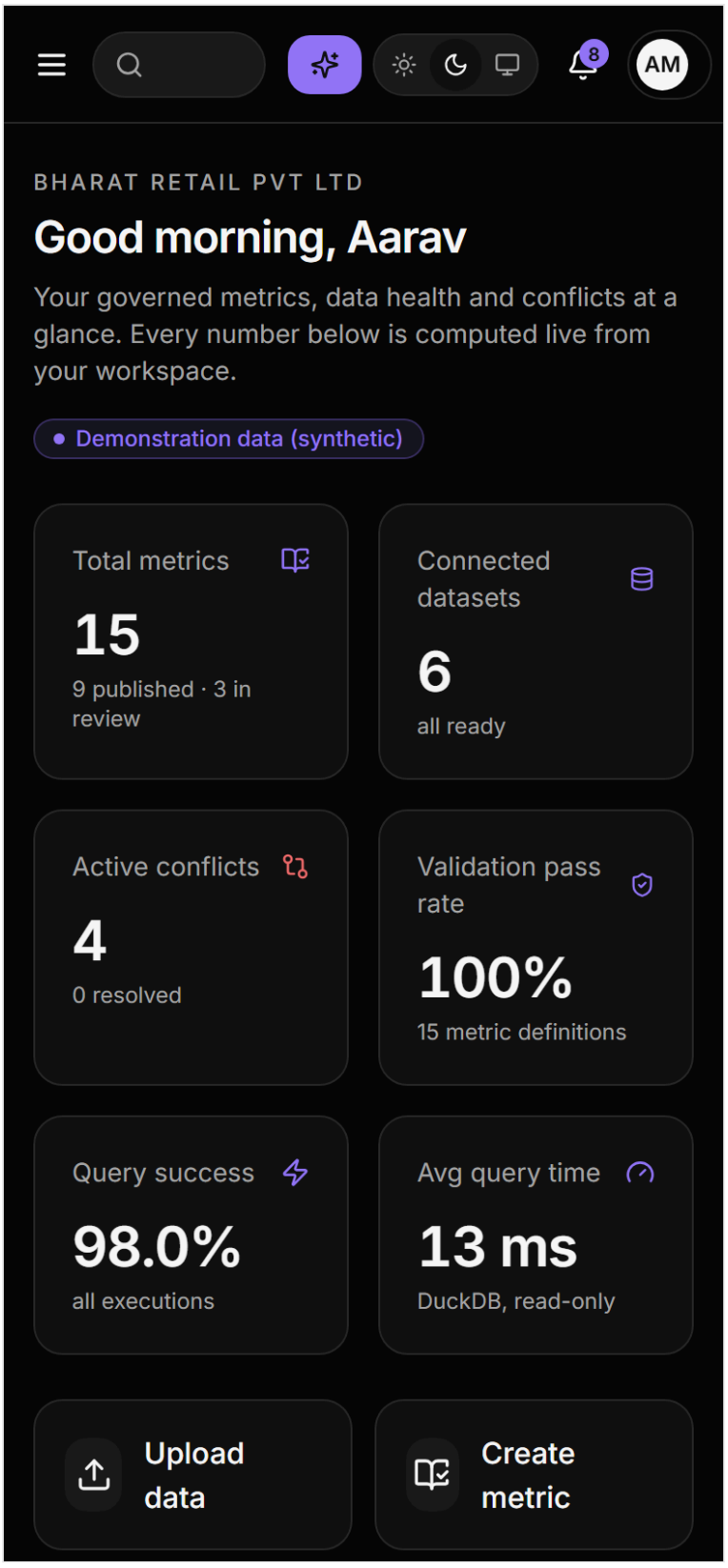


Mobile: landing page, dark mode

Mobile: dashboard

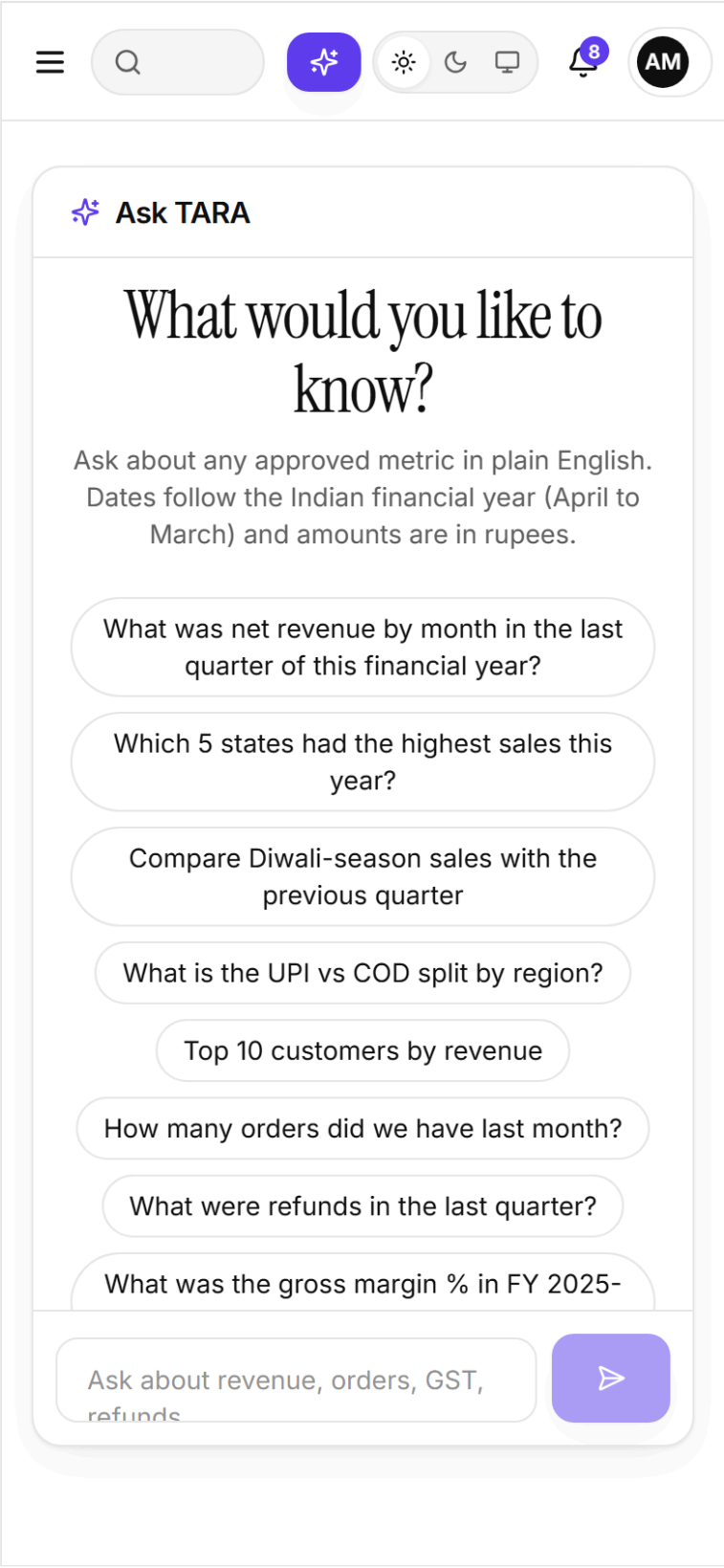


Mobile: dashboard, light mode

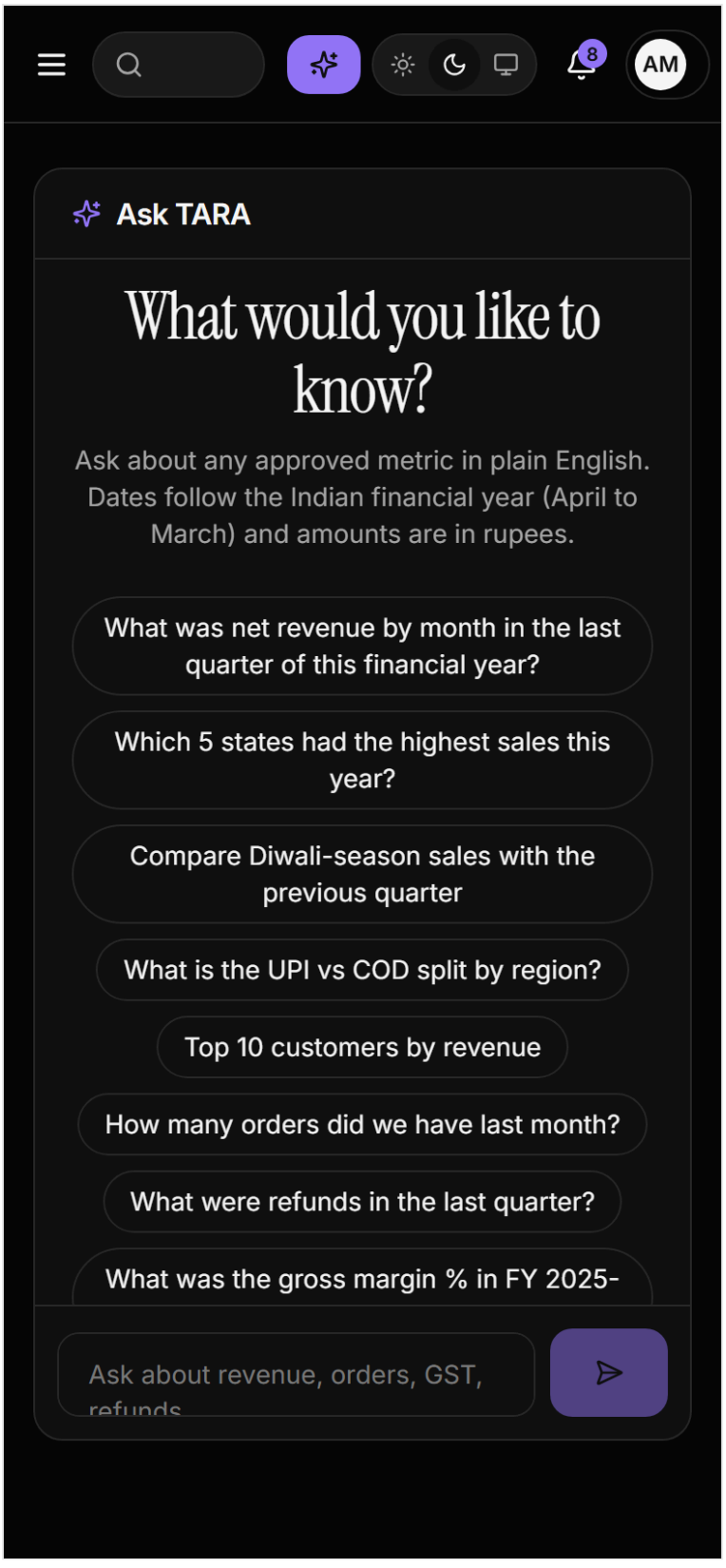


Mobile: dashboard, dark mode

Mobile: Ask TARA



Mobile: Ask TARA, light mode



Mobile: Ask TARA, dark mode

8 A ten-minute walkthrough

Use the desktop browser at http://localhost:5173 (or the port you chose).

Step	Sign in as	Do this	You should see
1	Meera Iyer (Reviewer)	Open *Conflict detection*, click **Run conflict detection**	Conflicts for **Revenue**: the Finance version subtracts refunds; the Marketing version uses the GST-inclusive invoice value
2	Meera Iyer	Open the Finance conflict; **Assign** to Meera Iyer, **Start review**, **Approve**, then **Resolve** by renaming the variant (for example to *Net Revenue*) and adding a note	The renamed metric becomes version 2, and the "Audit history" lists every step
3	Diya Reddy (Analyst)	Open *Formula Lab*, choose `sales_transactions`, enter `[Revenue] - SUM(refund_amount)`, group by `state`, **Execute**	A table of rupee values by state, with rows scanned and duration
4	Diya Reddy	Replace the formula with `SUM(net_sales); DROP TABLE users`	A red error, "Nothing was executed"
5	Rohan Sharma (Business User)	Open *Ask TARA* and click each suggested question in turn	See below
6	Rohan Sharma	Ask "What is the weather in Delhi?"	A refusal saying TARA could not match the question to an approved metric; no number is invented
7	Rohan Sharma	Open *Dashboards*, then *Sales Overview*	KPI cards, trend line and breakdowns in rupees
8	Aarav Menon (Admin)	Open *Audit log*	Every action above with user, time (IST) and request id
9	Suresh Naidu (other workspace)	Open *Metrics* and *Datasets*	Only Sagar Foods data; nothing from Bharat Retail

The suggested questions in step 5 are:

1. *What was net revenue by month in the last quarter of this financial year?* A monthly line or bar of the last completed April–March quarter.
2. *Which 5 states had the highest sales this year?* A ranked table and bar chart; "this year" means the financial year to date from 1 April.
3. *Compare Diwali-season sales with the previous quarter.* Two labelled date ranges, both totals, and a per-day note because the periods differ in length.

4. *What is the UPI vs COD split by region?* A grouped chart with each payment method's share of the region.
5. *Top 10 customers by revenue.* A ranked table.

Every answer lists the metric and version used, the date range in DD-MMM-YYYY, and the filters applied.

9 Understanding Indian conventions in the application

Topic	How the application behaves
Currency	Rupee symbol, lakh/crore digit grouping (12,45,300), compact forms ₹45 L and ₹1.2 Cr
Financial year	1 April to 31 March. Q1 = April–June, Q2 = July–September, Q3 = October–December, Q4 = January–March. "This year" means the financial year to date
Dates and time	DD-MMM-YYYY, times in IST (Asia/Kolkata)
GST	`net_sales` is the taxable value **excluding** GST; `net_amount` is the invoice value **including** GST. The two Revenue definitions differ by exactly the GST, which is why the GST conflict is detected
Festivals	Diwali, Dussehra, Pongal/Sankranti, Eid, Holi and the End-of-Season sale have date windows that TARA understands ("Diwali-season")
Payment methods	UPI, card, net banking, wallet and cash on delivery (COD)

10 Troubleshooting

Problem	Cause and fix
`python -m app.seed` says the database is locked	The back end is still running. Stop it, then seed again
Browser shows “Failed to fetch” or a blank page	The back end is not running on port 8000. Start it and reload
Page loads but images are grey gradients	Offline, or the photo host is blocked. Harmless; the application is fully usable
Dataset stays on *pending*	In the no-Docker route a background thread processes it; check the back-end console for an error. With Celery, make sure a worker and Redis are running
Everything works but TARA says “offline deterministic mode”	Expected without an AI key. Set `AI_PROVIDER` and the key in the back-end environment to enable live AI
Login fails for a demo account	The database was seeded before the accounts changed. Run `scripts\reset-demo.ps1`
Port already in use	Change `--port` for uvicorn (and `VITE_API_PROXY` for the front end), or stop the other program
`npm install` is slow	First install downloads about 300 MB of packages; later runs are cached

11 Running the tests

```
cd backend;  python -m pytest tests -q
cd frontend; npx vitest run
cd frontend; npx playwright install chromium; npx playwright test
```

The Testing Report lists what each suite covers and the measured results.